equilibrium, i.e. to the equilibrium of the lattice is characterised by a time constant referred to as T_1. A measurement of T_1 is frequently an important part of the NMR experiment, and is required for the following reasons: (a) To select the most suitable radio-frequency pulse interval, the T_1 values of the various resonances need to be known. (b) Knowledge of T_1 values helps in determining the rates of chemical reactions through NMR. (c) T_1 provides information about the molecular structure. (d) In modern medicine T_1 is the most useful parameter used in NMR imaging. Frequently the image of the tissue seen is nothing but a map of the values of T_1 at different positions within the tissue. T_1 may be measured by FTNMR by applying a sequence of radio-frequency pulses. There are several different pulse sequences that may be used. A commonly used sequence is the inversion recovery sequence in which a 180° pulse is first applied. (As already explained, a 180° pulse changes the angle of the bulk magnetisation vector by 180°). The magnetisation is allowed to relax back to its equilibrium value for a time interval τ and a 90°pulse is applied and the FID is collected soon after. This signal will contain information about the strength of M_z after time τ. If the process is repeated for different values of τ, the spin-lattice relaxation time constant T_1 may be obtained.

Spin-spin relaxation is responsible for the return of the M_{xy} component of the magnetisation to its equilibrium value of zero. This is also known as transverse relaxation and is characterised by a time constant T_2 known as the spin-spin or transverse relaxation time. Consider Figure 8.6. Immediately after the application of the radio-frequency pulse, the net magnetisation is given by some maximum value (Figure 8.6(a)). If there is only a single spin in the sample resonating at precisely one frequency, the magnetisation (as considered in the rotating frame of reference) remains static and of constant magnitude. However, if there is spread of frequencies, different nuclei will precess at different frequencies. In the rotating frame of reference the magnetisation vectors for the spins will fan out, resulting in a reduction of the net magnetisation (Figures 8.6 (b) and (c)). Finally M_{xy} becomes zero, the equilibrium value (Figure 8.6(d)). Since T_2, the time constant characterising this process depends on the spread in frequencies, the width of the resonance peak is a measure of T_2.

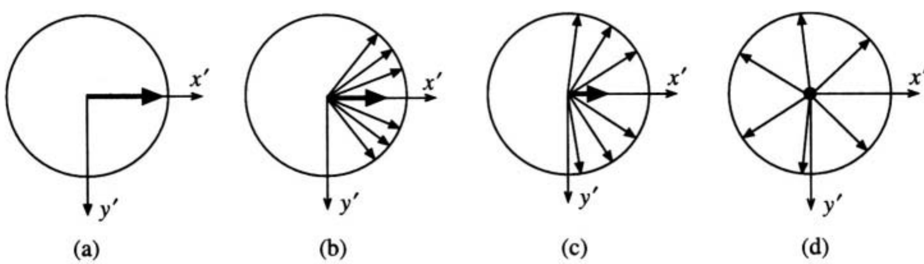


Fig. 8.6 Spin-spin relaxation—the return of M_{xy} to its equilibrium value.

Spin-spin relaxation processes are therefore those that cause line broadening. One component of this is the dipole-dipole interaction, which also causes spin-lattice relaxation, and therefore T_2 cannot be longer than T_1. Additional contributions to T_2 come from the z components of the small perturbing fields of the neighbouring spins. T_2 measurements provide information about molecular structure. They can also be used to determine rates of molecular diffusion. Information about T_2 helps to design pulse sequences that may simplify the spectra and enhance the resolution.

8.5.5 Spin-Spin Coupling

This is different from the spin-spin relaxation process explained above. Resonances in the NMR spectrum often do not occur as a single peak but are observed to be split into doublets, triplets, quadruplets or higher multiplets. The splitting is due to the neighbouring spins that are transmitted via the electrons of the covalent bonds between the atoms. Spin-spin coupling is thus a 'through-bond' coupling and occurs only due to nuclei within the molecule. Splitting of the peak can be caused only by nuclei with spin $I \neq 0$. The magnitude of splitting is designated by the spin-spin coupling constant *J*. It is measured in Hz and is independent of the applied magnetic field strength. Figure 8.7(a) is the spectrum of acetaldehyde (CH_3CHO) in which the CH_3 signal is split into a doublet due to spin-spin coupling of the protons with the neighbouring CHO proton through the three bonds between them (Figure 8.7(b)). The CHO resonance itself is split into a quadruplet owing to coupling with the three methyl protons. A complete theoretical treatment of spin-spin coupling is beyond the scope of this book. We substitute such a treatment by the following discussion.

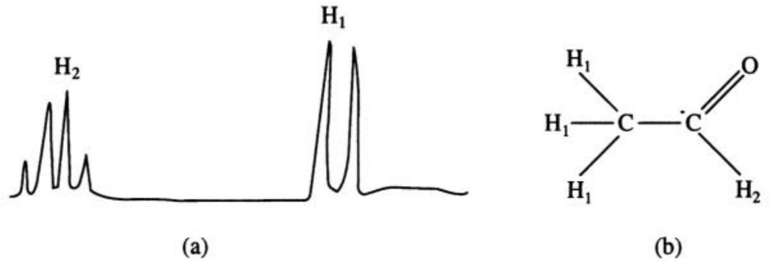


Fig. 8.7 (a) The NMR spectrum of acetaldehyde showing the splitting of the peaks (schematic) (b) Chemical structure of acetaldehyde.

Spin-spin coupling can be first order or second order. In the case of first order splitting the difference in frequency between the resonance peaks involved in the coupling is much greater than the coupling constant *J*. In the case of second order spectra this is not so. Obviously second order spectra will be much more difficult to interpret than first order spectra. First order multiplicities can be predicted by the following rules. (a) A singlet is a peak undisturbed by the presence of neighbouring nuclei. (b) Each neighbouring nucleus with spin $I \neq 0$ will split the peak into $(2I+1)$ components. If there are *n* equivalent nuclei, then the peak is split into $(2nI + 1)$ separate peaks. In Figure 8.7, the CHO peak is split by the three equivalent methyl protons (spin 1/2) into $(2 \times 3 \times (1/2) + 1) = 4$ components. (c) The components of a multiplet are evenly spaced and the spacing is a measure of the coupling constant *J*, in Hz. (d) The intensities of the components of the multiplet are proportional to the coefficients of binomial expansion. In the case of spin 1/2 nuclei, for example, the possible multiplets and the ratio of the peak heights are singlet, doublet 1:1; triplet 1:2:1; quartet 1:3:3:1; etc. (e) Since it is only required that the neighbouring nucleus possesses a spin $I \neq 0$, coupling between ^{13}C and ^{1}H (i.e. heteronuclear coupling) can occur as well as homonuclear coupling. However the relative abundance of ^{13}C is so small that such effects are usually not noticeable.

Coupling constants are affected by the following factors, among others; (a) dihedral angle between the nuclei (b) the electronegativity of the substituents on the moeity (c) valence angles (d) bond lengths. Figure 8.8 illustrates these parameters for a H-C-C-H molecule.

An empirical equation called the Karplus equation gives the dependence of the coupling constant on the dihedral angle of the above system.

$$J = 8.5 \cos^2 \phi - 0.28 \qquad \text{for } \phi = 0\text{-}90°$$

$$J = 9.5 \cos^2 \phi - 0.28 \qquad \text{for } \phi = 90°\text{-}180°$$

Clearly, measurement of the coupling constants is a sensitive method of determining molecular conformation as will be illustrated later.

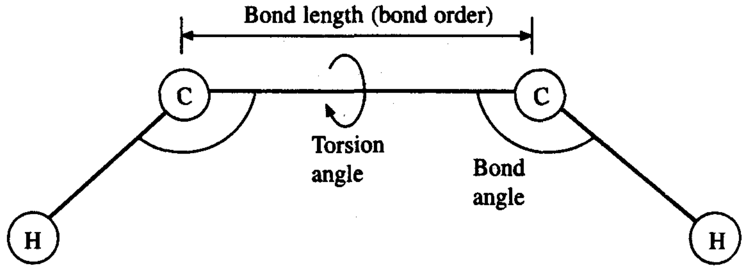


Fig. 8.8 The geometrical factors that affect the coupling constants for a H—C—H system

8.6 The Nuclear Overhauser Effect

The nuclear Overhauser effect or NOE illustrates the so-called 'double resonance' technique in which a second radio frequency field is applied to the sample in addition to the observing radio frequency field. The two fields may be denoted as H_2 and H_1, respectively. The second, strong, H_2 field is usually used to saturate particular resonances and is used in spin decoupling, saturation transfer or NOE experiments. H_2 is maintained at the resonance frequency of one resonance, to saturate it while another resonance is monitored by the non-saturating H_1 field. The monitored resonance may change in intensity if affected by the saturated resonance due to a change in the populations of the nuclear energy levels. Under normal conditions the relative populations of the spin states (or nuclear energy levels) is given by the Boltzmann distribution. If this distribution is altered by a saturation of some of the nearby resonances, and the consequent transfer of magnetisation by dipole-dipole coupling, it can lead to an enhancement of the observed signal. Strictly speaking a theoretical framework for NOE enhancement can be constructed only when the coupling is entirely a dipole-dipole coupling. Nevertheless it is possible to obtain an excellent indication of the degree of dipolar coupling between the nuclei by a measurement of the degree of NOE enhancement. The magnitude of dipolar coupling is strongly distance dependent and is proportional to $1/r^6$ where r is the distance between the nuclei. NOE is a 'through-space' effect and unlike spin-spin coupling, it does not need the interacting nuclei to be covalently linked to each other. Thus NOE is a powerful aid to the conformational analysis of molecules. A typical NOE experiment would consist of identifying all the resonances in the molecule and then saturating them one by one, each time measuring the enhancement of the intensity of the other resonances. In this way, it is possible for example to identify those portions of a polypeptide chain which may fold back to be close together in space though they are far apart in sequence.

8.7 NMR Applications in Chemistry

8.7.1 Chemical shift

The proton and ^{13}C chemical shifts in an NMR spectrum are a sensitive measure of the chemical environment of the nucleus. The shifts are influenced by a number of chemical factors, which are slightly different for protons and for ^{13}C nuclei. These are discussed separately below.

Table 8.2 gives the proton chemical shifts (in ppm from TMS) for some common chemical moieties. Some of the factors which influence the proton chemical shifts are: the state of hybridisation of the atom to which the proton is attached, van der Waals effects, anisotropy induced by the bond order of the neighbouring bonds or aromatic rings, solvent effects, effects of concentration and temperature, hydrogen bonding, contact shifts in organometallic compounds and chemical shift due to charged species.

The state of hybridisation of the atom (usually carbon atom) to which the proton is attached influences the amount of deshielding of the proton and hence the chemical shift. As the *s* character of the hybridisation increases, the resonance shifts more and more downfield (i.e. in the direction of increasing values of ppm or to the left of a spectrum as normally drawn). Thus a proton attached to a '*sp*' hybridised carbon will have its peak 5 to 8 ppm downfield of one attached to sp^3 hybridised carbon.

Non-bonded nuclei when pushed close to a proton distort the electronic shield around it, leading to an influence on the chemical shift.

Anisotropy effects on the chemical shift arise due to the fact that the electrons around neighbouring nuclei are asymmetrically distributed. This results in a variation of the chemical shift when the position of the resonating spin is changed with respect to the neighbouring nucleus. Thus in a molecule such as cyclohexane, protons which are equatorial to the ring will have different chemical shifts from those which are axial. Anisotropy effects are dependent on the order of the neighbouring bonds. Single bonds lead to less anisotropy than double or triple bonds. In aromatic rings, the circulation of π electrons around the ring results in a shielding of nucleus above and below the plane of the ring, leading to an upfield shift, and a deshielding of protons in the plane of the ring, leading to a downfield shift of the resonance. Van der Waals or anisotropy effects are also produced by the solvent in which the sample is dissolved.

Hydrogen bonding produces a downfield shift in the resonance of the protons involved in the hydrogen bonds, since such a bond is between electronegative atoms, which would withdraw the electron away from the proton leading to a deshielding.

8.7.2 *Spin-spin coupling*

Spin-spin coupling is dependent on the nature of the neighbouring nuclei and coupling constants carry information about the chemical structure of the molecule. It is however necessary to first of all distinguish between chemical equivalence of atoms and the magnetic equivalence of atoms. In the case of chemical equivalence the spins have identical chemical environments and thus identical chemical shifts. Magnetically equivalent nuclei will not only have the same chemical shift but will also possess the same coupling constant with every other nucleus in the molecule. Figure 8.9(a) shows tert-butanol in which the methyl hydrogens are both chemically and magnetically equivalent. In Figure 8.9(b) the methylene hydrogens of diflouroethylene are chemically equivalent but magnetically non-equivalent since each hydrogen is coupled to each of the two flourines with a different coupling constant.

Coupling constants in molecules can be either positive or negative in sign and up to a few tens of Hz in magnitude. Of the chemical factors affecting spin-spin coupling, the dihedral angle, the bond

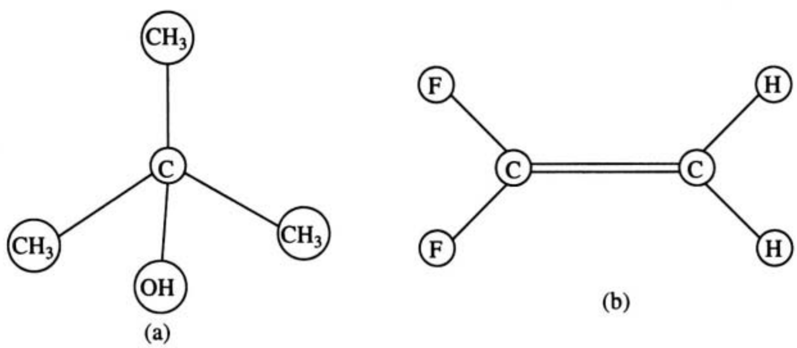


Fig. 8.9 Chemical and magnetic equivalence. (a) Tert-butanol. (b) Difluoroethylene

length and bond angle in a H-C-C-H system has already been discussed. Other factors are: (1) the electronegativity of the substituents, (2) valence angles and (3) bond lengths. Vicinal protons, i.e. protons such as those in the system H-C-C-H, show a decrease in the coupling constant if an electronegative substituent is attached to the carbon atoms. If more than one such substituent is present, the decrease is greater. In cyclic systems the steric disposition of the electronegative substituent also affects the coupling constant, being larger when it is equatorial and smaller when it is axial. The effect is the greatest when the substituent is trans-coplanar to one of coupling protons. For geminal protons, i.e. protons attached to the same carbon atom, the coupling is usually negative. An additional electronegative substituent decreases the values further (magnitudes increase, but the sign remains negative). Neighbouring π bonds such as those found in aromatic systems decrease coupling constants still further. Vicinal coupling constants are sensitive to the angles between the C–H and C–C bonds, as well as to the C–C bond lengths. In both cases the coupling constants decrease as the values of the angles or bond lengths increase. Long range spin-spin coupling extending over four or five bonds is also possible, through the magnitude is usually small, of the order of a few Hz.

8.7.3 ^{13}C NMR

Among the spin 1/2 nuclei ^{13}C is of interest in chemistry, especially organic chemistry. However the isotope ^{13}C has a natural abundance of only 1.11 % as compared, for example to 99.98 % for ^{1}H. ^{12}C has an abundance of greater than 90 % but spin $I = 0$. In spite of its low abundance ^{13}C NMR is frequently used in chemistry. ^{13}C chemical shifts are in the range 0-230 ppm with reference to TMS. Many of the factors that affect the ^{13}C chemical shifts are the same as for ^{1}H NMR. The state of hybridisation of the observed ^{13}C nucleus is an important factor. sp^3 carbons resonate between -20 and 100 ppm with respect to TMS; sp^2 carbons between 120 and 240 ppm and *sp* carbons 70 and 110 ppm. The electronegativity of the substituent is another important factor influencing the chemical shifts, as are steric and van der Waals effects. Mesomeric effects arise from electron-donating groups attached to benzene rings that shield the ortho and para carbon atoms (electron withdrawing groups would deshield these positions). Hydrogen bonding, attachment of heavy atoms, bond conjugation, attachment of deuterium atoms instead of hydrogen, *pH* and the solvent, all contribute to the ^{13}C chemical shift.

Spin-spin coupling in ^{13}C NMR can be observed between the ^{13}C nucleus and neighbouring protons, as well as between two ^{13}C nuclei. The latter effect is more difficult to observe, since due to the low natural abundance of ^{13}C isotopes, the chances of two such nuclei occuring as neighbours in the molecule is very small.

^{13}C NMR also lends itself to the study of sample in the solid state. In solutions, local magnetic fields due to the nuclei of neighbouring molecules will be averaged out due to the rapid tumbling motion and will not have any effect on the width of the resonance peak. In solids however, each individual molecule would be surrounded by a different static environment. This leads to a widening of the resonance. Until the advent of recent high resolution solid state NMR methods, ^{13}C NMR was only sparsely applied in the solid state, and many interesting aspects of solid state chemistry, polymer chemistry, etc. could not be studied. High-power decoupling cross polarisation (CP) and Magic Angle spinning (MAS) are two of the relatively recently developed techniques which make it possible to obtain high resolution even in solids. So much so, at least for low molecular weight compounds, solid state NMR gives a resolution comparable to the solution state.

8.8 NMR Application in Biochemistry and Biophysics

In addition to the fact that biologically important molecules are usually of very large molecular weight and size, biological systems also reveal more complex interactions between the molecules or groups of molecules. The study of such complex systems through NMR has in recent times become possible in much greater detail and precision with the advent of very high-field magnets and techniques such as multi-dimensional NMR. Even before these developments, however, NMR was being used in biochemistry to determine concentrations, study the pH behaviour of biomolecules and to elucidate the conformation of small biologically important molecules.

8.8.1 Concentration Measurement

The determination of relative concentration by NMR is made simply by measuring the area under the NMR absorption signal. If the spectrum has been calibrated against a signal of known concentration, absolute concentrations can be measured in this way. Precautions however have to be taken against several sources of error that may creep in. One of these is the fact that if, due to different relaxation times, one signal has become saturated, while another is not, the measurements of the relative concentrations may be wrong. Another source of error is the possible uneven intensity of the excitation pulse which may lead to different frequencies being excited with different intensities. It is however possible to choose signals which do not suffer from such drawbacks, enabling fairly precise measurements of concentrations.

8.8.2 pH titration

The protonation or deprotonation of functional groups in biological molecules has a detectable influence on the chemical shift. The rates of proton exchange with the solution are much faster than the NMR time scales and the average effect is observed. This effect is not only sufficiently strong to be detected at the protonation/deprotonation site, but even several bonds away in the molecule. The protonation of a particular site depends on its *pK*. pH titration is the measurement of the chemical shift as a function of pH and helps to measure the *pK* of the site. The protonation or deprotonation of the site at its *pK* value is marked by a sharp change in its chemical shift at that value of the pH. In proteins, histidine residue resonances are especially dependent on the pH of the solution and together with other such resonances can be used as a monitor of denaturation. This is because protonation sites in the interior of the proteins will be progressively exposed to the solution as denaturation proceeds, which can be detected by the dependence of the spectrum on pH or temperature (or other denaturing conditions). Inorganic phosphates participate in many enzymatic reactions, and exist, during the course of the reaction, in many different protonated states such as H_3PO_4, HPO_4^-, HPO_4^{2-}, and PO_4^{3-}. By using ^{31}P NMR and pH titration, the course of such a reaction can be followed.

8.8.3 Conformation of Biomolecules

The NMR spectra of biomolecules contain a large amount of structural and conformational information. The problem is to extract this information. Even at relatively high resolution, most of the peaks in a ^1H NMR spectrum of a protein are broad. The causes for such broadening in macromolecules are as follows. (a) A biopolymer usually contains more than one unit of any particular monomer, and each unit would be in a slightly different chemical environment, leading to a widening of the resonance. This cause for the widening is the same as that occurring in solids. (b) The presence of other nuclei close-by leads to dipolar relaxation in excess of what would be observed for the monomer and contributes to the broadening of the resonance. (c) Chemical exchange of protons between different close-by sites can also cause line broadening. (d) Paramagnetic ions are frequently found in biological molecules and are a source of line broadening. The consequence of all this is that detailed conformational studies are possible by the conventional one dimensional and or continuous wave NMR only in the case of small organic molecules (as discussed earlier). However, certain aspects of biomolecular conformation can be studied by conventional NMR. One such is the helix to random-coil transition in polypeptide chains. Similarly one can use NMR to follow the denaturation-renaturation transition in proteins. In either case the NMR spectrum of the random coil would be a composite of the spectra of the constituent amino acids. The spectrum of the ordered structure would deviate markedly from the simple random-coil spectrum. The transition can thus be followed as function of temperature, *pH* or any other relevant parameter.

8.8.4 Two-dimension NMR

As mentioned earlier, detailed conformational studies of macromoleules are possible now with the development of two-dimensional techniques. A one-dimensional FTNMR experiment involves the application of a RF pulse after the sample has been allowed to achieve equilibrium during the 'preparation' period (Figure 8.10(a)). Immediately after comes the detection phase where the FID (i.e. the signal S is collected as a function of time (here represented by the variable t_2). A Fourier transform of the signal into the frequency domain results in the NMR spectrum.

$$S(t_2) \rightarrow S(\omega_2)$$

In the case of two dimensional NMR spectroscopy an additional time period (denoted t_1) and called

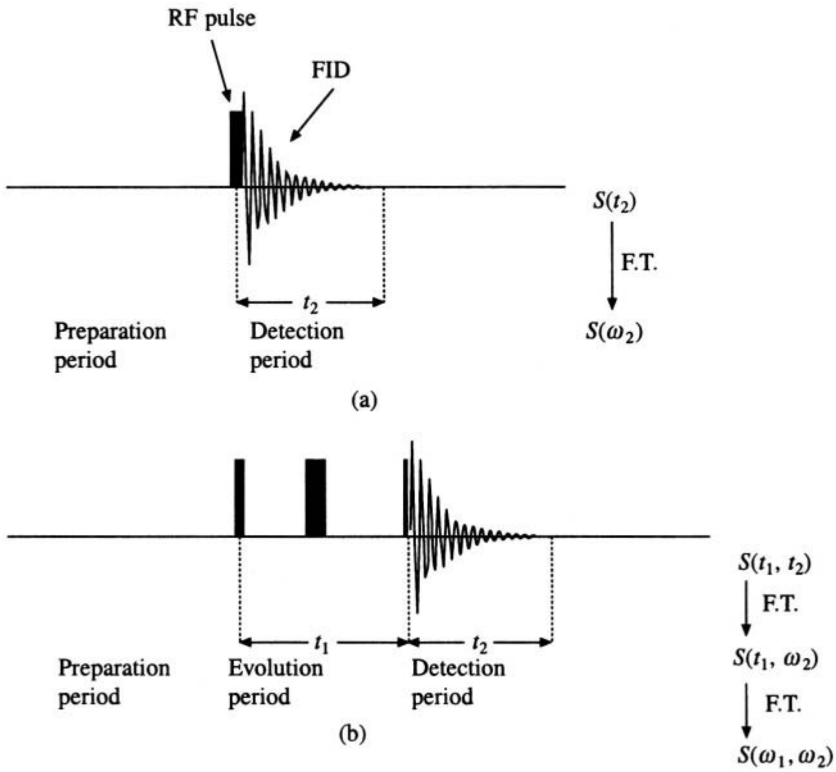


Fig. 8.10. A schematic representation of the NMR experiment. (a) 1-D experiment. (b) 2-D experiment.

the evolution period is introduced, after which the FID is collected (Figure 8.10(b)). During the evolution period another RF pulse may be applied and a further period of time, called the mixing period, may be allowed to elapse before the FID is collected. In each experiment, the time t_1 may be given a different value, ranging from zero to some maximum value depending on the relaxation times. Each time the FID is collected and Fourier transformed to yield $S(t_1, \omega_2)$. The several spectra so obtained are arranged in a data matrix which can be subjected to a second Fourier transform, column wise

$$S(t_1, \omega_2) \rightarrow S(\omega_1, \omega_2)$$

The spectrum thus will have two dimensions. ω_1 is along one of the dimensions and ω_2 along the other. The ω_2 axis contains information about coupling interactions with other nuclei during the evolution period t_1, and the ω_2 axis contains information about the chemical shifts. The precise pulse sequence applied to the samples, gives rise to different types of coupling information being displayed along the ω_1 axis. There are currently more than 50 different pulse sequences that have been published. The two most commonly used sequences in biological research are abbreviated as COSY and NOESY.

COSY is an abbreviation for correlated spectroscopy. There are different types of COSY pulse sequence. The so-called homonuclear shift correlated spectrum displays the J coupling, or the spin-spin coupling through bonds as a two-dimensional plot. Figure 8.11 is a simple COSY spectrum in which the connectivity of spins in the molecule can be mapped out. COSY spectra are of great use in assigning the peaks to the nuclei that cause them. Solving the chemical structure of the molecule can be accomplished through a COSY spectrum. The assignment of peaks in the spectrum of a large biomolecule such as a protein is relatively easily accomplished.

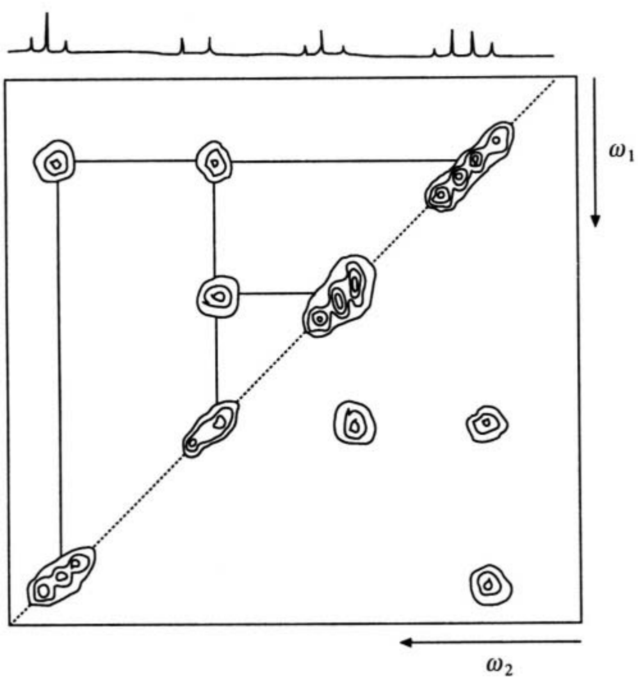


Fig. 8.11. A simple COSY contour plot shows the connected atoms.

NOESY is an abbreviation for Nuclear Overhauser Enhancement spectroscopy. The nuclear Overhauser effect is a 'through-space' effect and can be used to determine the spatial separation between nuclear spins. NOESY spectra are similar to COSY spectra except that in this case the off diagonal peaks correspond to 'through-space' connectivity of spins. The strength of the effect varies as $1/r^6$ where r is the distance between the spins, and measurement of the strength of the off diagonal NOE peaks in a 2D NOSEY spectrum can be used to estimate distances between spins. As explained in the next section, such measurements are of great use in arriving at the three-dimensional folding patterns of biopolymers such as proteins in which the polypeptides chain would fold back on itself to bring into close proximity nuclei which are widely separated along the chain.

8.8.5 Determination of macromolecular structure

The experiment to determine the three dimensional structure of a macromolecule usually consists of two steps. The first step is to assign all the resonance peaks that occur in the spectrum. The high-resolution spectrum of a macromolecule such as a protein is very complex and only through a 2-D spectrum such as COSY is it possible to trace the connectivity in the molecule and assign the peaks. In proteins, especially, but also in nucleic acids, the interpretation of the 2-D COSY spectrum is sometimes made easier by the existence of recognisable patterns of connectivity which can be attributed to specific amino acids or base-sequences. The second step in the structure determination is to identify secondary and tertiary structures from the J-connectivity patterns and from NOE data. A NOESY spectrum will give information about the 'through-space' couplings between the spins in the molecule. The NOE intensities may be divided into classes as strong, medium and weak NOEs corresponding to a distance less than 2.5 Å, 3.0 Å and between 3.0 and 4.0 Å, respectively. (Usually no NOEs are observable beyond 4.0 Å). The distance data so obtained may be used to attempt a reconstruction of the three-dimensional folding of the biopolymer chain. This is mathematically a complicated problem and no general algorithm exists which can lead from the distances to the geometry in a reliable way. There are however several approaches to solving this problem, each being successful to a lesser or greater degree. The NOE distances alone are not sufficient to define the structure. Oxygen and nitrogen atoms, for example, are usually not seen in the NMR spectra and their positions must be inferred from the positions of connected protons or ^{13}C nuclei. Distance geometry algorithms therefore use all chemical information available and may be performed in torsional-angle space where the degrees of freedom are the dihedral angles of the main chain and the side chains. Additional constraints in the form of non-bonded van der Waals contacts, hydrophobic interactions and so on, may also be used in arriving at the structure. A commonly used technique is restrained molecular dynamics. As described elsewhere in the book the observed experimental data, in this case the NOE distances, are written as a pseudo potential, along with other semi-empirical potential terms such as bond length energy, bond angle energy etc. This method has been successfully used in many instances in arriving at precise three-dimensional structures of small proteins (Mw ~20,000). Structures of larger proteins (Mw > 50,000) are still somewhat inaccessible to current techniques.

8.9 NMR in Medicine

Some of the most spectacular applications of modern NMR techniques have been in the area of medical imaging, namely Magnetic Resonance Imaging or MRI. Unlike NMR spectroscopy, where the NMR signals are obtained as a single spectrum from the entire sample, imaging requires that the signal be resolved in space. The different signals from the different parts of the object are reconstructed on the computer screen as an image of the object. The electromagnetic fields and waves used in NMR pass easily through living bodies and signals from any interior portion can be detected and processed.

In this way it is possible to build up clear images which will be of great use in medicine. Three NMR measurable parameters are commonly used in imaging. The first is the spin-density or the number of a particular type of nucleus (usually the water protons) per unit volume. This parameter can be deduced from the intensity of the corresponding resonance peaks at all the different portions of the object. The second parameter that can be used in imaging is the spin-lattice relaxation time T_1. Different types of tissue within the object would have different T_1 values and if these values are mapped out, an image is obtained. The third commonly used parameter; viz. the spin-spin relaxation time T_2 is also treated the same way. Thus, in contrast to X-ray imaging (including the CAT scan), where the X-ray absorption is the only detectable parameter, there are at least three parameters which can be detected in NMR imaging and NMR tomography. This allows greater detail and also different types of tissues to be imaged. In addition, NMR does not use any highly ionising radiation to produce its images.

The most basic principle in the imaging technique is the selection of a slice to be imaged from the entire volume of the object. Unlike as in NMR spectroscopy, where the magnetic field is kept steady over the entire sample, in the case of NMR imaging, spatial selection is achieved by applying a magnetic field gradient on top of the static magnetic field. This would mean that particular spins, such as for example protons (^{1}H) come into resonance at different frequencies at different parts of the sample along the axis of the field gradient. If the sample is irradiated by a radio-frequency excitation pulse in a very narrow range of frequencies, then only the spins in a particular spatial slice of the object, perpendicular to the direction of the magnetic field gradient would come into resonance and contribute to the NMR signal. If a 2D image of such a slice is obtained, many such images can be stacked one on top of the other to yield a complete image of the object.

In order to obtain a 2-dimensional image of the slice, a 2D-Fourier imaging technique is most commonly used. The slice selection gradient is assumed to be applied along the z axis. Thus the slices are perpendicular to this axis and parallel to the xy plane (Figure 8.12). After the first excitation pulse, a gradient $G(x)$ is applied for a time t_x along the x-axis. Immediately after, another gradient $G(y)$ is applied along the y-axis for a time t_y. The NMR signal is then collected. The process is repeated for different values of t_y ranging from 0 to the maximum, each time obtaining the signal for different values of t_y. Thus a matrix of signal of size n^2 is obtained where n is the number of different values of t_x and t_y. It can be shown that the matrix when Fourier transformed will result in 2D image of the slice. The time taken to obtain such an image is usually less than 10 minutes. While this is sufficiently short for imaging live animals, if the process were to be repeated several times in order to obtain the complete three dimensional image, it is difficult to keep the object still and motionless for the few hours that it might take. Other methods have been developed where additional slices are measured almost simultaneously and the entire image reconstructed in one go. This may entail a loss in resolution, contrast, etc.

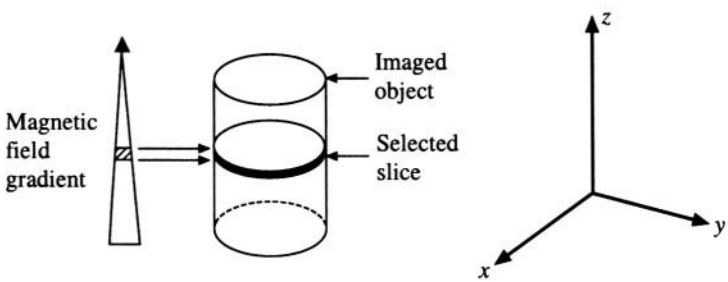


Fig. 8.12 The geometry of the MRI experiment.

One of the most important applications of NMR tomography is the imaging of sensitive parts of the human body such as the head. X-rays are highly attenuated by the skull, making X-ray imaging of the brain quite difficult, in addition to the fact that X-rays are ionising radiation and therefore can cause damage. The radio frequency radiation used in NMR passes almost unhindered through the skull and has no as yet known ill effects on the tissues. Certain images of the skull, such as the so-called sagittal view are simply not possible with X-ray tomography. In addition, pathological conditions are easily identifiable especially with the possibility of using different modes of imaging (i.e. spin density, T_1 or T_2). Apart from these three parameters it is also possible to image the organs by means of the difference in the chemical shifts. NMR imaging may also be used to study continuous flow processes within the body such as blood flow. The excitation pulse sequences may be adjusted so that only the flow is visible, leading to a clear image of the blood vessels alone.

NMR imaging at very high spatial resolution is called NMR microscopy. At present the maximum obtainable resolution is about **20 μM,** mainly because below this limit there are simply not enough protons in the sample volume to give a signal with sufficiently signal-to-noise ratios. Thus NMR microscopy, while potentially of great use, since the sample is preserved intact even while observing interior portions, has yet to reach the resolution of the best light microscopes.

A related application of NMR in medicine is magnetic resonance spectroscopy (MRS) also known as 'in vivo spectroscopy'. Once again a spatial selection technique is used to obtain a reasonably high resolution NMR spectrum from a small volume of the object, such as typically, the human body. MRS differs from MRI in that in the latter, information is available regarding only one aspect of the sample such as the spin density. MRS looks at the entire information content of the spectrum. The spatial selection techniques are somewhat different in detail from those used in imaging. This is because the portion of the sample volume may be defined with respect to either the laboratory frame of reference or with respect to certain biological structures, i.e. whether the molecule investigated is inside or outside a defined closed structure such as a cell. Typically in vivo spectroscopy is used to monitor the state of various identifiable metabolites as a function of the progress of some biological condition, e.g. disease. This method is thus in the process of developing into a sophisticated early diagnostic tool as well as a way of following the results of therapy. The method however is only now moving out of the laboratory and into the clinics.

9

Molecular Modelling

9.1 Introduction

X-ray analysis or NMR or other methods of structural characterisation of the molecules give, as output, a set of numbers that specify the relative positions of each of the atoms of the molecules in space. In order to use the information present in that list, human capabilities almost always demand that it be converted to a form in which it can easily be visualised. Before the advent of powerful computer graphics, the co-ordinates were used to build models in wood or plastic. These suffered from the following disadvantages: (1) They were tedious to build. (2) They were static; they could not be easily altered to try out new ideas about the structure. (3) They were not accurate. (4) They were fragile. (5) They were heavy, prone to collect dust and were difficult to maintain. (6) They took up a lot of space. (7) They were expensive.

Despite these disadvantages they were widely used and several new ideas in biology can be credited to such models, the most notable being the double-helical structural model of DNA, constructed by J.D. Watson and F.H.C. Crick with metal parts.

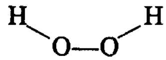
Computers have made all the difference to molecular modelling. They possess none of the disadvantages listed above. The chief problem with computer graphics is that they are two-dimensional pictures of 3-dimensional objects. Nevertheless one can get over this problem to a great extent by using techniques such as stereopsis. Most programs also have a facility which allows the user to 'rotate' the picture on the display screen so that different parts of the molecule come into view continuously giving the illusion of watching a real three-dimensional model. Thus the advantages of using computer graphics far, far, outweigh the single disadvantage.

9.2 Generating the Model

If the co-ordinates of the molecule are already available, from, say, an X-ray crystallographic analysis, then they are just fed into the computer program which will interpret each data point as an atom and display it according to the mode and orientation chosen by the user. If the co-ordinates are not available, but if the chemistry of the molecule is known (i.e. the chemical formula for a small molecule; and the amino acid sequence for a protein), it is possible to generate the co-ordinates for the molecule, *ab initio,* on the computer. As an example we demonstrate the generation of the co-ordinates for a very simple molecule, namely, hydrogen peroxide

9.2.7 *Building the structure of H_2O_2*

Hydrogen peroxide, H_2O_2 has the chemical structure



Chemical studies have established that the molecular dimensions are as given below:

Distances	Angles
O–H = 1.1 Å	**H–O–O = 120°**
O–O = 1.5 Å	

The one variable in this molecule, which is left to the choice of the modeller, is the dihedral angle[8] or the torsion angle H–O–O–H. If this is changed, it implies a rotation of one hydrogen atom about the O–O bond, with the other hydrgen atom kept fixed. In order to generate the co-ordinates for such a molecule, we first define the frame of reference as in Figure 9.1. Note that it is conventional to

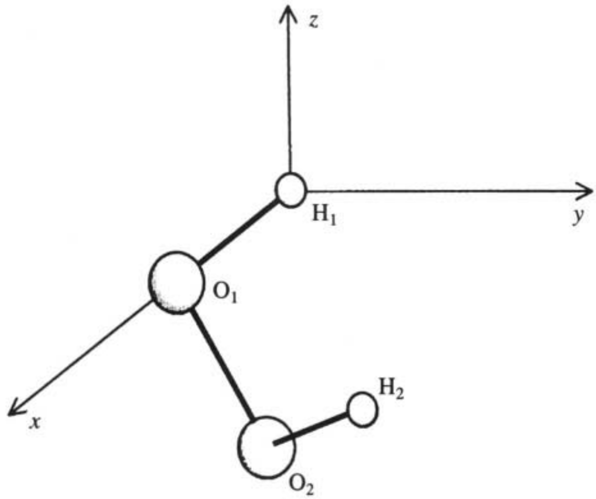


Fig. 9.1. Frame of reference chosen to generate H_2O_2

[8] A dihedral angle is the angle between two planes, measured as the angle between the normal vectors to the two planes. A torsion angle is the angle made between the two outermost bonds when three consecutive bonds are considered. For example, consider the atoms *P, Q, R* and *S* bonded in a chain, *P* to *Q, Q* to *R* and *R* to *S.* The torsion angle between *P–Q* and *R–S* is measured by first viewing the four atoms along the central *Q–R* bond. In this view, atoms *Q* and *R* superpose on one another. The angle made by the far bond, say *R–S,* with respect to the near bond, *P–Q,* measured in the clockwise direction, is the torsion angle. This is the same as the angle between the plane *PQR* and the plane *QRS,* and hence is sometimes called the dihedral angle.

choose a right-handed Cartesian co-ordinate system. We first place one of the hydrogen atoms, which we call $\mathbf{H_1}$, at the origin. Co-ordinates of $\mathbf{H_1}$ are therefore

H1 0.0 0.0 0.0

Next we place the O1 atom along the x-axis, at a distance of 1.1 Å from orign. The co-ordinates of O1 are therefore

$\mathbf{O_1}$ **1.1 0.0 0.0**

The third atom $\mathbf{O_2}$ placed in the *xy* plane, at a distance of 1.5 Å from O1 and such that the $\mathbf{O_2–O_1}$ bond makes 120° with the $\mathbf{H_1–O_1}$ bond. An elementary geometrical analysis shows that the co-ordinates of $\mathbf{O_2}$ are

$\mathbf{O_2}$ **1.5 1.3 0.0**

Finally in order to determine $\mathbf{H_2}$, we place it at a $\mathbf{O_2–H_2}$ distance of 1.0 Å and $\mathbf{O_1–O_2–H_2}$ angle of 120°. This still leaves us the freedom to place the atom anywhere on a circle. If we choose the torsion angle to be + 60, the co-ordinates of $\mathbf{H_2}$ are

$\mathbf{H_2}$ **1.7 2.0 0.8**

Thus the spatial positions (co-ordinates) of all the atoms in hydrogen peroxide molecule can be obtained by simple geometrical considerations. The lengths between the bonded atoms, the angles and the torsion angles, are called the internal parameters or internal co-ordinates of the molecule. A complete set of internal co-ordinates is exactly equivalent to the set of Cartesian co-ordinates and one can be derived from the other in the complete and non-redundant way.

9.2.2 Building protein structure

The geometry of each amino acid in the protein is known, i.e. the relevant bond lengths and angles are known. Also known is the geometry of the linking peptide bond. Despite this, building a protein chain is not a trivial task since the torsion angles are not known. In the amino acids with the longer side chains, especially, the orientations of the various chemical groups depend on the side chain torsion angles. Apart from this, in the main chain, or the peptide backbone, the ϕ and the ψ angles are not fixed by chemistry. Hence, unless the fold of the protein is known or guessed at, a three-dimensional model of the entire protein molecule cannot be built.

It is known that information regarding the final three-dimensional structure of the protein is inherent in its amino acid sequence. Hence knowledge of the latter should, in principle, lead us to the former. However, the rules of protein folding are still unknown. It is not yet possible to go from the sequence to the structure without additional experiemental information. This is called the 'Protein Folding Problem' and is one of the great, unsolved problems in molecular biology. It may be succinctly stated as follows: Given the amino acid sequence of a protein, how does one predict its three-dimensional structure. Several methods of structure prediction are available, though none of them have been completely successful. One of the most popular prediction methods was developed by P.Y. Chou and G.D. Fasman. The Chou and Fasman method has been quite successful in predicting the secondary structure of proteins. The method is based on a statistical analysis of the known protein structures that have been collected in a computer databse.

Extensive collections of published three dimensional structures are now available in computer readable format. A collection of all the published protein structures is available at the Research Collaboratory for Structural Biology in USA and is called the Protein Database or PDB for short. A typical entry in this databse would consist of the name of protein, the authors of its structure, the methods used to determine the structure, and a list of the coordinates and thermal parameters of each atom in the structure including solvent atoms, ligands, etc. There are now over fifteen thousand entries in the database. Most of the structures deposited in the database have been solved by X-ray crystallography. There are also about 500 structures determined by NMR techniques. The number of structures in the database is increasing rapidly and about 2000 new structures are added to the database every year. The database can be accessed through computer networks or can be obtained on CD-ROMS. A similar high quality database is the crystallographic structural database or the CSD at the University of Cambridge, UK. This contains more than 1,00,000 entries of the co-ordinates of the crystal structures of small organic molecules, i.e. those with a molecular weight less than about 1000.

Chou and Fasman used the protein structural database to estimate the frequency with which each amino acid occurred in one of the three main secondary structural motifs, i.e. **α-helix, β-strand** and reverse-turn. The normalised values that they obtained are given in Table 9.1.

Table 9.1 Chou-Fasman propensity values of the 20 amino acid residues for each of three secondary structures

Residue	α-helix	β-strand	Reverse turn
Ala	1.45	0.97	0.15
Cys	0.77	1.30	0.31
Asp	0.98	0.80	0.33
Gln	1.53	0.26	0.12
Phe	1.12	1.28	0.19
Gly	0.53	0.81	0.45
His	1.24	0.71	0.18
Ile	1.00	1.60	0.1[illegible]
Lys	1.07	0.74	0.27
Leu	1.37	1.22	0.14
Met	1.20	1.67	0.18
Asn	0.73	0.65	0.45
Pro	0.59	0.62	0.41
Glu	1.17	1.23	0.15
Arg	0.79	0.90	0.27
Ser	0.79	0.72	0.41
Thr	0.82	1.20	0.27
Val	1.14	1.65	0.08
Trp	1.14	1.19	0.30
Tyr	0.61	1.29	0.33

The numbers represent the propensity for each amino acid to take up the respective conformation. Thus glutamine has a very high propensity to occur in a helix, i.e. to be a helix-former. Similarly isoleucine strongly prefers to exist in a **β-strand.** The propensities for the reverse turn are not as well expressed. However, it may be noticed that some residues like glycine or asparagine have a fair degree of propensity for the reverse turn, without a strong preference either for the **α-helical** or **β-strand** conformation. The way this table is used to predict the secondary structure of a polypeptide chain is illustrated by means of an example.

Consider a 18-residue sequence such as shown in Figure 9.2. The **α-helical, β-strand** and reverse turn propensities for each residue is also indicated as taken from Table 9.1. Also indicated in the

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Sequence	Lys	Ile	Thr	Val	Val	Gly	Ala	Gly	Ala	Asn	Gly	Leu	Ala	Gln	Ala	His	Ser	Leu
Propensities																		
α-helix	1.07	1.00	0.82	1.14	1.14	0.53	1.45	0.53	1.45	0.73	0.53	1.37	1.45	1.53	1.45	1.24	0.79	1.37
β-strand	0.74	1.60	1.20	1.65	1.65	0.81	0.97	0.81	0.97	0.65	0.81	1.22	0.97	0.26	0.97	0.71	0.72	1.22
β-turn	0.27	0.15	0.27	0.08	0.08	0.45	0.15	0.45	0.15	0.45	0.45	0.14	0.15	0.12	0.15	0.18	0.41	0.14
Average Propensity (window = 6)																		
α-helix	1.01	1.03	0.95	0.96	0.91	0.97	0.92	0.82	0.96	1.01	1.18	1.18	1.26	1.31	1.31	1.28	1.21	1.13
β-strand	1.30	1.37	1.28	1.43	1.30	1.26	1.09	0.95	1.02	0.91	0.81	0.81	0.82	0.81	0.81	0.78	0.91	0.88
β-turn	0.19	0.17	0.22	0.19	0.24	0.22	0.28	0.34	0.29	0.30	0.24	0.24	0.20	0.19	0.19	0.20	0.22	0.24
	β-strand (1–6)						β-turn (7–10)				α-helix (11–18)							

Fig. 9.2 Secondary structure prediction of a 18-mer oligopeptide using the Chou-Fasman method

figure are the average values of the propensities taken six at a time. (Hexa-peptides are the normally used probe length, though other lengths also may be used. For the **α-strand** especially the probe length can be as short as five peptides). An examination of the averages shows that the average strand propensity is the highest for the hexa-peptide that starts at residue 1. A **β-strand** can be considered to nucleate at this point, since the average propensity for an **α-helix** is rather low at the same point. The strand can be extended in both directions until a region is reached where the helix is a more probable structure. Thus on comparing the propensities for both types of structure it can be determined that the residues 1 to 6 are most likely to be **β-strand,** while residues 11 to 18 are most likely to be **α-helical.** The determination of possible reverse turns is more complicated since each residue has a propensity value for its occurrence at each position of the reverse turn. (A reverse turn has four residues i.e. four positions). The principle however is the same, and, after a thorough analysis, the entire polypeptide chain can be classified as **α-helix, β-strand,** reverse turn or none of the above, i.e. random coil. Such methods of secondary structure prediction are now available as easy-to-use computer programs. Given the sequence of a protein one can use the program to quickly predict its secondary structure.

The problem of arriving at the structure of the protein from knowledge of its sequence is nevertheless far from solved. Firstly the secondary structure prediction methods, such as the Chou and Fasman method are still quite approximate, and all the regular structural elements of the protein chain cannot be identified. Predictions made have often turned out to be wrong and the overall rate of success does not exceed 60% in the best test cases. Methods are still evolving and combinations of methods are expected to have much higher success rates.

The other major obstacle in protein structure prediction is that the arrangement of secondary structural elements in space follows rules that are not completely understood. This comes in the way of arriving at the tertiary fold of the chain from a knowledge only of the sequence. The prediction of the three dimensional structure of a series of proteins with homologous sequences is much easier if the structure of one of them is known. Thus almost the first task that is done when the sequence of a protein is known to compare it with all other known protein sequences and identify those which are similar. If the structure of one of the similar proteins is known through other experimental or theoretical techniques, it can be used as a guide in generating the structure of the new protein. The principle which is followed is that stretches of similar sequences must have similar or the same secondary structure. If the two sequences show a high degree of overall similarity (homology), then the tertiary fold of the two sequences can also be taken to be similar, and the co-ordinates of the unknown protein can be generated by appropriately modifying the co-ordinates of the known protein. This method of generating protein structures is especially used to build the initial trial model in the molecular replacement method in X-ray crystal structure analysis.

9.2.3 Building nucleic acid structure

To build a DNA double helix is relatively simple as compared to building a protein chain. All one needs are the co-ordinates of one nucleotide unit and the helical transform that generates the next repeat along the strand. A rotation about the two-fold axis perpendicular to the helix axis generates the other strand. Monomer co-ordinates for A, B and Z type DNA are available in the literature. Other unusual, regular DNA structures such as tetraplexes, triplexes, etc., can also be generated by following the same principle. "Real" DNA molecules are not perfectly regular and show a sequence dependent variation from the standard structure at each base-pair step. Since no data on the correlation between the sequence and the microstructure exists, it is difficult to build models that show this variation. Thus most DNA models are regular helices.

Models of RNA structures are much more difficult to build. Only a few structures of large single stranded RNA molecules exist and the principles of RNA structure are not well known. An analysis of the sequence of the RNA molecule, if known, enables prediction of secondary structure, specifically regions of the molecule which base pair with each other and therefore can be thought to exist as short double-helical stretches. The way in which these double helices are put together to form the three-dimensional RNA structure is an unsolved problem.

9.2.4 Displaying and altering the generated model

There are many ways of representing the three dimensional molecule on the two dimensional computer display. Choosing the appropriate mode of representation depends on the particular information that is required to be emphasised.

Small molecules are typically represented by the so-called ball and stick model (Figure 9.3), where each atom is shown as a sphere and each covalent bond as a stick. The radius of the sphere is commonly taken to be proportional to the Van der Waals radius of the respective atom. In colour representations, a commonly used code is hydrogen—white, carbon—black, nitrogen—blue, oxygen—red, phosphorous and sulphur—yellow. If the anisotropic thermal vibrations of the atom also need to be represented, then ellipsoids are used instead of spheres to depict the atoms (Figure 9.4). The radius of the atoms may be increased to an extent where the sticks representing the bonds are hidden. Such a model is called a space-filling representation. A chemically accurate version of a space-filling model was first designed in plastic by Corey, Pauling and Koltun and is called the CPK model and may be thought to represent the electron could of the molecule. A computerised version of this draws the CPK model on the screen. Large molecules, such as most proteins, are effectively seen when drawn as line diagrams rather than ball and stick models. Even then, there is too much detail to be fully appreciated (Figure 9.5). One way of getting over this, problem may be to show the picture in stereo (Figure 9.6). It may be sufficient to show only the atoms of backbone, if the interest is only in

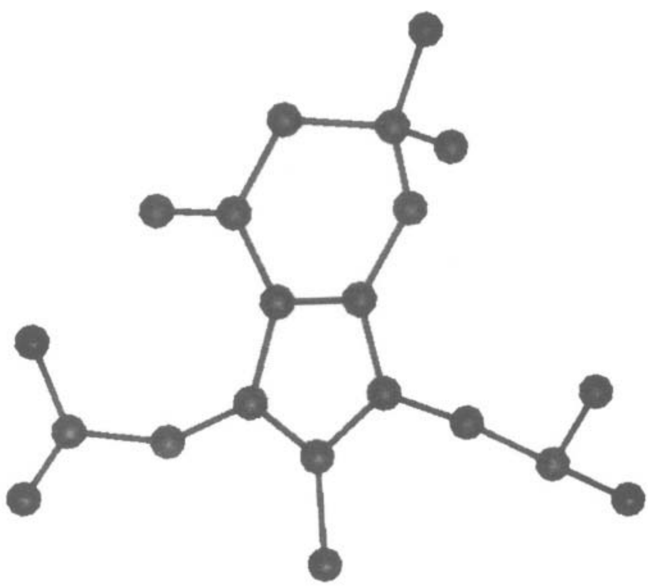


Fig. 9.3. A ball-and-stick representation of an indole molecule showing all the atoms by spheres of equal radius.

the protein fold. A further abstraction is to plot and join only the C^α atoms (Figure 9.7). Cartoon diagrams (Figure 9.8) are useful to show the disposition of the seondary structural elements. It is conventional to depict **α-helices** as cylindrical rods and **β-strands** as flat arrows. If the interest is only in the surface of the protein, diagrams such as Figure 9.9 may be used.

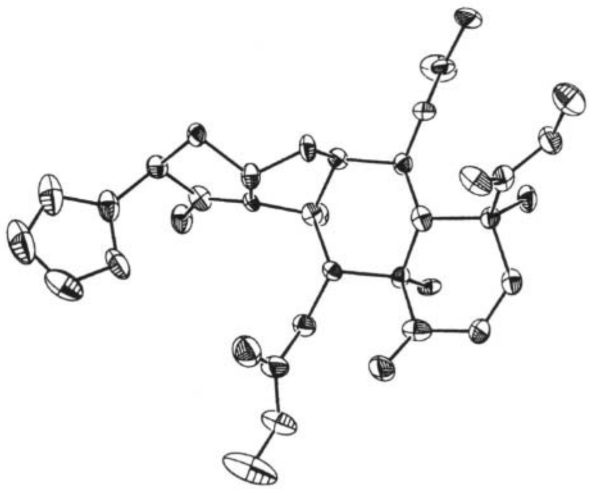


Fig. 9.4 A thermal ellipsoid representation of a molecule of thermal vibration of each atom, indicating the extent (drawn using the computer program ORTEP).

9.3 Optimising the Model

The generated molecule needs to be checked for compliance with the laws of physics and chemistry before it can be accepted for further processing. The stereo-chemical criteria used during the construction of the model are usually insufficient to decide upon the final correct structure. The dynamics of the molecule, i.e. the various forces acting on various parts of it have to be explicitly calculated and the structure has to be modified to optimise the action of these forces. A simple first step in the structure optimisation is to ensure that the bond lengths and bond angles fall within the ranges measured in accurate spectroscopic or X-ray diffraction experiments. Atomic sizes are also known from experiment and the fold of the generated molecule should not be such that the distance between two non-bonded atoms is less than the sum of the atomic radii, i.e. there should be no inter-penetration of the atomic spheres. These chemical and physical data can be checked and optimised using computer programs. It is usually not possible to exactly match the requirements. However the difference between the expected and observed geometrical values is minimised using least squares techniques.

Such geometrical optimisation is sufficient to give a plausible model. It is however not sufficient to serve as a basis for new interpretation regarding the structure-function relationships in the molecule, i.e., no new 'science' can be expected from it. Energy minimisation and structural optimisation using molecular dynamics are at the next higher level of modelling. The free energy of the molecule is an experimentally measurable thermodynamic quantity. This quantity can also be computed. A very important physical principle requires that at the equilibrium state of any system, the free energy be the least. In other words, the minimum energy state is the equilibrium state and all systems tend to move towards this state. This process of structure optimisation is used to determine the minimum energy configuration. In order to calculate the free energy for various states or various conformations

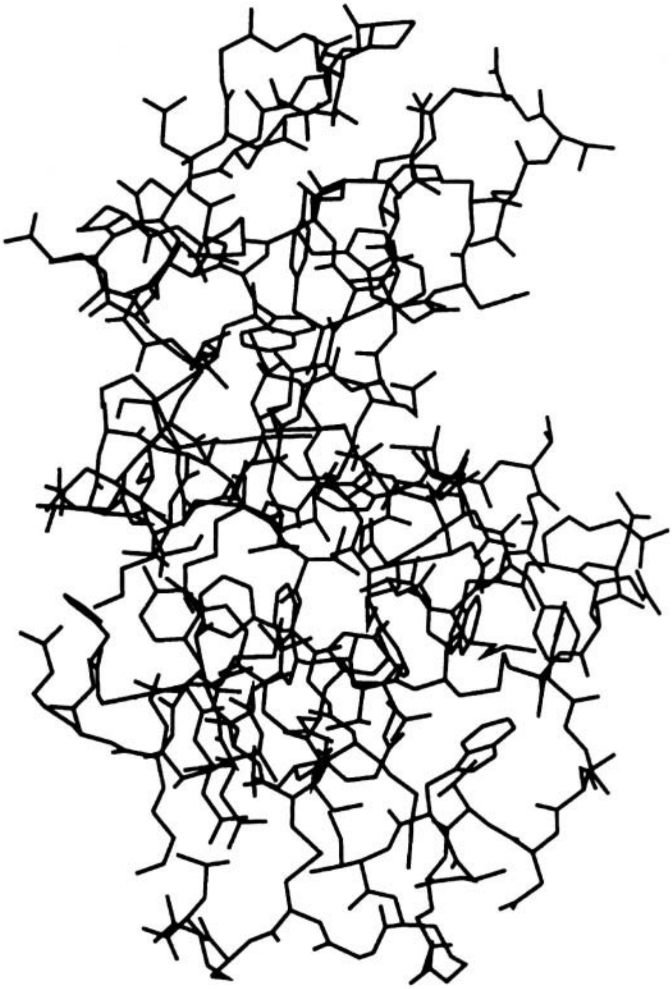


Fig. 9.5. Line diagram of lysozyme

of the molecule it is necessary to set up the quantum mechanical equations for the whole molecule and to solve them. This task is beyond the scope of currently available mathematical and computational techniques. However, the value of the free energy can also be calculated from a semi-empirical expression such as

$$E_{\text{tot}} = E_{\text{bond-length}} + E_{\text{bond-angle}} + E_{\text{tor}} + E_{\text{non-bonded}} + E_{\text{elec}} + E_{\text{hydrogen-bond}}$$

where E_{tot} is the total semi-empirical free energy, $E_{\text{bond-length}}$ is the contribution to the energy due to the deviation of the bond lengths from their expected values, $E_{\text{bond-angle}}$ is the contribution due to the deviation of the angles from their standard values, E_{tor} is the contribution due to the torsion angles and is dependent on the disposition of the four atoms that make up the torsion angle (e.g. an eclipsed configuration has a very high energy). $E_{\text{non-bonded}}$ is the contribution due to the van der Waals contacts between the atoms. Interpenetration will raise the energy enormously. E_{elec} is the contribution due to the electrostatic interaction of attraction and repulsion between the atoms. Even though the molecule as a whole may be electrically neutral, quantum mechanics tells us that each atom will posses a 'partial' charge, positive or negative, and it is these partial charges which take part in the electrostatic interactions. This is in addition to interactions between charged residues such as Glu or Asp. Finally $E_{\text{hydrogen-bond}}$ is the energy contribution due to hydrogen bonds. Other terms, such as one for the 'anomeric effect' are sometimes added to this expression for the energy. The conformation of the molecule is optimised with respect to the energy. In other words, the conformation is changed so that the value of the energy is a minimum. The minimisation is carried out by some well-known mathematical techniques such as least squares or gradient search methods. Such a procedure will give the final 'correct' conformation of the molecule. In addition new interactions and structures not built into the original model may appear.

Fig. 9.6. Stereo picture of protein. (Though only a two-dimensional image of the object falls on each retina, the brain is able to synthesize the three-dimensional picture from the two images since they are produced by views at slightly different angles. Stereo pictures mimic this effect by providing views of the molecule as seen from two different angles. In order to obtain the 3-D effect it is necessary to allow the two eyes to simultaneously focus only on one picture each, to the exclusion of the other. This may be done by placing a card between the eyes and blocking out the second image. Many people are able to obtain the 3-D effect without any such aids. The reader may try to stare at the example given above, focusing and defocusing his/her eyes unitl he/she obtains the 3-D effect. The first attempt may take hours before it is successful.)

The molecule may also be subjected to the molecular dynamics calculations to study its dynamical behaviour. In this procedure each atom is thought to be moving randomly at a high velocity, depending on the temperature used in the calculation. Each atom is influenced by forces such as bonding forces,

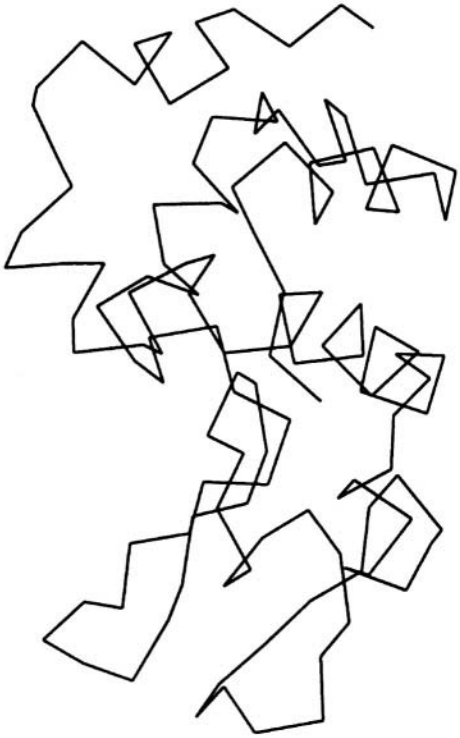


Fig. 9.7. The C^{α} atoms of lysozyme

Fig. 9.8. A cartoon diagram of a protein

angle distortion forces, etc. Since the force on each atom depends on the positions of all the other atoms, and since the positions are continuously changing, the technique is carried out in iterative cycles. First step is to assign the velocities to all the atoms and calculate the initial force felt by each atom in the initial configuration of the molecule. The next step is to calculate the new positions of the atoms after a very short time interval, usually of the order of 0.01 picoseconds (1 picosecond = 10^{-12} second). The new positions of the atoms are then used to calculate the new force field. Thus the motion of the various parts of the molecule can be followed. The results are usually checked every 100 steps or so and give 'snapshots' of the molecule at intervals of about 1 picosecond. The simulation usually cannot be carried on for longer than about 5000 picoseconds for the following two main reasons: (a) Enormous amounts of computer time are necessary even on the fastest computers. (b)

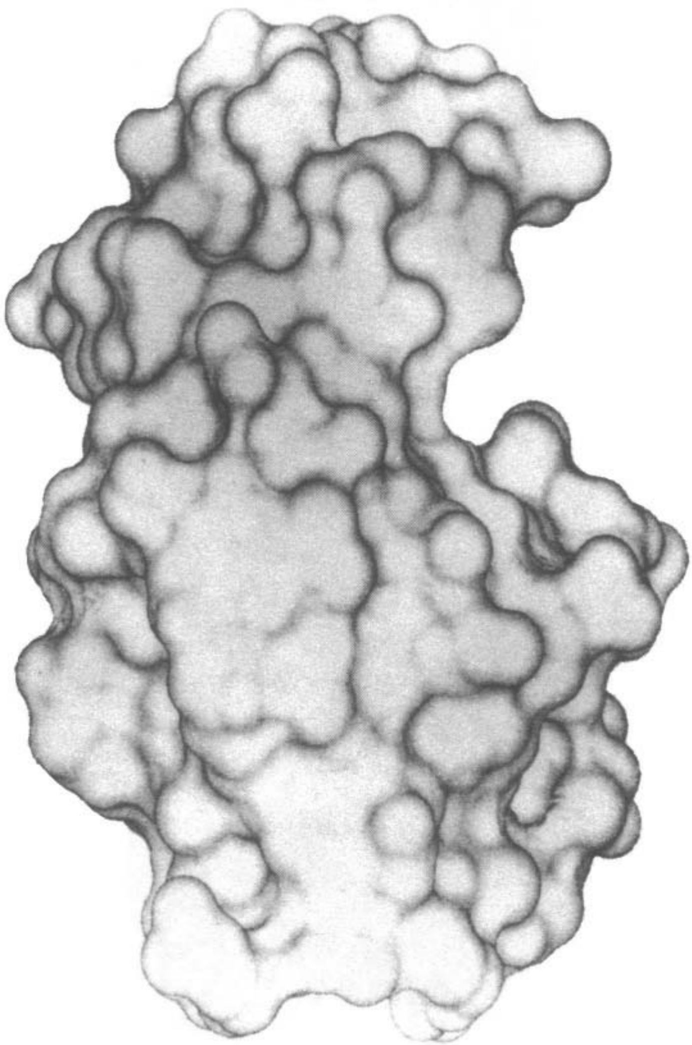


Fig. 9.9. Surface representation of lysozyme.

The force fields are not realistic enough and the molecule tends to assume unrealistic configurations if the simulation is carried on for longer than about 5000 picoseconds. However, large advances have been made in the past five years both in computer technology and in modelling the force-field accurately. This has allowed several interesting simulations of small proteins and DNA fragments to be carried out for as long as 1 millisecond or 10^6 picoseconds. The great advantage of the molecular dynamics technique is that it allows a much more natural picture of the molecule. Molecules are always in motion, but X-ray crystalography and many other techniques give only a static picture.

Apart from building models of molecules, molecular modelling, optimisation and molecular dynamics are used to model interactions between molecules. Successful applications include studying the docking of proteins to substrates, virus assembly, etc. Modelling studies on drug-DNA interactions and drug-protein interactions are used extensively in the rational design of new drugs. Many of the drugs now in the market are the result of such studies.

10

Macromolecular Structure

10.1 Introduction

The central paradigm of biophysics, indeed of physics in general, is

Structure → Function

Thus in order to understand and perhaps modify the behaviour of biomolecules, it is necessary to know and understand their structure. One of the most dramatic illustrations of this principle came with the discovery of the double-helical structure of DNA. It was immediately clear how DNA could function as the storage and carrier of genetic information. The vast advances in biotechnology since then stem almost entirely from this single discovery. Protein structure and virus structures also illustrate the truth of this principle. In this chapter we shall consider in detail the structural principles of some the important classes of biological molecules.

10.2 Nucleic Acid Structure

The storage, transmission and expression of genetic information in the cell is carried out by nucleic acids. Deoxyribonucleic acid or DNA is responsible for storing genetic information and passing it on to the progeny cells. DNA is therefore involved in replication and transcription. The other type of nucleic acid present in the cell is ribonucleic acid or RNA. It is involved in many different functions. The genetic information in the DNA is transcribed into messenger RNA or mRNA, which is then translated into the protein in the ribosome. Transfer RNA (tRNA) molecules mediate this process. RNA is also a part of the ribosome and is then called ribosomal RNA or rRNA. Small RNA molecules (~ 100 nucleotides) can fold into structures called ribozymes, which have catalytic activity. During each of these different functions the nucleic acid molecules have different three-dimensional structures.

10.2.1 The chemical structure of nucleic acids

Both DNA and RNA are polymers. The repeating unit, i.e. the monomer in the case of DNA is a deoxyribonucleotide, and a ribonucleotide in the case of RNA. A mononucleotide can be thought as made up of three parts, (a) a phosphate group, (b) a sugar moiety and (c) a nitrogenous base.

The phosphate group is simply PO_4^- as illustrated in Figure 10.1. When it is a part of the polynucleotide, the single extra negative charge is distributed between the two so-called 'pendant' oxygen atoms. It is this negative charge that is responsible for the acidity of nucleic acids. The phosphate group is a rigid unit and has no internal flexibility. The four oxygen atoms occupy the corners of a tetrahedron, with the phosphorous atom at the centre. When the group becomes a part of a polynucleotide, the geometry changes only slightly, with the lengths of the P–O bonds increasing by about 0.2 Å.

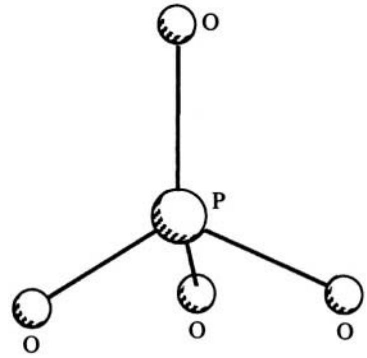


Fig. 10.1 Chemical structure of the phosphate group.

The sugar moiety consists of a five-membered furanose ring with exocyclic alcohol and hydroxyl groups (Figure 10.2). If both the 2′ and 3′ position have a hydroxyl attached, the sugar is ribose (RNA). If there is a hydroxyl group only at the 3′ position it is deoxyribose (DNA). Though it is cyclised, a substantial amount of conformational variation can still take place. Rotation about the C'_4–C'_5 bond produces the three preferred conformations shown in Figure 10.3. In addition, the five-membered ring is not planar, but puckered. Due to the sp^3 hybridised bonds at the carbon atoms of the ring, the energetically favourable conformation is one in which only three or four of the five atoms lie in the same plane. Further, in order to avoid short contacts between the pendant hydrogen atoms, the best way to relieve such energy strains is to push the C'_2 atom or the C'_3 atom out of the plane (Figure 10.4). Thus, in nucleic acids, one comes across C'_2-endo or C'_3-endo sugar puckers, indicating that the displacement of the atom out of the plane of the other four atoms is in the same direction as the C'_5 atom. If the displacement is in the direction opposite to the C'_5 atom, the conformation is termed C'_2-exo or C'_3-exo.

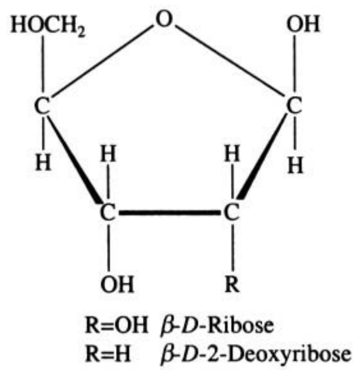


Fig. 10.2 Chemical structure of furanose

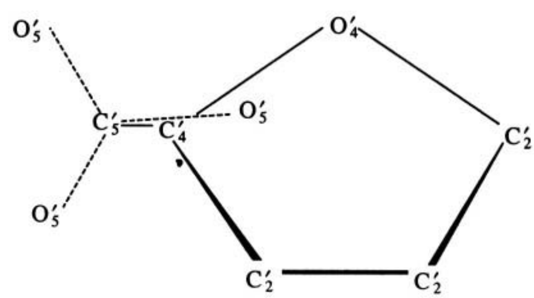


Fig. 10.3 The preferred conformations about the C'_4-C'_5 bond.

The planar bases constitute the chemically variable portion of nucleic acids. Bases can be

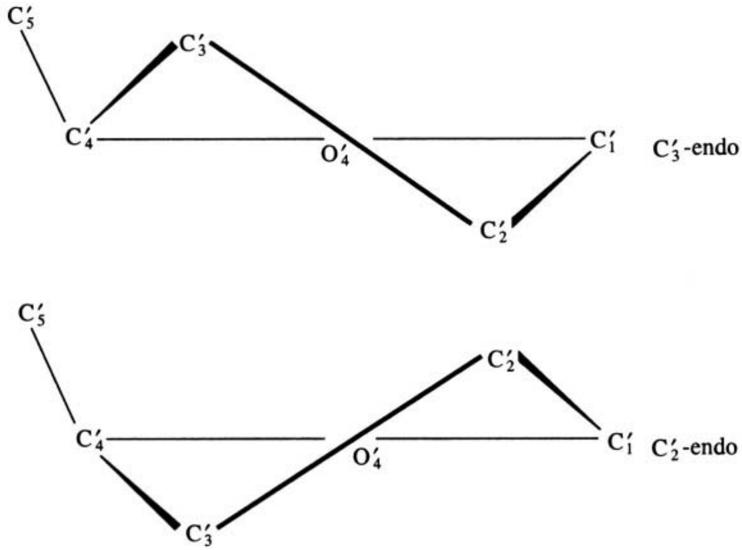


Fig. 10.4 Puckering of the sugar ring.

either pyrimidines or purines (Figure 10.5). The bases guanine, cytosine, adenine and thymine are found in DNA. In RNA the pyrimidine thymine is replaced by uracil. Like the phosphate group, the bases have no internal conformational flexibility.

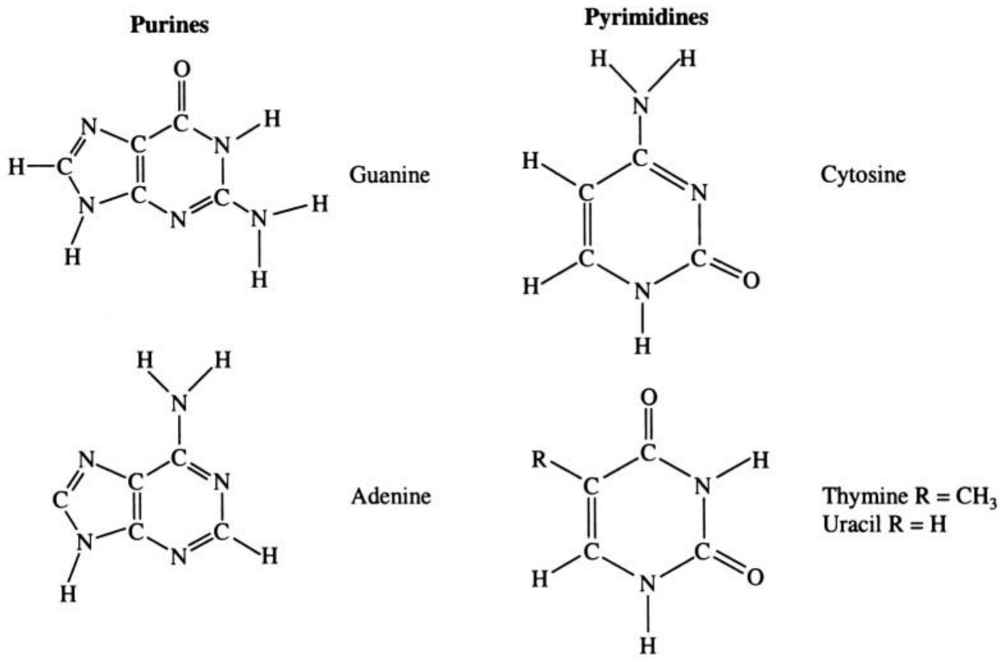


Fig. 10.5 Chemical structure of the nucleic acid bases.

10.2.2 Conformational possibilities of monomers and polymers

A nucleoside is formed by the covalent attachment of one of the bases to a sugar moiety. The bond is called a glycosidic bond and occurs between the C_1' atom of the sugar and either the N_9 atom of a purine or the N_1 atom of a pyrimidine. The plane of the base is almost perpendicular to the approximate plane of the sugar ring. As compared to its components considered separately, a nucleoside has additional conformational possibilities by way of rotation about the glycosidic bond leading to the *anti* or *syn* orientations of the base with respect to the furanose (Figure 10.6).

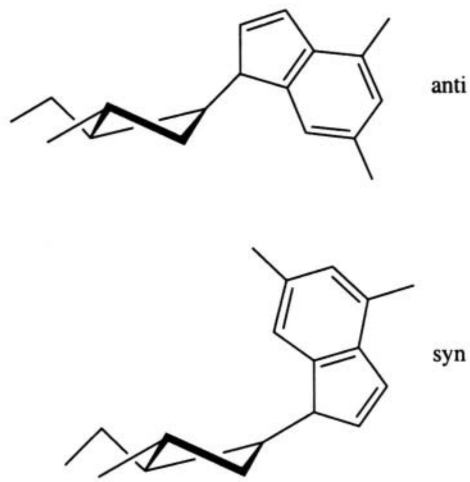


Fig. 10.6 Conformation of the base about the glycosidic bond

The formation of a nucleotide by the addition of a phosphate group to the sugar, at the 5′ end or the 3′ end (Figure 10.7) further increases the conformational possibilities. However, model building, theoretical calculations and experimental evidence from NMR and X-ray crystallography have shown that all conformations are not equally likely.

Mononucleotides link to each other through phosphodiester bonds to form the polynucleotide chain (Figure 10.8). In a dinucleotide there are a total seven bonds about which rotation is possible. In addition, the sugar puckering is also a variable. However, only a few combinations of these angles lead to regular structures when repeated along the chain. Upon formation of the polynucleotides, certain new chemical interactions are introduced which do not exist in the mononucleotide. The most powerful of these are the stacking interactions between the planar bases. These interactions occur between the π-electron clouds and lead to the bases being arranged in parallel planes one over the other like a stack of coins. The inclusion of such

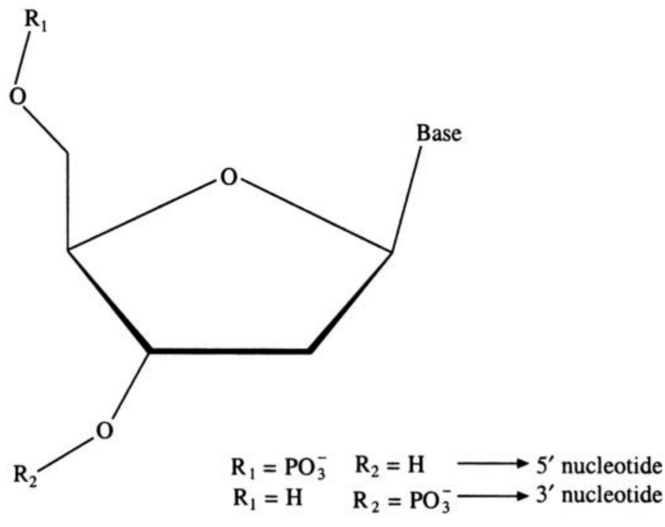


Fig. 10.7 A nucleotide unit

interactions in the calculation of the optimum three-dimensional structure results in a further decrease in the number of possibilities. Some of these structures are discussed below.

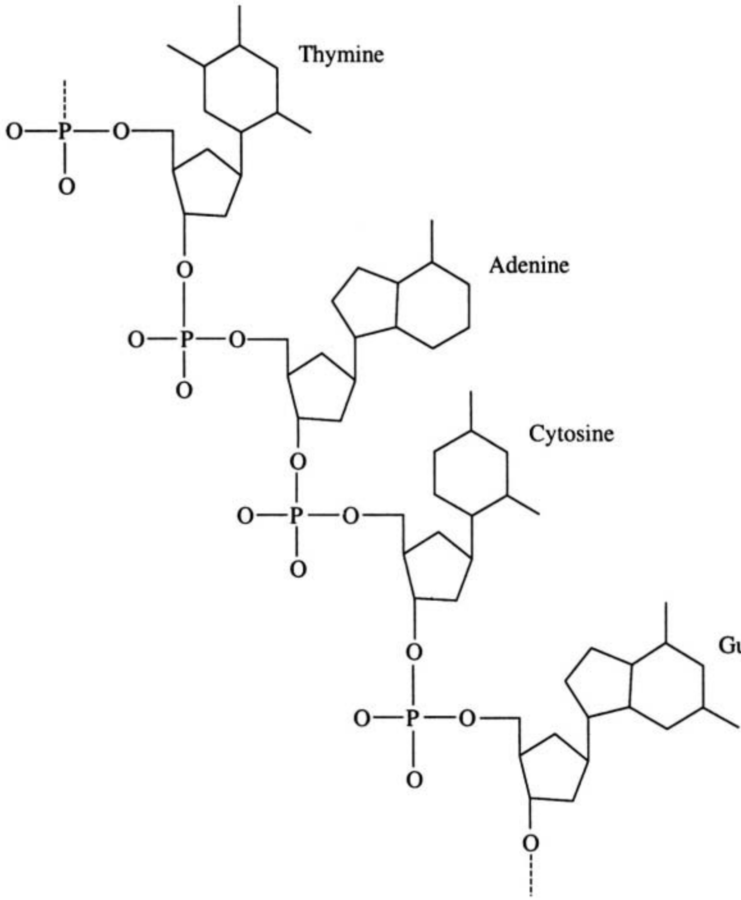


Fig. 10.8. The chemical structure of polynucleotide chain

10.2.3 The double helical structure of DNA

DNA in the cell is one of the largest molecules found in nature. It is a polymer whose length can range from a few thousand nucleotides in bacteria to more than a billion **(10^9)** nucleotides in cells of higher mammals. The discovery of the double helical structure of DNA in 1953 by J.D. Watson and F.H.C. Crick was one that marked an epoch and is the basis for all of modern biotechnology.

It had been noticed that in the DNA molecules, the number of adenine nucleotides was exactly the same as the number of thymine nucleotides and the number of guanine nucleotides equalled the number of cytosine nucleotides. Furthermore, X-ray diffraction experiments on DNA fibres had suggested that each DNA molecule consisted of two strands which wrapped around each other to form a helix and that these two strands were arranged antiparallel to each other (Figure 10.9). (Note that the attachment of the phosphate group at the 5′ position of the sugar ring is different from its attachment at the 3′ position. This gives rise to a directionality in the polynucleotide chain.) The concept of base pairing explained the base ratio by suggesting that adenine formed specific hydrogen

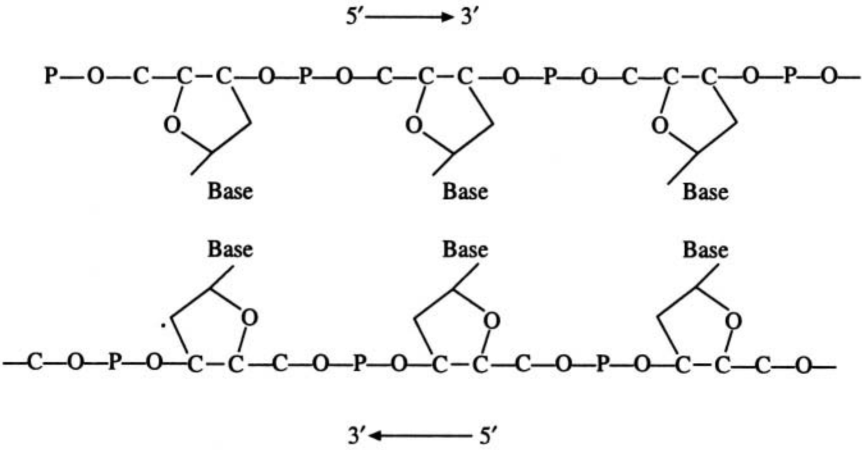


Fig. 10.9 Antiparallel arrangement of the two nucleic acid strands

bonds with thymine and guanine with cytosine (Figure 10.10). Other hydrogen bonding schemes between the bases were much less favourable. Additionally, both the A:T and the G:C base pairs were of the same size. These facts led to the proposal of the structure shown in Figure 10.11. One may visualise the DNA double helix as a flexible rope ladder wrapped around a central pole. The sugar phosphate chains of the two polynucleotide backbones form the sides of the ladder. The base pairs placed perpendicular to the central helix axis form the rungs. The geometrical details of the structure are given in Table 10.1.

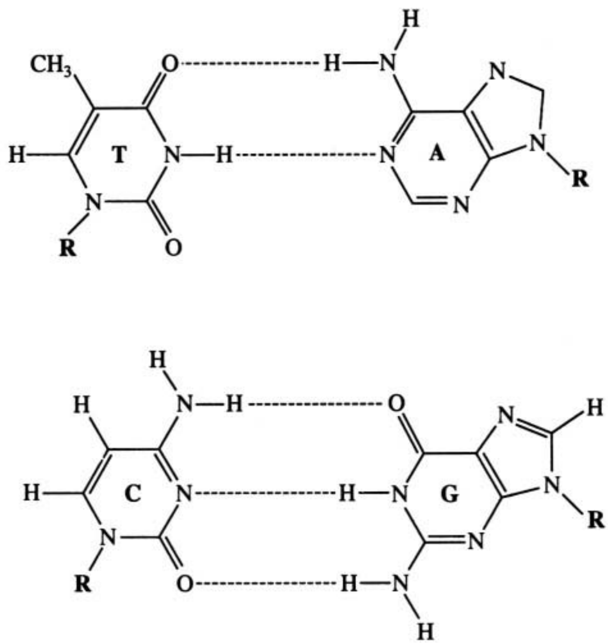


Fig. 10.10 Base pairing in nucleic acids (DNA). In RNA, T is replaced by U

Fig. 10.11 The double-helical, base-paired structure of DNA

Table 10.1 Structural parameters of three polymorphic forms of the DNA double helix

	A	B	Z
Axial rise per residue (Å)	2.56	3.38	3.7
Twist angle per residue (°)	32.7	36.0	–30.0*
Pitch (Å)	28.2	33.8	45.0
Number of base pairs per turn	11	10	12
Handedness of the helix	right	right	left
Number of nucleotides in the repeat unit	one	one	two
Tilt of base pairs from helix axis (°)	20.0	–6.0	–7.0
Dislocation of base pairs into groove (Å)	4.4	–0.14	–5.5#
	minor groove	major groove	major groove
Major groove width (A)	2.7	11.7	8.8
Major groove depth (Å)	13.5	8.5	3.7@
Minor groove width (Å)	11.0	5.7	2.0
Minor groove depth (Å)	2.8	7.5	13.8

*Average of –60° for a dinucleotide repeat. The twist is actually –10° and –50° for the first and second steps respectively of the dinucleotide.
This is an approximate value.
@ This groove is filled up with exocyclic base atoms.

The complementary base-pairing scheme immediately suggests how genetic information is stored and passed on from generation to generation. If the information is ‘written’ as the sequence of bases, then it may be seen that, when the two strands of the DNA molecule separate, if a complementary strand is built on each of the two strands, it results in two identical copies of the original double helical molecule. This can be repeated again and again and each time the cell divides and in each daughter cell the DNA would be identical to that in the parent cell (Figure 10.12).

10.2.4 Polymorphism of DNA

The structure of DNA proposed first by Watson and Crick explained the X-ray diffraction patterns produced by ‘wet’ DNA fibres or the ‘B’ type patterns. This model has thus come to be called B-DNA. Fibres at much lower humidity produced another pattern, with sharper spots and therefore indicating a higher degree of order. They were called the A type patterns and Watson and Crick proposed a somewhat altered model of the double helix to explain them (Figure 10.13). The differences in the helical parameters are quite substantial, as may be seen from Table 10.1. The crucial features that remain unchanged in the two models of DNA are their double helical nature, the antiparallel orientation of the strands and the base pairing scheme. The A type structure also may be imagined as a rope ladder wrapped around a central pole, but the imaginary pole is far thicker in this case. A way of recognising this is to observe the amount of displacement of the base pairs from the helix axis (Figure 10.14). In B DNA, the axis passes through the base pairs, while in A DNA, the axis is displaced by about 2 Å into the major groove. Also the B type model is slimmer than the A type model.

Both A and B type models are right-handed helices in which the helix proceeds as in a right hand screw. A dramatic example of the polymorphism of DNA was discovered in the left-handed Z type DNA (Figure 10.15). It was first observed in sequences of the type 5′ … CGCGCGCG … 3’, i.e. alternating pyrimidine-purine. Circular dichroism and UV absorption studies indicated a change in the helical sense from right handed to left handed in sequences of this type. Later a single crystal

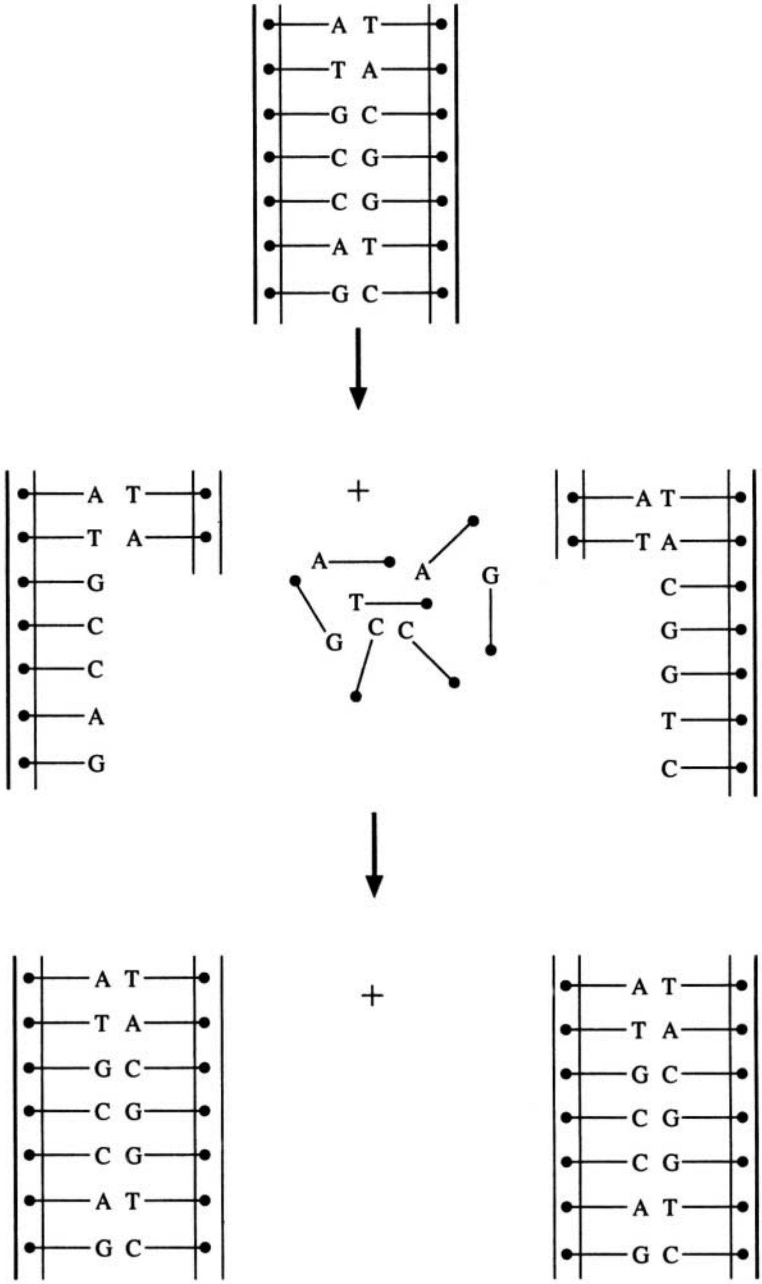


Fig. 10.12 A schematic representation of how the double helix explains DNA replication

structural study of the hexanucleotide 5′-CGCGCG-3′ showed that, not only was the helical sense left-handed but also the phosphate groups followed a zigzag pattern, hence the name Z DNA. The helical parameters of Z type DNA are also given in Table 10.1. It may be noted that left-handed Z DNA has been conclusively and positively identified only *in vitro*. Though antibodies raised against

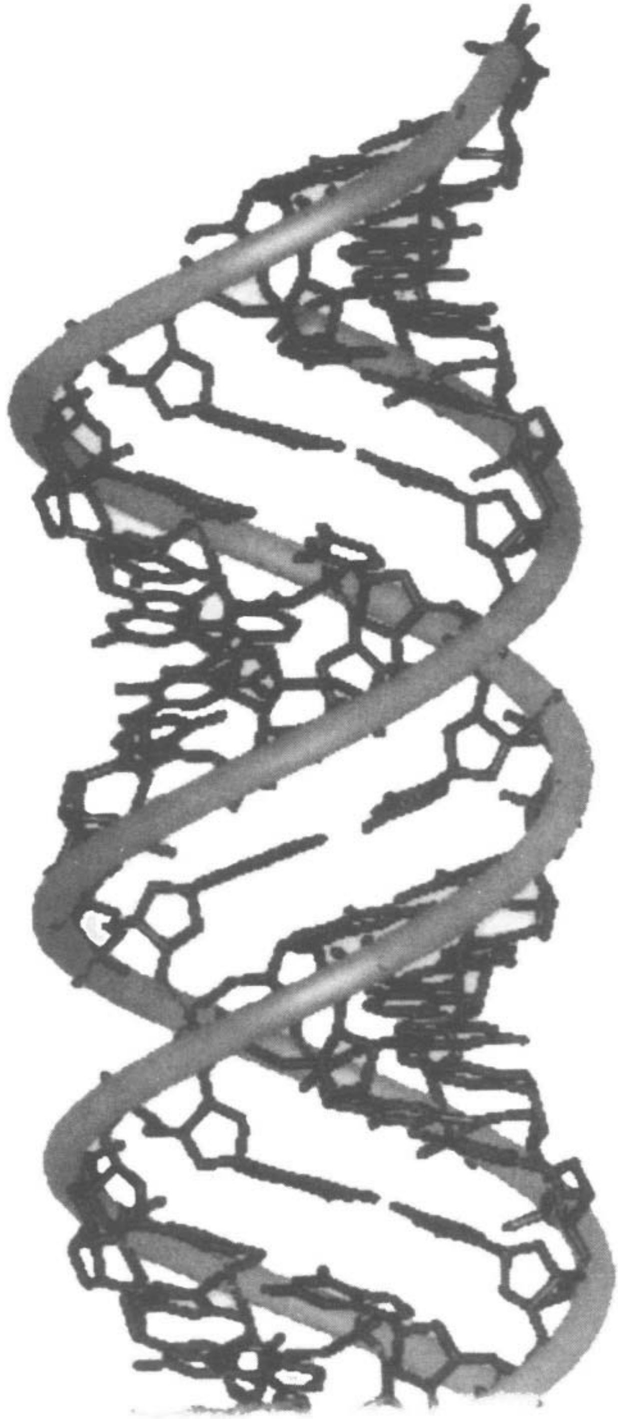


Fig. 10.13 The A type double-helical structure

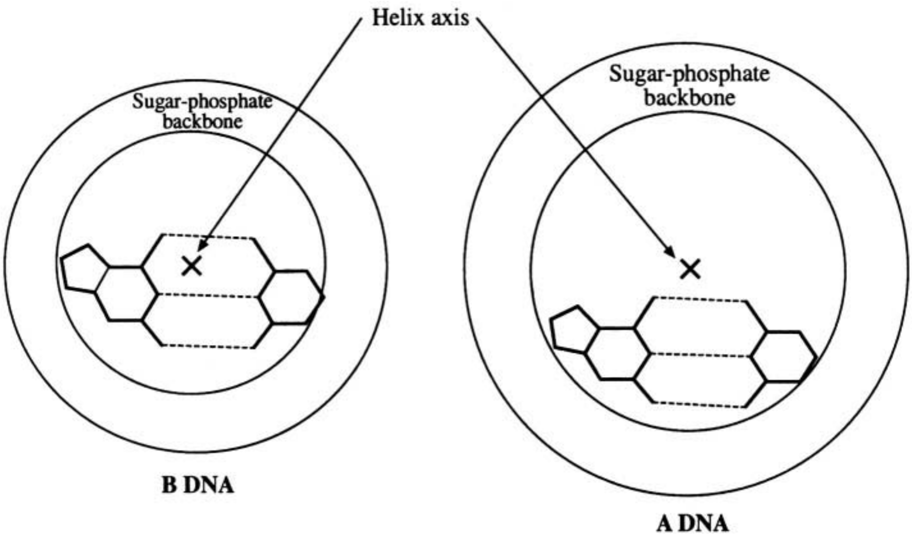


Fig. 10.14 Displacement of the base pairs into the minor groove in A type and B type DNA

such sequences have been shown to cross-react with rat chromosomes, definitive evidence for a biological role for Z DNA is still lacking. Since 1978, single crystal studies of small DNA fragments have yielded high-resolution data on DNA structure. One of the most important findings is that, the structure of DNA deviates from the 'standard' models described above in a sequence dependent manner. The regular models of A, B, and Z DNA, which are described above and in Table 10.1, have been constructed in such a manner that, except for the change in the base, the structure of the molecule, especially the sugar-phosphate backbone, is exactly the same in one part of the molecule as in another. Given the co-ordinates of one monomer and helical transform it is possible to generate the co-ordinates for any length of the double helix.

The crystal structure results go against this and show a microheterogeneity (Figure 10.16). The micro level structure in each portion of the helix appears to depend on the sequence in that part, though the correlation between the sequence and structure is unclear. Sequence specific structure has implications for biology in that it could be another way of transferring information from the DNA molecule (Figure 10.17).

10.2.5 DNA supercoiling and unusual DNA structures

As we have seen, owing to the nature of its stereochemistry, the energetically most favourable state of the DNA molecule is two strands coiled around each other. If, for some reason, the number of times one strand goes around the other is changed from the optimum value, supercoils (Figure 10.18) relieve the extra strain. Exactly the same effect may be seen, for example, in household electrical wires made of two or three twisted strands. In supercoiled DNA, two double helices further wind around each other, like two strands of a rope which are themselves made of coiled fibres. Another way of relieving the strain is by increasing the so-called writhe. In this the DNA molecule wraps round and round an imaginary or real object. In these terms, for example, A type DNA has a higher degree of writhe than B type DNA. The structure of chromatin consists of DNA coils wrapped around histones (Figure 10.19). Each such histone-DNA unit is a nucleosome and in the chromatin state, the nucleosomes are arranged like beads on a string. In its more condensed state, the nucleosome 'necklace' is further wound around to form a solenoid like structure.

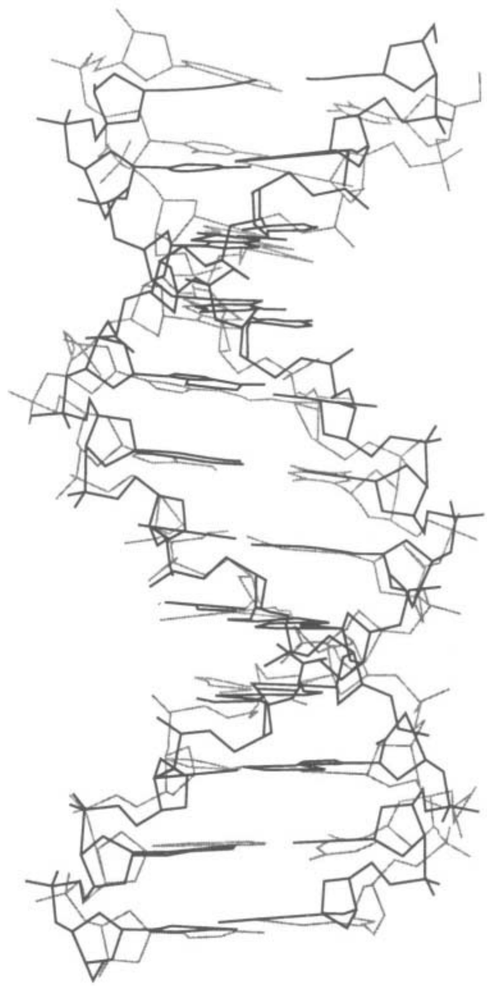


Fig. 10.16 The microheterogeneity observed in the three-dimensional structure of DNA. Dark lines represent the regular helical model derived from fibre diffraction. Light lines correspond to the single crystal structure of the same sequence

Fig. 10.15 Left-handed Z type DNA

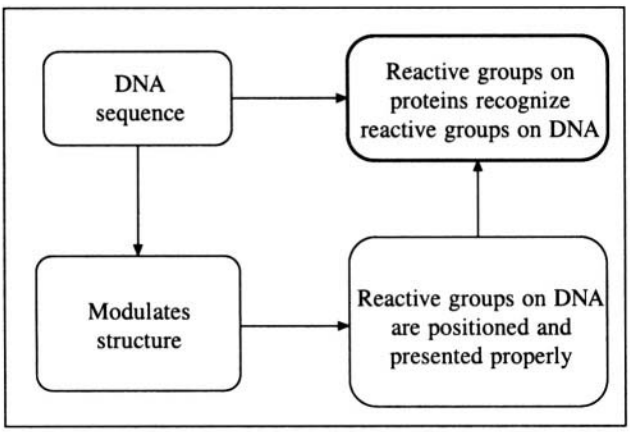


Fig. 10.17 Information transfer from DNA to proteins

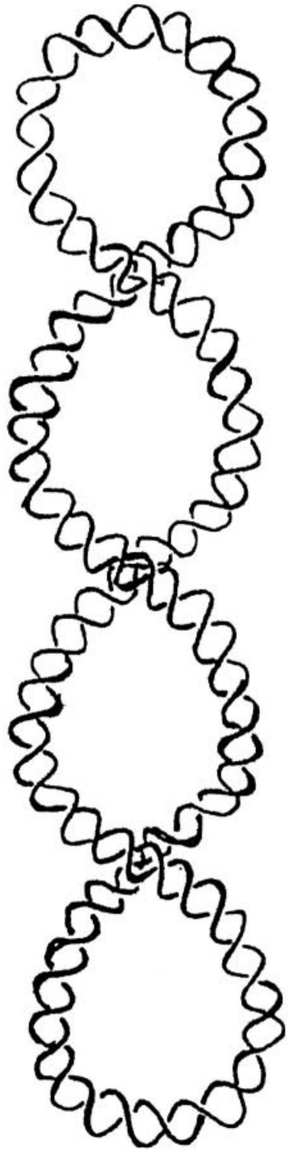


Fig. 10.18 Schematic diagram of supercoiling in DNA

Apart from the double helix, certain DNA sequences have been shown to assume other types of structures. One of these is when stretches of guanine bases come together to form a G-quartet structure (Figure 10.20). Here the guanine bases form hydrogen bonded tetrads that are stacked one over the other. The 'i'-motif is another unusual DNA structure (Figure 10.21) formed by cytosines and which has two distinctive features: (i) The cytosine-cytosine base pair. (ii) The intercalated structure of the two duplexes which make up the tetraplex.

Other unusual DNA structures, observed using various biophysical techniques, include triplexes (Figure 10.22), loops, cruciforms, slipped DNA, etc.

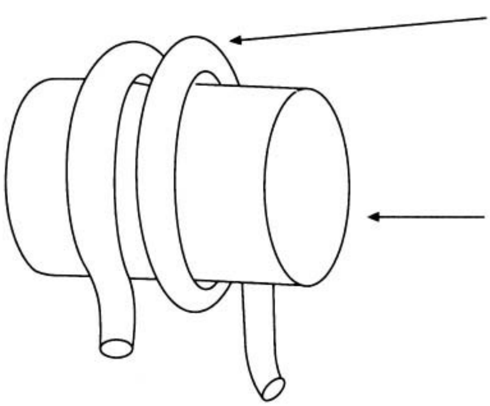


Fig. 10.19 The structure of a nucleosome core particle

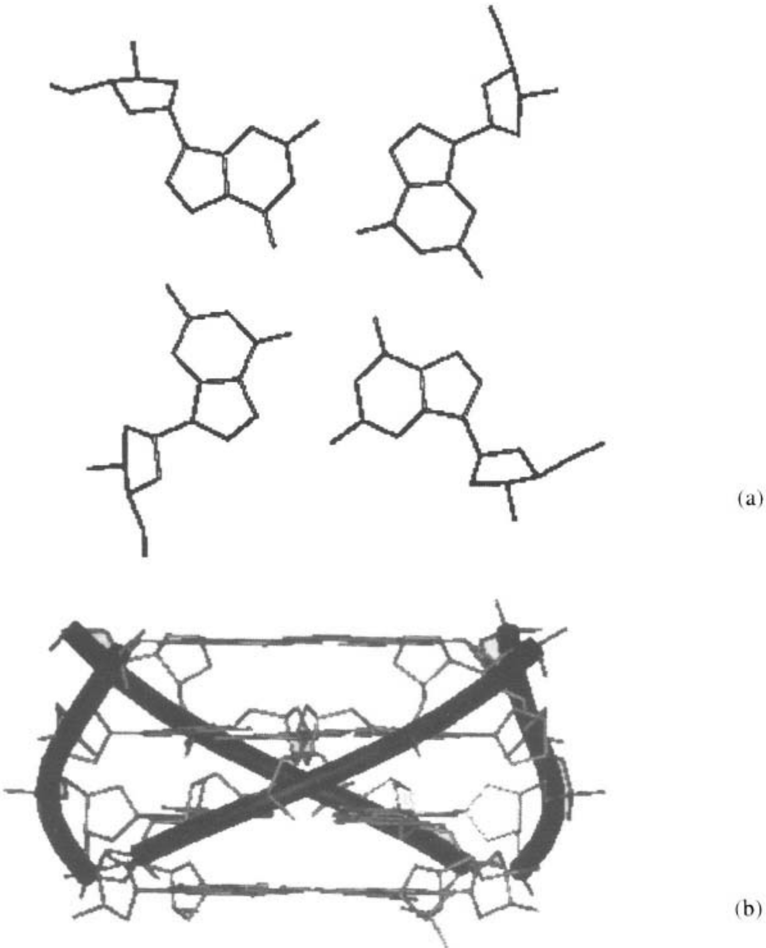


Fig. 10.20 Structure of G-plates. (a) View down the helical axis showing the arrangement of four guanine residues in a plane. (b) View perpendicular to the helical axis

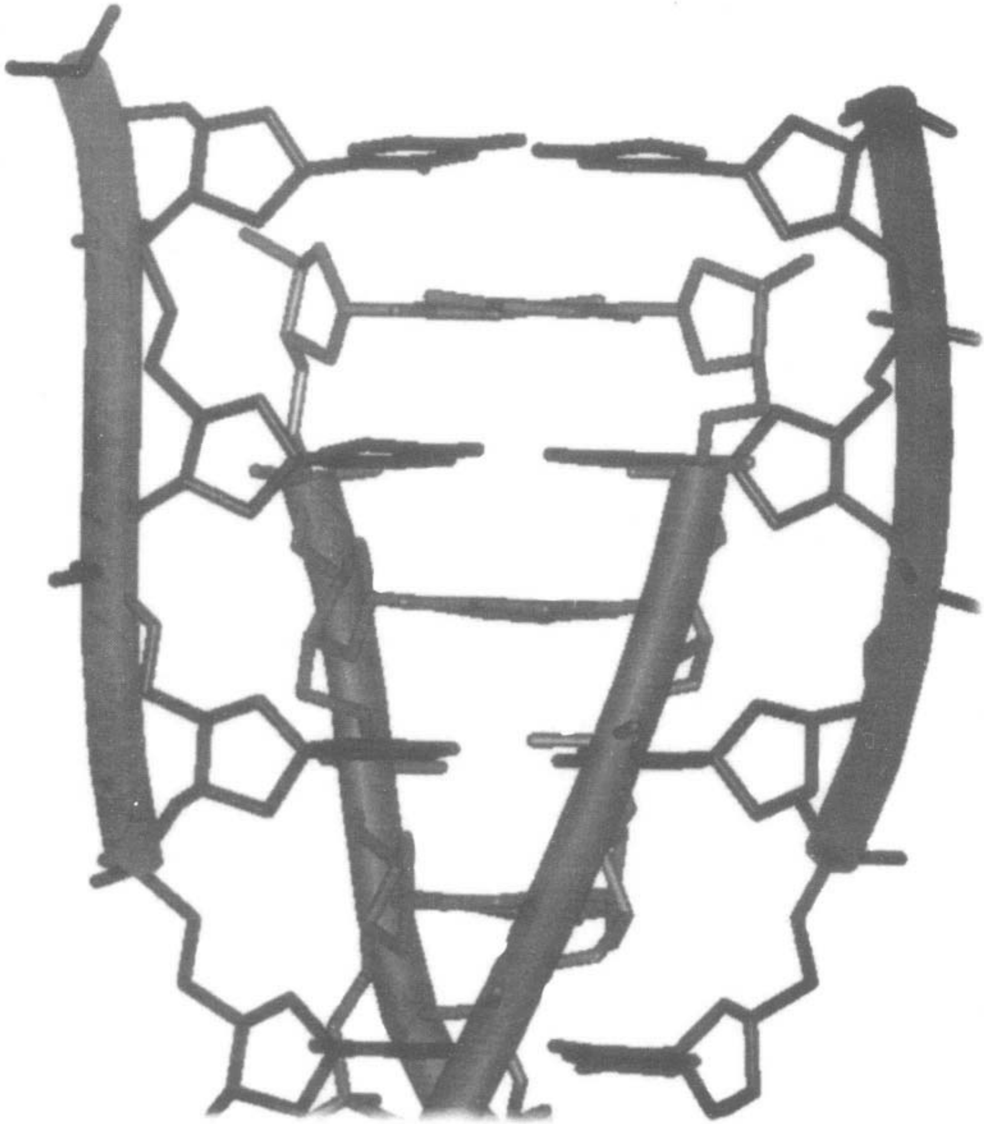


Fig. 10.21 The 'i'-motif structure

10.2.6 The structure of transfer RNA

Most of the ribonucleic acid in the cell exists as rather short single strands, compared to the very long double stranded form assumed by DNA. For example, transfer RNA (tRNA) molecules are only about 70-80 nucleotides long. Some of these nucleotides consist of unusual bases such as psuedouridine and dihydrouracil. Messenger RNA (mRNA) molecules have varying lengths, depending on the code they carry. Ribosomal RNA (rRNA) molecules, which form a part of the ribosome, are also long nucleotides.

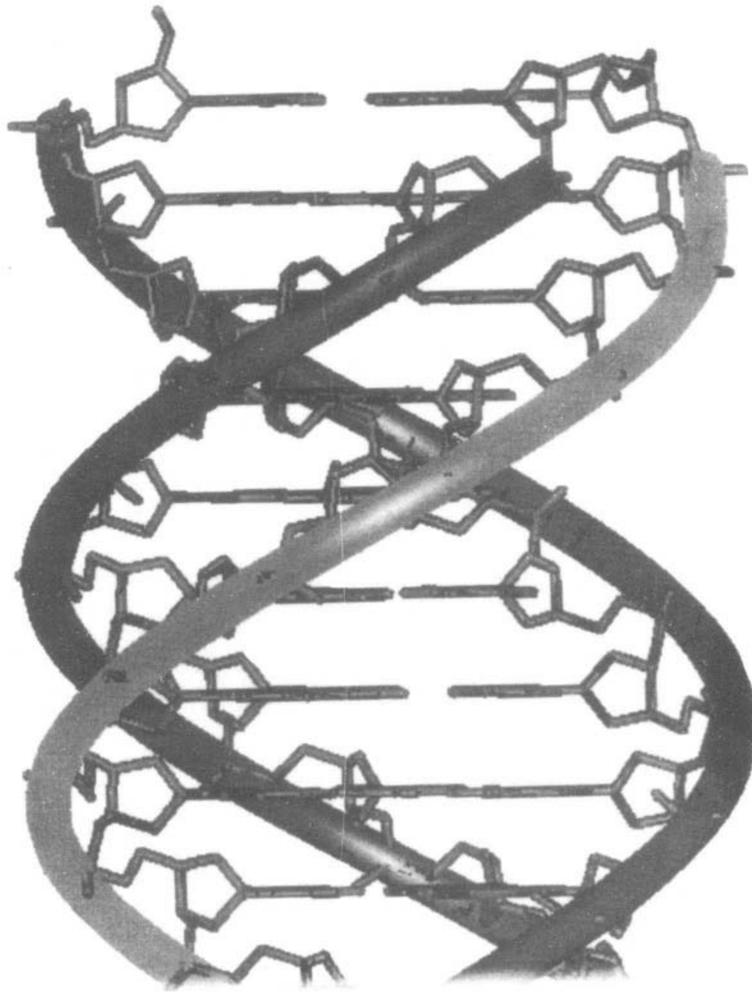


Fig. 10.22 The structure of triple-stranded DNA.

Only the structure of tRNA is known in some detail. Being a single strand, the structure is not a double helix. However portions of the sequence can form Watson-Crick base pairs with other portions of the molecule and these regions are double helical. An analysis of the sequences of tRNAs suggests that they can be arranged to form four base-paired regions so that the general shape is that of a clover leaf (as seen in playing cards) (Figure 10.23). Earlier theoretical and modelling studies, supported by diffraction and other physico-chemical studies, had shown that RNA double helices must necessarily be of the A type. This is because steric hindrances due to the extra oxygen atom at the 2′ position of the sugar which would prevent the other structural types of helices. In tRNA too, the base paired stems form short A type helices. The three dimensional arrangement of these short helices and the extra loops is such that the structure looks like the letter 'L' (Figure 10.24). The anticodon region, i.e. the portion that carries the code is at one end and the amino acid is attached at the other end.

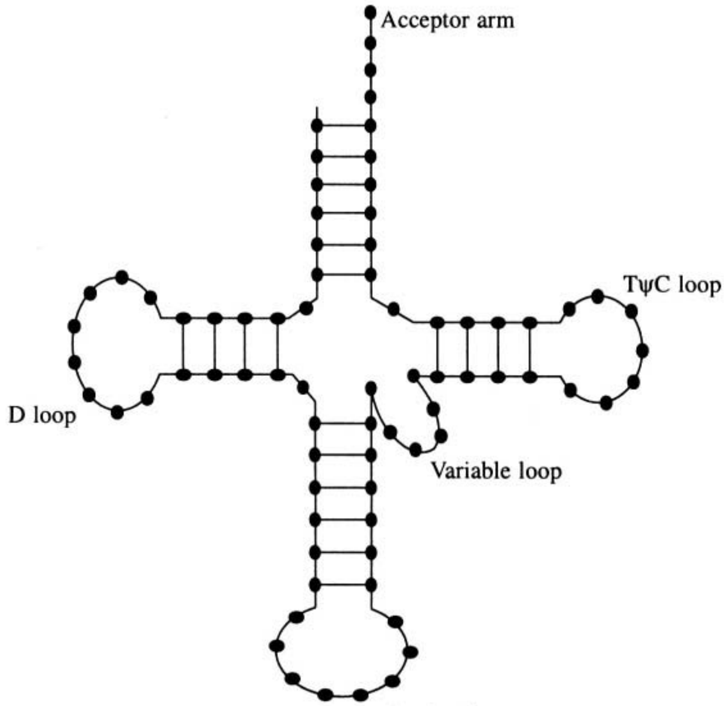


Fig. 10.23 Clover leaf secondary structure of tRNA.

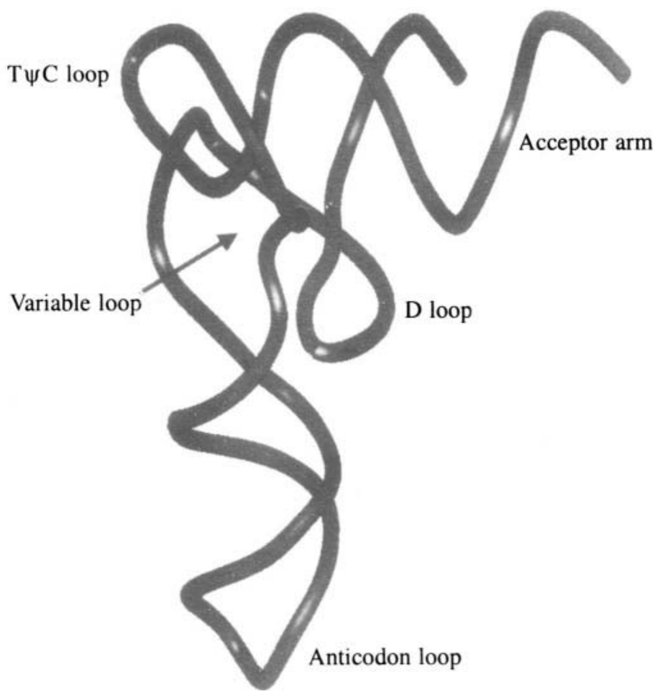


Fig. 10.24 Three-dimensional structure of tRNA. Only the trace of the sugar-phosphate backbone is shown.

10.3 Protein Structure

Proteins are an important class of biological polymers. They are the molecules that actually do almost all the work in the cell and are largely responsible for expressing the information present in the DNA molecule. They range in size from a few kilodaltons to hundreds of kilodaltons. All proteins are built up from 20 amino acids, which constitute the monomers. The sequence in which these amino acids are arranged differs from protein to protein and is referred to as the primary structure of the protein. Some sequences of amino acids are known to arrange themselves into regular three-dimensional structures that are referred to as the secondary structure of the proteins. At the next level of structural organisation, the secondary structural units arrange themselves into globular shapes. This is called the tertiary structure and is stabilised mainly by interactions between the amino acid side chains. Thus the side chain conformations of the amino acids also form part of the tertiary structure. Also discernible at this level are domains corresponding to compactly arranged groups of secondary structural features. In the next higher level of structure, globular folded polypeptide chains come together in specific arrangements called the quartenary structure. Thermodynamic experiments by Anfinsen have shown that the functional three-dimensional structure of the protein is encoded in its primary amino acid sequence. Thus a polypeptide chain in solution will automatically fold into its three dimensional structure given only that the conditions of temperature, ionic strength, pH, etc., are within a fairly wide range. Ultimately the structure and therefore the function of proteins is directly derived from the chemical nature of the amino acids.

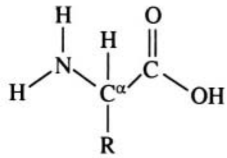
10.3.1 Amino acids and the primary structure of proteins

Alpha amino acids can be considered as being chemically put together around the central carbon atom called C^{α}. Carbon atoms have a valency of four and in amino acids this valency is satisfied by an amino group, a carbonyl group, a hydrogen atom and a side chain. The only exception relevant to proteins is the amino acid proline. Figure 10.25 gives details of the side chain of the twenty amino acids that are found in naturally occurring proteins. It is not known why only these twenty amino acids were chosen. Presumably the answer lies in the way in which life originated on earth.

The 20 amino acids have two major features in common (a) they are all α- amino acids and (b) they are all L-amino acids. They can be separated into groups with other common features. Glycine, alanine, valine, isoleucine, leucine and phenylalanine have hydrophobic, non-polar side chains. Phenylalanine, tryptophan and tyrosine have aromatic planar rings. Aspartic acid and glutamic acid carry a negative charge at neutral pH, while lysine and arginine each carry a positive charge. Histidine has a highly reactive 5 membered planar ring that also carries a positive charge at neutral pH. Cysteine, serine, threonine, asparagine, glutamine and tyrosine have neutral, polar (and therefore hydrophilic) side chains. The chemical nature and size of the side chain determines its activity and also its position in the protein. Histidine frequently occurs at the active site of enzymes. Hydrophobic side chains usually occur in the interior and hydrophilic groups on the surface of the proteins that are active in aqueous media. The situation is the reverse in the case of membrane proteins. The sequence in which the amino acids occur along the polypeptide chain therefore closely controls how the chain will fold into the three-dimensional structure. In other words the primary structure of the protein determines its secondary, tertiary and quaternary structures and ultimately the function of the protein. The polypeptide chain is built up when the amino acids join to each other by means of a peptide bond.

10.3.2 The peptide bond and secondary structure of proteins

The peptide bond is formed as shown in Figure 10.26. The C-N bond has a partial double bond character, which means that the molecule cannot rotate about it. The six atoms that form the peptide



The general formula of an amino acid
R = sidechain

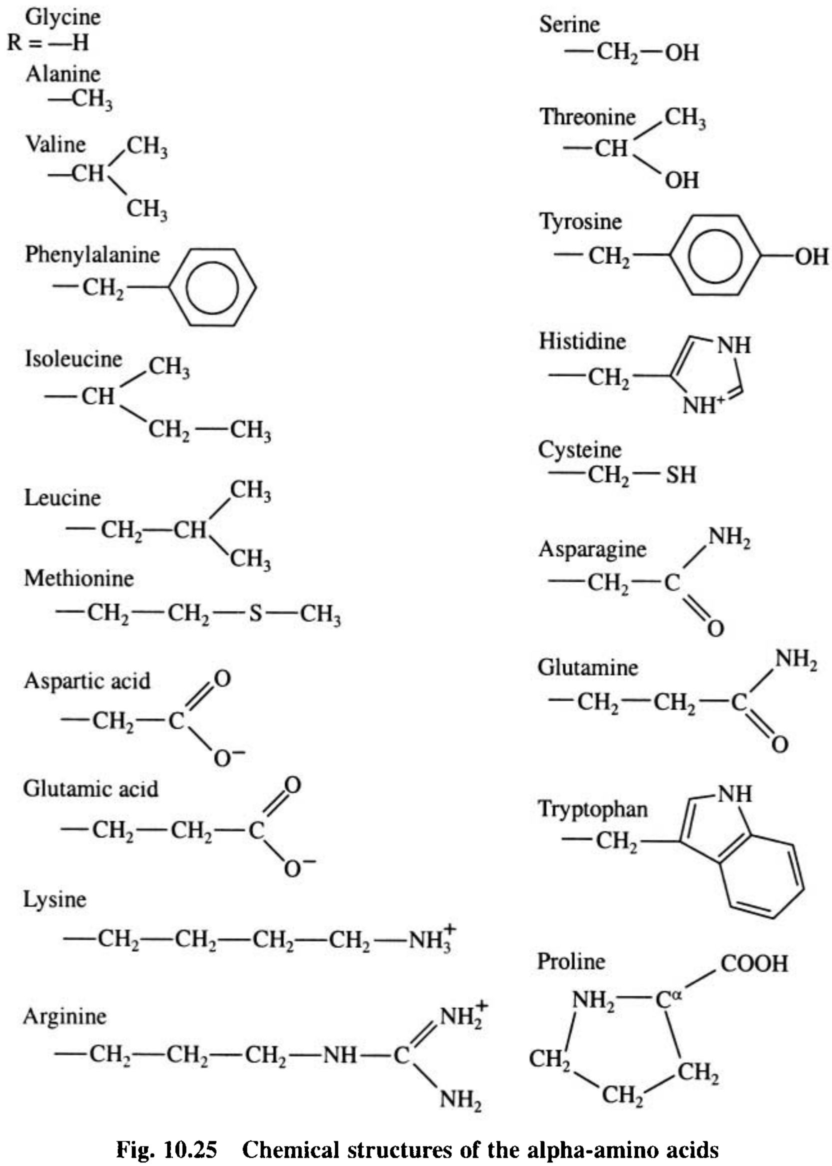


Fig. 10.25 Chemical structures of the alpha-amino acids

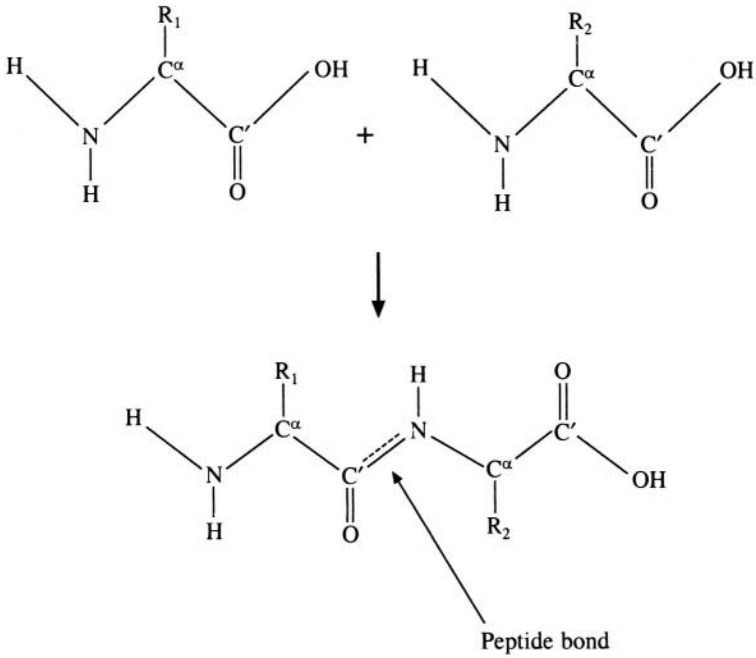


Fig. 10.26 The peptide bond. Dotted line indicates a partial double-bond character.

group are constrained to lie on a plane thereby forming the planar peptide group (Figure 10.27). Ramachandran and his colleagues at the University of Madras realised that this property simplified the geometrical analysis of the polypeptide chain.

If the rotational possibilities of the longer side chains are ignored and the backbone of the protein chain alone is considered, then each residue is described by two torsion angles ϕ and ψ (Figure 10.28). The freedom of rotation about even these bonds is not absolute, but is restricted by possible steric hindrances. For example, if the $C'_{i-1} - N_i$ bond is *cis* to the $C_i^\alpha - C'_i$ bond when the $C_i - N_{i+1}$ bond is *cis* to the $N_i - C_i^\alpha$ bond, i.e. when both ϕ and ψ are 0°, then a severe clash between H_{i+1} and O_{i-1} will occur, making this particular pair of values disallowed. A systematic search of all pairs of values of ϕ and ψ reveals that, only about 18% of the values are allowed. This is shown as the Ramachandran map (Figure 10.29). This map indicates the situation when the atoms are modelled as hard spheres allowing no interpenetration. The radii of the spheres were derived from the interatomic distances seen in the crystal structures of small molecules. Apart from these normal values, somewhat smaller values for the contact distances were also obtained. These were called the extreme limits, and when they were used, the allowed area on the Ramachandran Plot expanded to 22%. The map is correct for eighteen of the twenty side

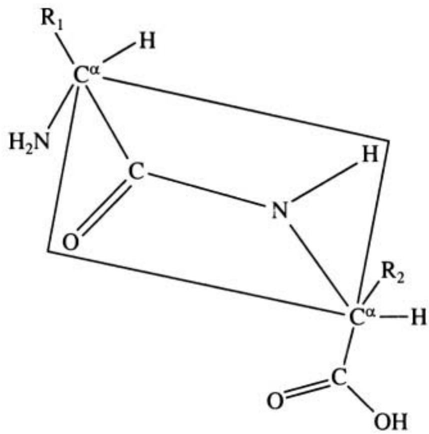


Fig. 10.27 The planar peptide unit

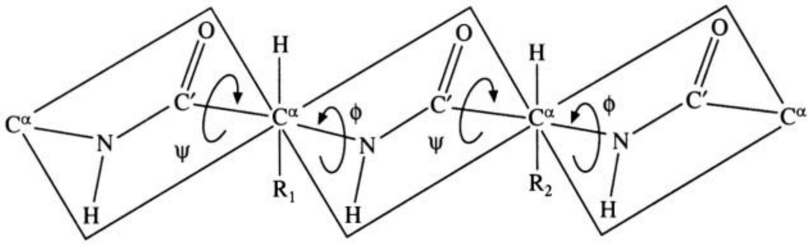


Fig. 10.28 The torsion angles ϕ and ψ used to describe the conformation of a polypeptide chain

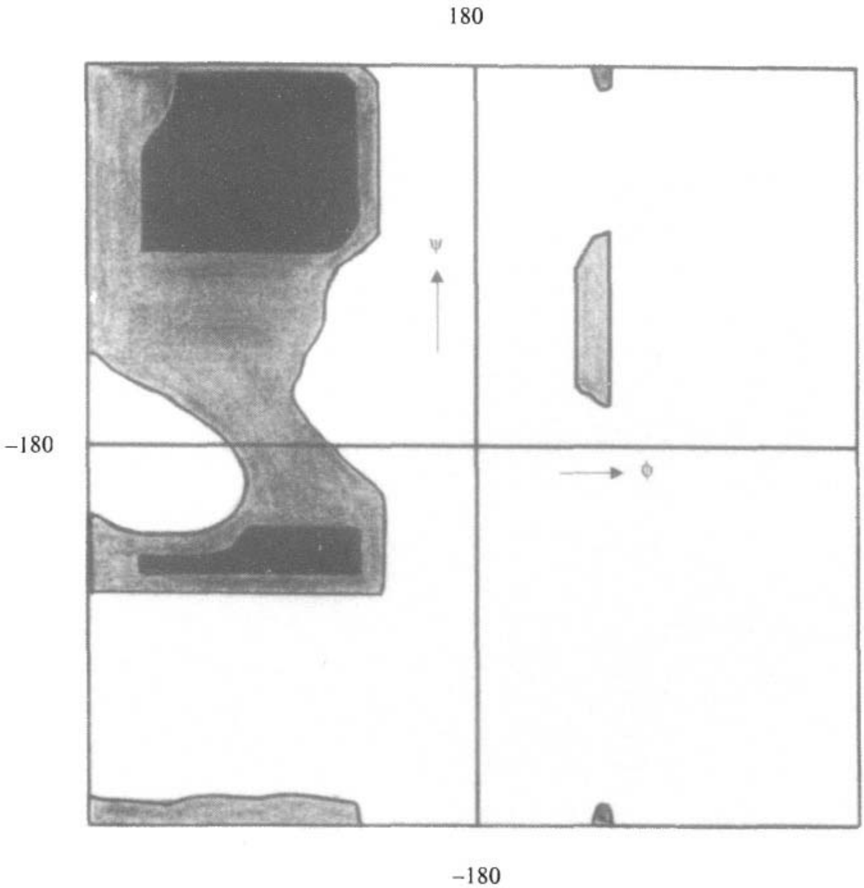


Fig. 10.29 The Ramachandran map. Strictly allowed regions are black while generously allowed are light grey.

chains. The exceptions are glycine and proline. Glycine has a much larger allowed region owing to the absence of the C^{β} atom (Figure 10.30). The side chain of proline binds to the peptide nitrogen atom and this limits the value of ϕ to $-60 \pm 20°$.

The peptide units are not absolutely planar, as assumed in constructing the above maps. The torsion angle ω about the peptide bonds can deviate by 10 to 20° from planarity. This changes the ϕ–

ψ map somewhat. The map also changes when the 'rigid sphere' model of the atoms is replaced by a semiempirical energy function involving interatomic distances. The changes are however quite small and over 98% of the experimentally determined values of ϕ and ψ fall within the allowed regions of the Ramachandran plot.

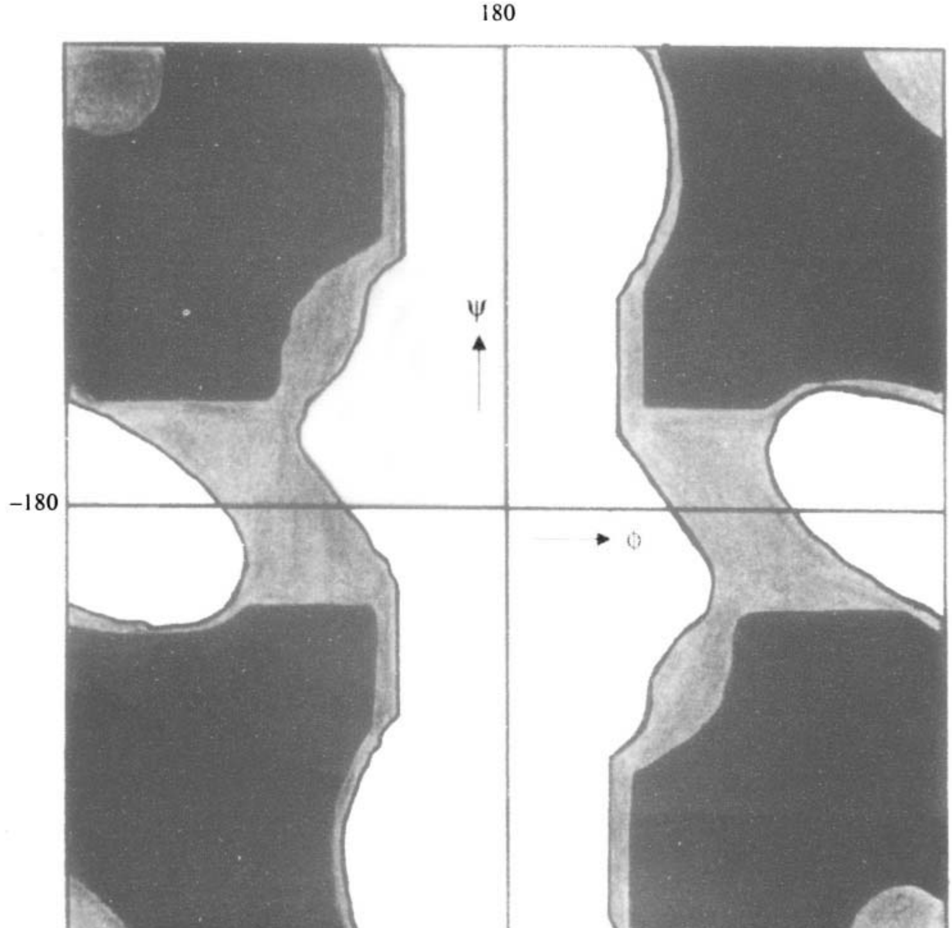


Fig. 10.30 The Ramachandran map for glycine. Strictly allowed regions are black while generously allowed are light grey.

If some of the pairs of values of ϕ and ψ which fall within the allowed region are repeated for many residues along the polypeptide chain, it leads to regular recognisable secondary structures, such as **α-helices** and **β-sheets**.

10.3.2.1 ***α-helix*** This is one of the most commonly occurring secondary structures in proteins, with **(ϕ, ψ)** values of (–57°, –47°) (Figure 10.31). The helix has 3.6 residues per turn. The occurrence of such a non-integral value for the helical repeat was a surprise when Linus Pauling first proposed it.

But subsequent experimental and theoretical studies have confirmed the structure overwhelmingly. The stability of the **α-helix** is enhanced by the hydrogen bonds between the C=O group of residue '*i*' and the NH group of residue '*i* + 4'. Each turn of the **α-helix** is 5.4 Å long, indicating a rise per residue of 1.5 Å. The width of the helix is about 4 Å. For *L*-amino acids only right handed **α-helices** are allowed and therefore almost all **α-helices** observed in protein structures have a right handed twist. Very short (3-5 residues) left-handed **α-helical** regions have also been observed in proteins, though very rarely. Owing to the fact that all the hydrogen bonds in the **α-helix** point in the same direction, the helix as a whole has a dipole moment such that the amino end or the *N* terminal has a net partial positive charge and the carboxy end or the *C* terminal has a net partial negative charge. This induces negatively charged ions such as PO_4^- to bind to the amino end. However, positively charged ions rarely bind to the carboxyl end. Alpha helices are preferentially made up of certain of the 20 amino acid residues. Among the strongest helix-formers are alanine, glutamine, leucine and methionine. Conversely proline, glycine, tyrosine and serine occur in helices only rarely. An **α-helix** often occurs on the surface of a protein. Then it is usually made up of hydrophobic residues on the side facing the interior of the protein and hydrophilic residues on the other.

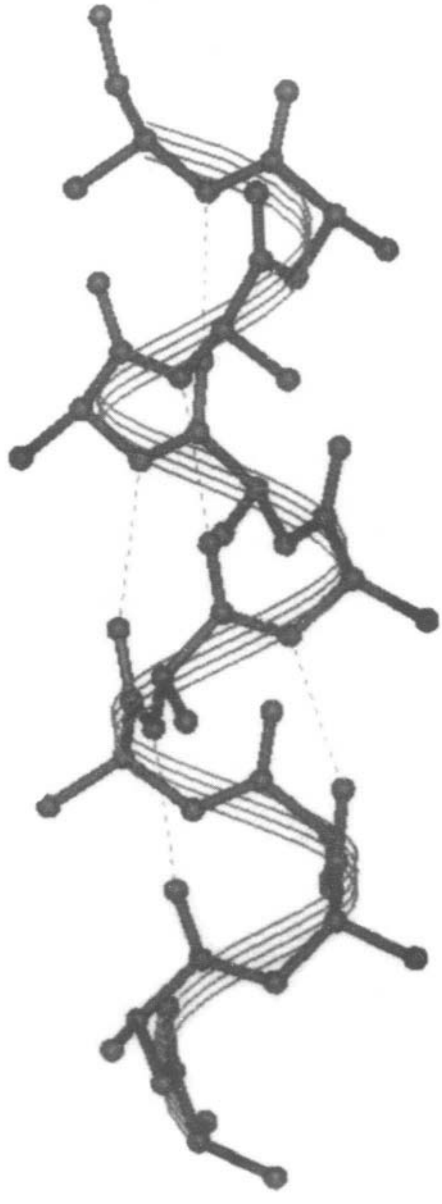


Fig. 10.31 The alpha helix

*10.3.2.2 **β-strands** and **β-sheets*** **β-strands** (Figure 10.32) are highly extended polypeptide chains and the value of (ϕ, ψ) are in the large allowed region in the top left quadrant of the Ramachandran Map. **β-strands** interact with other **β-strands** to form **β-sheets**. This interaction can occur in two ways. Firstly, all the interacting **β-strands** could run in the same direction, i.e. the amino terminii of all strands are on the same side. In this case, a parallel a **β-sheet** is formed (Figure 10.32(a)) in which somewhat distorted H-bonds are formed between the amino groups of one strand and the carboxy groups of the other. In the other arrangement, the strands run in opposite directions, resulting in an antiparallel sheet (Figure 10.32(b)). Here again N–H … O hydrogen bonds stabilise the structure. In both cases, the amino acid side chains project out of the sheet on either side. The sheets are usually not flat but have a twist (Figure 10.33) and always in a right handed sense. Mixed parallel and antiparallel **β-sheets** also occur, but only rarely.

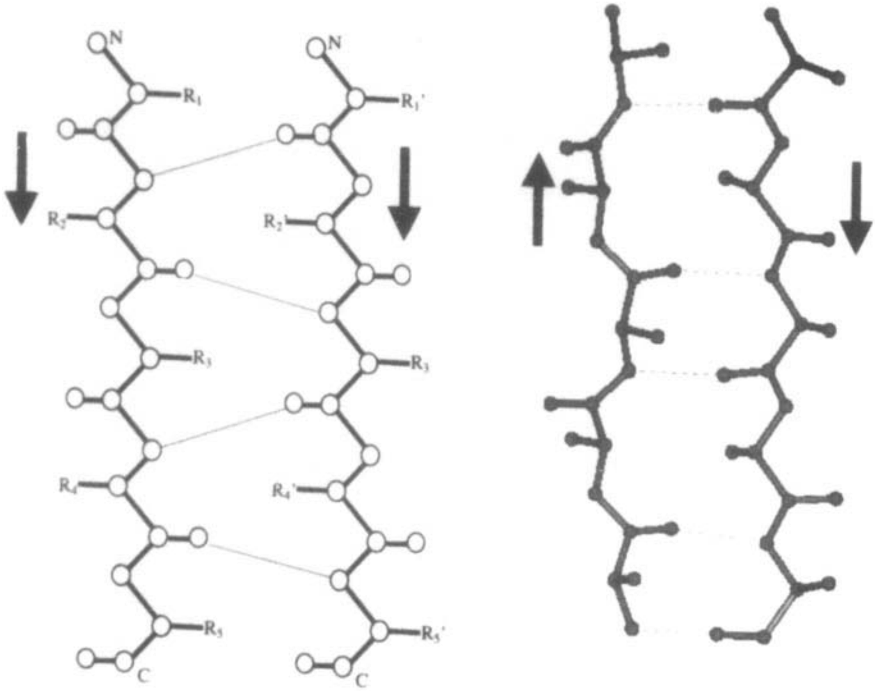


Fig. 10.32 Beta sheets: (a) parallel (schematic) and (b) anti-parallel (3-D representation)

*10.3.2.3 **β-turns*** Whenever the peptide chain has to reverse its direction, as frequently happens in globular proteins, it can sharply and efficiently do so through a structure known as the **β-turn** or a reverse turn. Three consecutive peptide units are involved in such a turn and the most prominent stabilising factor is the hydrogen bonds that are formed between $\mathbf{O}_i$ and $\mathbf{N}_{i+3}$ (Figure 10.34). The **(φ, ψ)** values of all the residues involved in this secondary structural unit do not fall in the same region of the Ramachandran plot. At the residue *i*+1, the values are (–60, –30°) and at the residue i+2, the values are (–90,0°), assuming that the turn is made up of the residues *i, i*+1, *i*+2 and *i*+3. This is called the Type I **β-turn.** In the Type II turn, the values are (–60,120°) and (80,0°) at residues *i*+1 and *i*+2 respectively. However, in this turn there is a steric clash between side chain $\mathbf{R}_{i+2}$ and $\mathbf{O}_{i+1}$, so that $\mathbf{R}_{i+2}$ can only be H, i.e. glycine. A type III reverse turn has regular structure with **(φ, ψ)** values of (–60, –30°) at both central residues. A repetition of these angles over a long stretch leads to a type of helix called a $\mathbf{3_{10}}$ helix, which is found in proteins, though rarely.

10.3.2.4 Collagen helix This is a special type of helix discovered by Ramachandran and his colleagues. It occurs in the fibrous protein collagen, which is the most abundant protein in animals. It is a left-handed helix, with **(φ, ψ)** values about (–60,140°). It is not a completely regular helix and the **(φ, ψ)** values are repeated exactly only at every 3 residue, i.e. at the glycine residue (the sequence of collagen is always (gly-*x*-*y*) where *x* and *y* are frequently proline and hydroxyproline respectively). The protein collagen itself consists of three of these single strand helices wound around each other to form a triple helical fibre (Figure 10.35). The structure was able to explain why such a repetitive sequence was required, since every third residue came close to the superhelix axis. Side chains larger than that of glycine would introduce a large degree of instability.

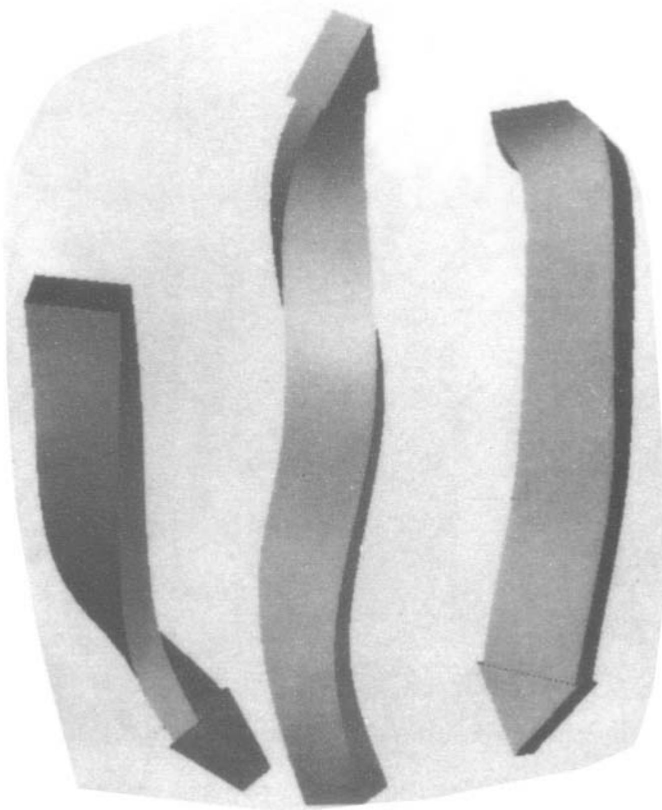


Fig. 10.33 The twist in the beta sheet.

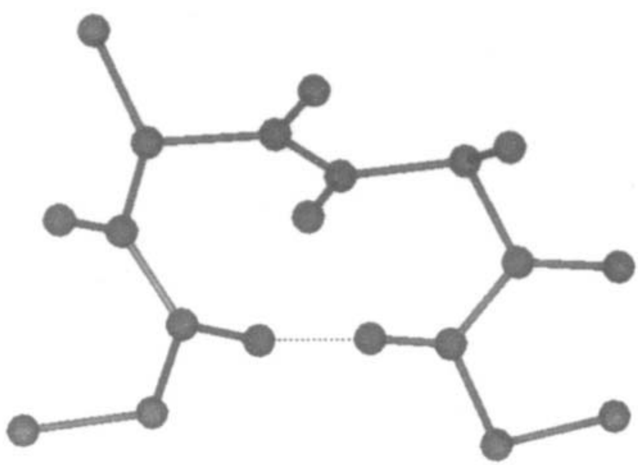


Fig. 10.34 A beta turn

10.3.3 Tertiary structure-supersecondary and domain structure

Various segments of a protein chain fold into various secondary structural units such as **α-helices** or **β-sheets.** These units further fold up into compact supersecondary structural units or domains. This is called the tertiary structure of the proteins and involves interactions between amino acid residues that are not necessarily close together in the primary sequence. One of the most important of such tertiary interactions is the disulphide bond.

The disulphide bond is formed between the sulphur atoms of two cysteine residues (Figure 10.36). Such bonds could occur within a single polypeptide chain or between two different chains. The chief function of the disulphide bridge is to provide mechanical stability to the protein structure. They may also determine chemical properties by stabilising the correct active conformation. In some proteins the disulphide bond may play a catalytic role.

Another major stabilising factor in the tertiary structure of proteins is the so-called entropic or hydrophobic forces. Since the protein is made up of both polar and non-polar side chains, in an aqueous solution the protein molecule behaves like a drop of oil, with polar groups on the surface, in contact with the solution and the non-polar groups in the interior of the molecule. The solvent molecules in the near vicinity of the protein also assume a certain amount of ordering around polar groups. For non-polar groups, however, the solvent molecules make a large negative contribution to the entropic energy, leading to the folded state being favoured over the denatured state. Other factors which stabilise the tertiary structure of proteins include salt bridges or strong Coulombic interactions between oppositely charged amino acids, such as glutamic acid and aspartic acid. Hydrogen bonds in the interior of the protein are another important stabilising factor not only of the secondary structural motifs but also of the tertiary structure.

The secondary structural units often form clearly identifiable domains or super-secondary

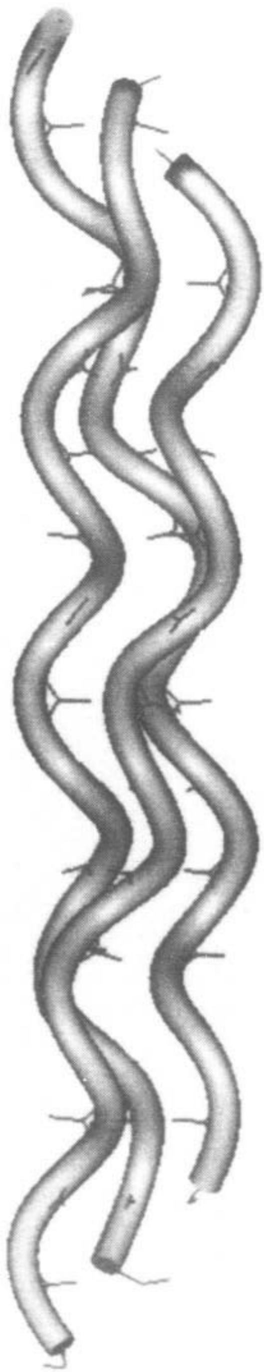


Fig. 10.35 Triple-helical structure of collagen

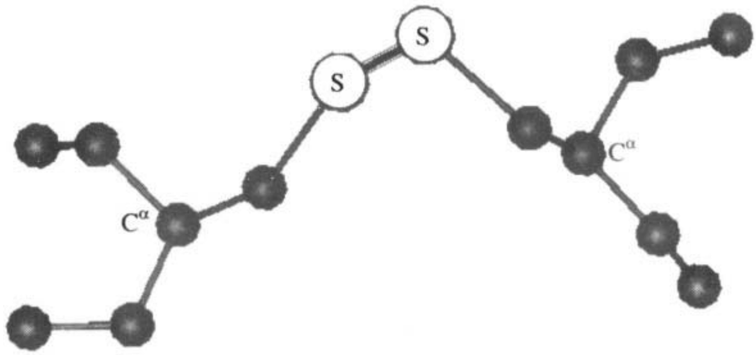


Fig. 10.36 The disulphide bond.

structures. In proteins like cytochrome b_{562}, four helices come together to form the so-called 4 helix bundle (Figure 10.37). This is a widely occurring super-secondary structure. Motifs containing both **β-sheets** and **α-helices** also occur in many proteins. An example is α/β barrel structure. It has been noticed that the core of such barrels is usually packed with hydrophobic side chains. Other super-secondary structural motifs include **β-barrels** such as the one seen in retinol binding protein. Domains in proteins can be loosely defined as an almost self-contained globular portion of the protein, loosely connected to other domains to form the complete molecule. A clear example of a domain structure is the protein troponin C (Figure 10.38). In other cases the domains may not be so clearly demarcated. However, calculations of the distances between pairs of residues can be performed and displayed on a so-called diagonal plot, which would clearly reveal structural domains in proteins.

10.3.4 Quaternary structure

Many proteins in their active or functional forms exist as aggregates of more than one folded polypeptide chain. The macromolecular structure so built up is called the quaternary structure of proteins. Usually the subunits of quaternary structure associate by non-covalent interactions, though disulphide linkages may exist between different subunits. The monomers, which form the quaternary structure, may be identical or may

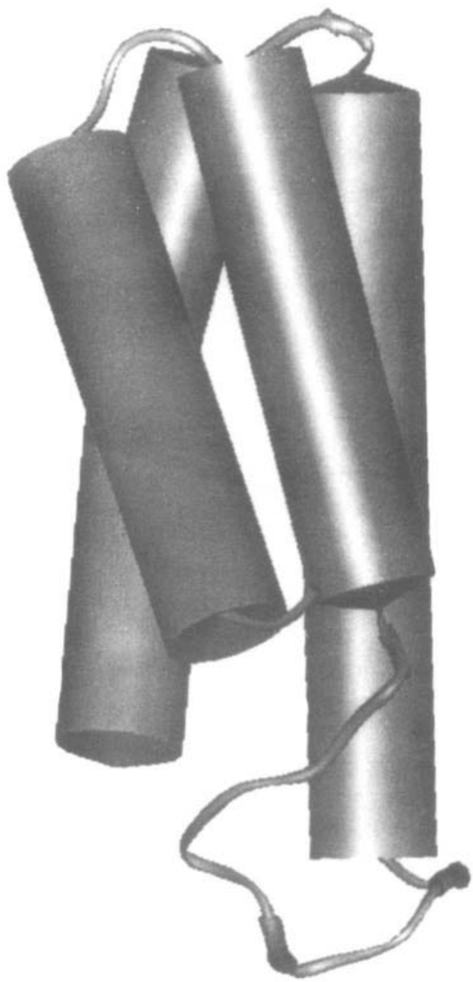


Fig. 10.37 A four-helix bundle.

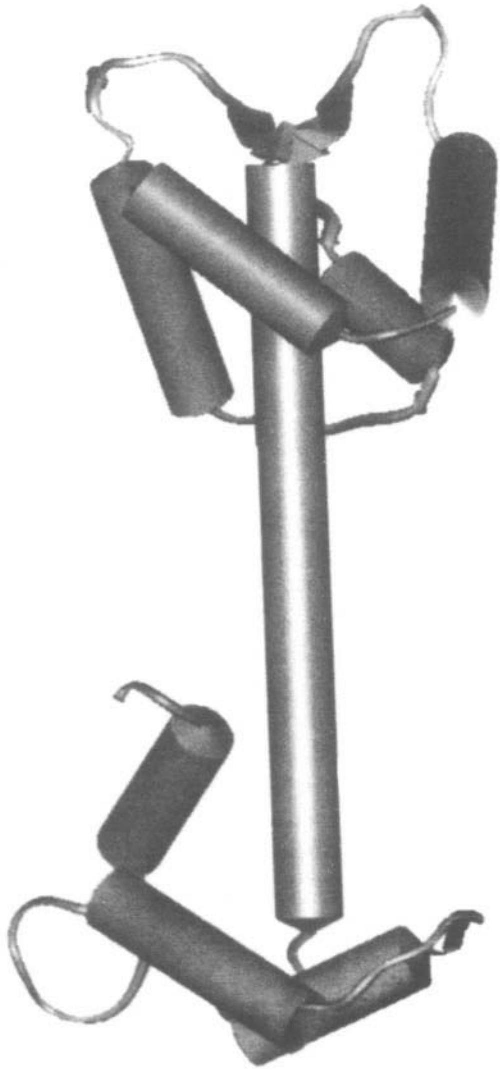


Fig. 10.38 Structure of troponin C showing the two domains

be different. For example, haemoglobin is a tetrameric protein consisting of two chains each of two different polypeptides called the **α-chain** and the **β-chain.** In most multimeric proteins, the number of subunits ranges from 2 to about 12, with a majority possessing 2 or 4 subunits. The exceptions are large enzyme complexes and viruses, which may have even 100 or more monomers. In proteins made up of only a few monomers or a small number of subunits, all the interactions that are possible by subunits must be completely fulfilled. Otherwise larger aggregates would occur. This implies that the subunits must be regularly packed around a central point. In other words most protein multimers possess point group symmetry consisting of one or more intersecting rotation axes. (Mirror inversion is also a part of point group symmetry but is not possible in protein assemblies, as it would require the presence of d-amino acids.) Some of the rotation axes found in proteins include two-fold, three-fold, four-fold, five-fold, six-fold and eight-fold axes. A tetrahedral symmetry is also seen where the monomers are placed at the vertices of an imaginary tetrahedron (Figure 10.39). Hexameric proteins may form trigonal prisms, hexagons or octahedra. Octameric proteins may form cubes or square prisms.

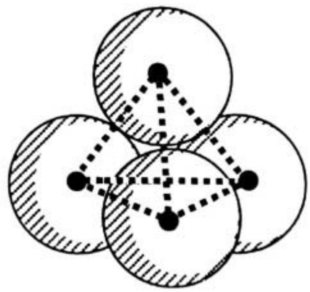


Fig. 10.39 Tetrahedral symmetry in the quaternary structure of the proteins. Each sphere represents one subunit of a tetrameric protein.

In order to attain a definite function, there are several reasons why an assembly of subunits is evolutionarily favourable, rather than giant covalently linked molecules. One of the reasons is that less DNA is needed to code such a protein, since the monomers also contain the information to self-assemble into the multimer. Again, it is easier to make smaller units in an error free fashion than large units. Similarly, defective units can be more economically discarded if they are small. A very important reason is probably that subunits with a particular function may be combined with different subunits to make up different proteins. This modular approach to building up a protein exists at the level of domains too. A particularly striking example is the structure of the nucleotide-binding region, which is almost identical in several proteins with different functions, such as glyceraldehyde-3-phosphate dehydrogenase, phophoglycerate kinase and phosphorylase.

10.3.5 Virus structure

Viruses are large macromolecular assemblies on the border between living and non-living things. The simplest virus conceptually consists of a genome (DNA or RNA) which codes for a protein that will build up a cover to protect the genome (Figure 10.40). However, the viral genome usually codes for a few other proteins too, which are necessary to allow the virus to infect its host. Even with a genome which codes for the coat protein alone, the nucleic acid molecule is too large to allow the protein to completely enclose it. If the protein were to be larger, so would the genome, requiring an even larger protein, and so on. The evolutionary answer to this problem is to construct the viral coat or 'capsid' from several copies of a single, relatively small protein. The capsid is roughly spherical in shape as for example in the common cold virus or in HIV. Other viruses are long cylinders or rod shaped, e.g. Tobacco Mosaic virus. More complex viruses such as bacteriophage *T*4 have more complex shapes.

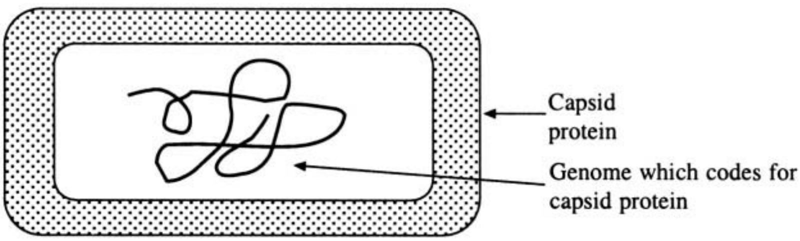


Fig. 10.40 Schematic representation of simple virus

Spherical viruses have the shape and symmetry of an icosahedron (Figure 10.41). Each of the twenty faces of the icosahedron is a triangle with a 3-fold symmetry and can accommodate 3 identical protein molecules. The viral capsid as a whole thus has 60 subunits. If the requirement for exact symmetry is somewhat relaxed, a larger number of subunits can be accommodated. However, only certain multiples (1, 3, 4, 7) of 60 subunits are likely to occur, under the assumption that identical subunits may have quasi-equivalent environments, rather than exact equivalence. More elaborate structures are also known for spherical viruses. The protein capsid of a class of viruses known as picorna viruses (which includes polio virus and common cold virus or rhino virus) contains 60 copies of each of four different polypeptide chains.

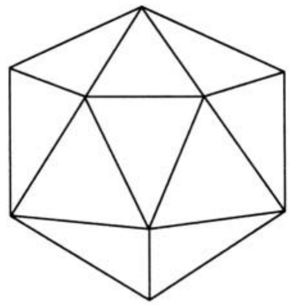


Fig. 10.41 The icosahedral shape of a 'spherical' virus

Rod-like viruses such as the tobacco mosaic virus (TMV) follow different structural principles. Here the main motif is a helix. In TMV a total of about 2130 subunits are arranged as a helix 3000 Å long and 180 Å in diameter. The pitch of the helix is 23 Å and there are 16.3 subunits per turn (i.e. 49 subunits in 3 turns) (Figure 10.42). The arrangement is reminiscent of a bunch of bananas with a long RNA molecule winding round the central stem of the bunch. Electron microscopy and electron diffraction studies indicate that the rod-like tail of bacteriophage *T*4 also consists of identical subunits arranged as a helix.

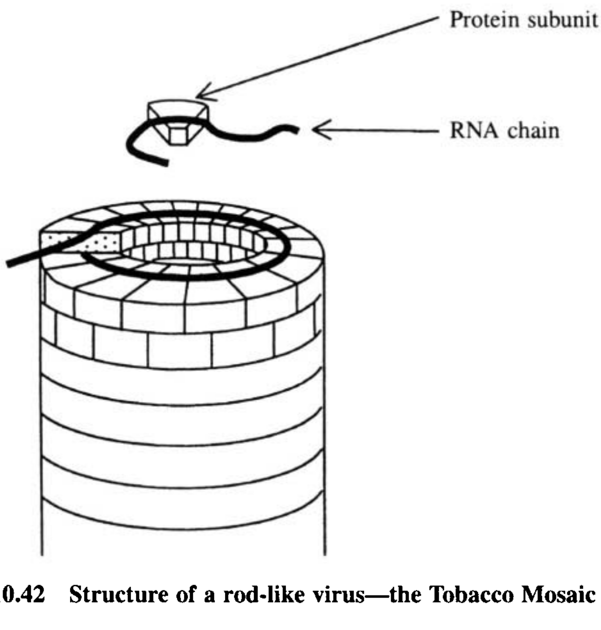


Fig. 10.42 Structure of a rod-like virus—the Tobacco Mosaic virus

11

Energy Pathways in Biology

1.1 Introduction

We studied in Chapter 1 that a process can occur spontaneously only if the sum of the entropies of the system and the surroundings increases. Therefore, for a natural process to take place, energy must be dissipated. Living systems continuously do work within themselves or on the environment. When work is being done by living system no temperature changes take place and hence these systems must spend their energy in some other form either internally or externally. The ultimate source of energy for all living systems is solar energy, which is harnessed through green plants, algae and certain types of bacteria. This process by which solar energy is converted into chemical energy that can be used by the plants and other living systems is known as photosynthesis. The various processes that are involved in the transfer of energy in the biosphere may be represented in a diagram called the energy cycle (Figure 11.1).

Living organisms do not undergo an increase in internal disorder (i.e. entropy) when they metabolise their nutrients. However this does not violate the second law of thermodynamics because the entropy of the surroundings increases. In other words living systems retain their internal order at the cost of the environment. Sometimes the entropy of a living system is said to be negative because it is rich in information content. According to information theory, information is a form of energy and is called negative entropy.

11.2 Free Energy

A continuous supply of free energy is required by living things for carrying out the active transport of ions and molecules, for muscle contraction and other mechanical work, and for the synthesis of macromolecules from their constituents. This free energy requirement is derived from the environment. Phototrops derive it from light while chemotrops obtain it in the form of foodstuff. The free energy derived from oxidation of foodstuff and from light is partly converted into a carrier molecule called

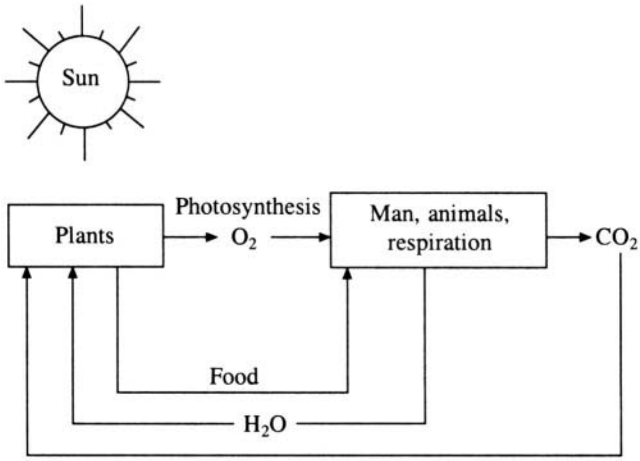


Fig. 11.1 The energy cycle

adenosine triphosphate or ATP (Figure 11.2). ATP is also known as the universal currency of free energy in biological systems. ATP consists of an adenine base, ribose sugar and a triphosphate unit. The energy rich part is the triphosphate unit. Free energy is liberated when ATP is hydrolysed to adenosine diphosphate (ADP) and orthophosphate P_i or when it becomes adenosine monophosphate (AMP) and a pyrophosphate (Figure11.3).

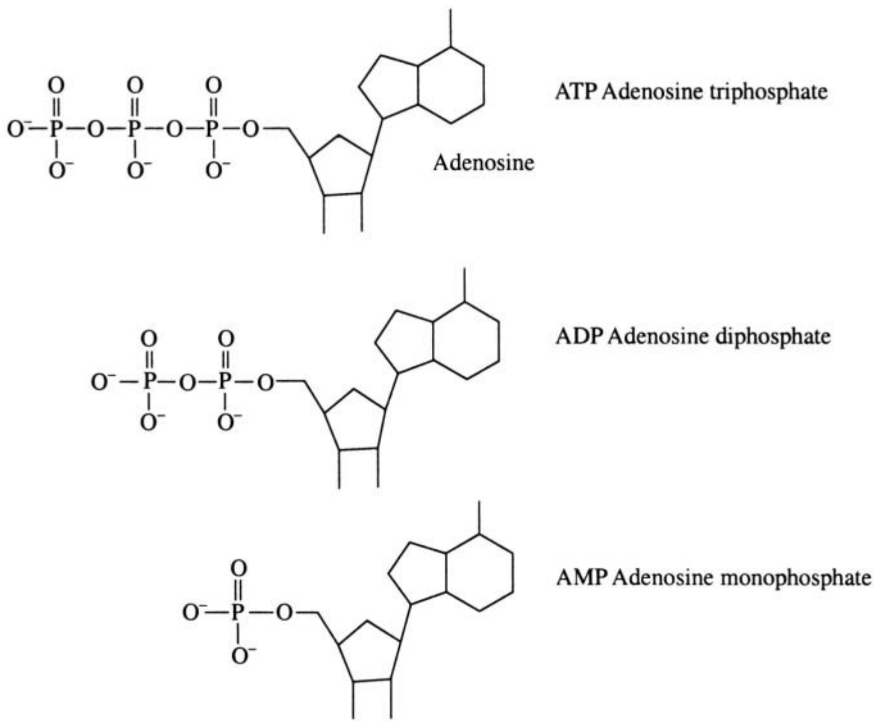


Fig. 11.2 Adenosine phosphates.

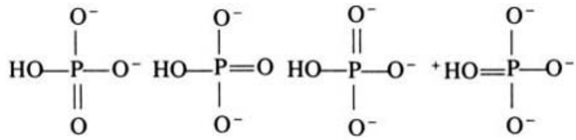


Fig. 11.3 Resonance forms of orthophosphate

$$\text{ATP} + H_2O \rightarrow \text{ADP} + P_i$$

$$\text{ADP} \rightarrow \text{AMP} + P_i$$

ATP, ADP and AMP are inter-convertible. ATP is formed from ADP when fuel molecules are oxidised in chemotrophs or when light is absorbed by phototrophs. The ATP-ADP cycle is the fundamental energy exchange in biological systems. The hydrolysis of ATP occurs as given by the following reaction

$$\text{ATP}^{4-} + H_2O \rightarrow \text{ADP}^{3-} + HPO_4^{2-} + H^+$$

At equilibrium, and under standard conditions the amount of energy released in this reaction is –7.3 K cal/mole. The function of ATP as an energy carrier molecule is dependent on the fact that ΔG^0 (change of free energy at pH 7.0 and under standard conditions) of the above reaction is negative. It is important to note that ATP is not the only phosphate compound with these features. There are others like phosphoenolpyruvate (**ΔG^0 = –14.8 K cal/mole**), 1,3-diphosphoglycerate (**ΔG^0 = –11.8 K cal/mole**), glucose-1-phosphate (**ΔG^0 = –3.3 K cal/mole**), and glycerol-1-phosphate (**ΔG^0 = –2.2 K cal/mole**). The free energy of hydrolysis of ATP falls in the middle of the range of the values for the other phosphate compounds occurring in biological systems. Hence it is an ideal candidate for energy transfer by coupled reactions with common intermediates. The hydrolysis of ATP releases free energy, which becomes useful only when it is coupled with other cellular processes that require free energy. In a typical cell, ATP molecules are consumed within a minute of their formation. Thus, ATP acts as an immediate donor of free energy rather than as an energy storage molecule. Different processes produce ATP in different biological systems. The two main processes are by (1) oxidation-reduction reaction in the inner membranes of mitochondria and (2) photosynthesis which takes place on the membranes of the grana in chloroplasts of green plants. The processes can be divided into those which depend on the oxygen availability i.e. respiration and those which are light driven i.e. photosynthesis.

11.3 Coupled Reactions

A thermodynamically unfavourable reaction can be driven by a favourable reaction, both reactions together having a net negative (and therefore favourable) difference in the free energy. These reactions are coupled by a common intermediate.

For example:

$$A \Leftrightarrow B + C \qquad \Delta G^0 = +6 \text{ K cal/mole} \qquad (1)$$

$$B \Leftrightarrow D \qquad \Delta G^0 = -9 \text{ K cal/mole} \qquad (2)$$

$$A \Leftrightarrow C + D \qquad \Delta G^0 = -3 \text{ K cal/mole}$$

Under standard conditions reaction (1) can not proceed, as the ΔG^0 value is positive. But B can be converted to D as reaction (2) is thermodynamically favourable (ΔG^0 is negative). As free energy changes are additive, the conversion of *A* to C and *D* is possible as the net ΔG^0 is negative when reactions (1) and (2) are coupled through *B,* the intermediate. The hydrolysis of ATP shifts the equilibria of coupled reactions. Therefore any thermodynamically unfavourable reaction can be converted into a favourable one by coupling it with the hydrolysis of a sufficient number of ATP molecules. For example the hypothetical reaction

$$A \Leftrightarrow B \quad (\Delta G^0 = 3 \text{ K cal/mole})$$

can be made to proceed when it is coupled to the hydrolysis reaction of ATP to ADP. The coupled reaction may be written as

$$A + \text{ATP} + H_2O \Leftrightarrow B + \text{ADP} + P_i + H^+$$

with $\Delta G^0 = -4.3$ **K cal/mole.** This final ΔG^0 value is obtained by adding the ΔG^0 of the $A \rightarrow B$ conversion (+ 3 K cal/mole) with that of the **ATP → ADP** reaction (–7.3 K cal/mole).

11.4 Group Transfer Potential

Compare the following reactions

$$\text{ATP} + H_2O \Leftrightarrow \text{ADP} + P_i + H^+ \qquad \Delta G^0 = -7.3 \text{ K cal/mole}$$

$$\text{Glucose-6-phosphate} + H_2O \Leftrightarrow \text{Glucose} + P_i \qquad \Delta G^0 = -3.3 \text{ K cal/mole}$$

The ΔG^0 for the hydrolysis of glucose-6-phosphate is much smaller than that of ATP. This means that ATP can transfer its terminal phosphoryl group to water more readily than glucose-6-phosphate. ATP is said to have a higher phosphate group transfer potential than glucose-6-phosphate. This higher potential has a structural basis. At biological pH 7:0, ATP is almost completely ionised and exists as ATP^{4-}. On hydrolysis we get

$$ATP^{4-} + H_2O \rightarrow ADP^{3-} + HPO_4^{2-} + H^+$$

Under standard conditions, ATP^{4-}, ADP^{3-} and HPO_4^{2-} will be present at concentrations of the order of 1.0 M, while the H^+ ion concentration at pH 7.0 is only 10^{-7} M. On the basis of the law of mass action, the low concentration of H^+ pulls the reaction towards the right. Further, at neutral pH there are four closely spaced negative charges in the ATP molecule. These charges repel each other and the dissociation of the terminal phosphoryl group is required, in order to release at least some of the stress. This repulsive force also prevents the products of the reaction coming together again to reform ATP. A third factor contributing to the high group transfer potential of ATP is resonance stabilisation. Orthophosphate exists in several significant resonance forms (Figure 11.3) while ATP has fewer resonance forms. In the hydrolysis of ATP the products HPO_4^{2-} and ADP^{3-} are in more stable resonance forms and contain less free energy than when they are still combined as ATP^{4-}. Thus once again the forward reaction is favoured over the reverse reaction.

In the case of glucose-6-phosphate, the reaction is

$$\text{Glucose-6-phosphate} + H_2O \rightarrow \text{Glucose} + HPO_4^{2-}$$

In this case there is no extra H^+ ion formed and chemical pressure on the reaction to proceed towards the right does not exist. Also, no repulsive force exists between glucose and HPO_4^{2-} and hence these two compounds can recombine to form glucose-6-phosphate readily. This molecule thus has a much lower transfer potential as compared to ATP.

11.5 Role of Pyridine Nucleotides

Chemotrophs get their energy from oxidation of fuel molecules like fatty acids, glucose etc. These reactions are catalysed by enzymes, which in turn require cofactors. The most common cofactors are nicotinamide adenine dinucleotide (NAD), nicotinamide adenine dinucleotide phosphate (NADP) and flavin adenine dinucleotide (FAD) (Figure 11.4). The cofactors act as carriers of electrons in the oxidation reduction reactions. In aerobic organisms electrons from the fuel molecules are transferred first to these carriers. The reduced form of these carrier molecules transfers the electrons to O through an electron transport chain located in the inner membrane of the mitochondria. ATP is formed from

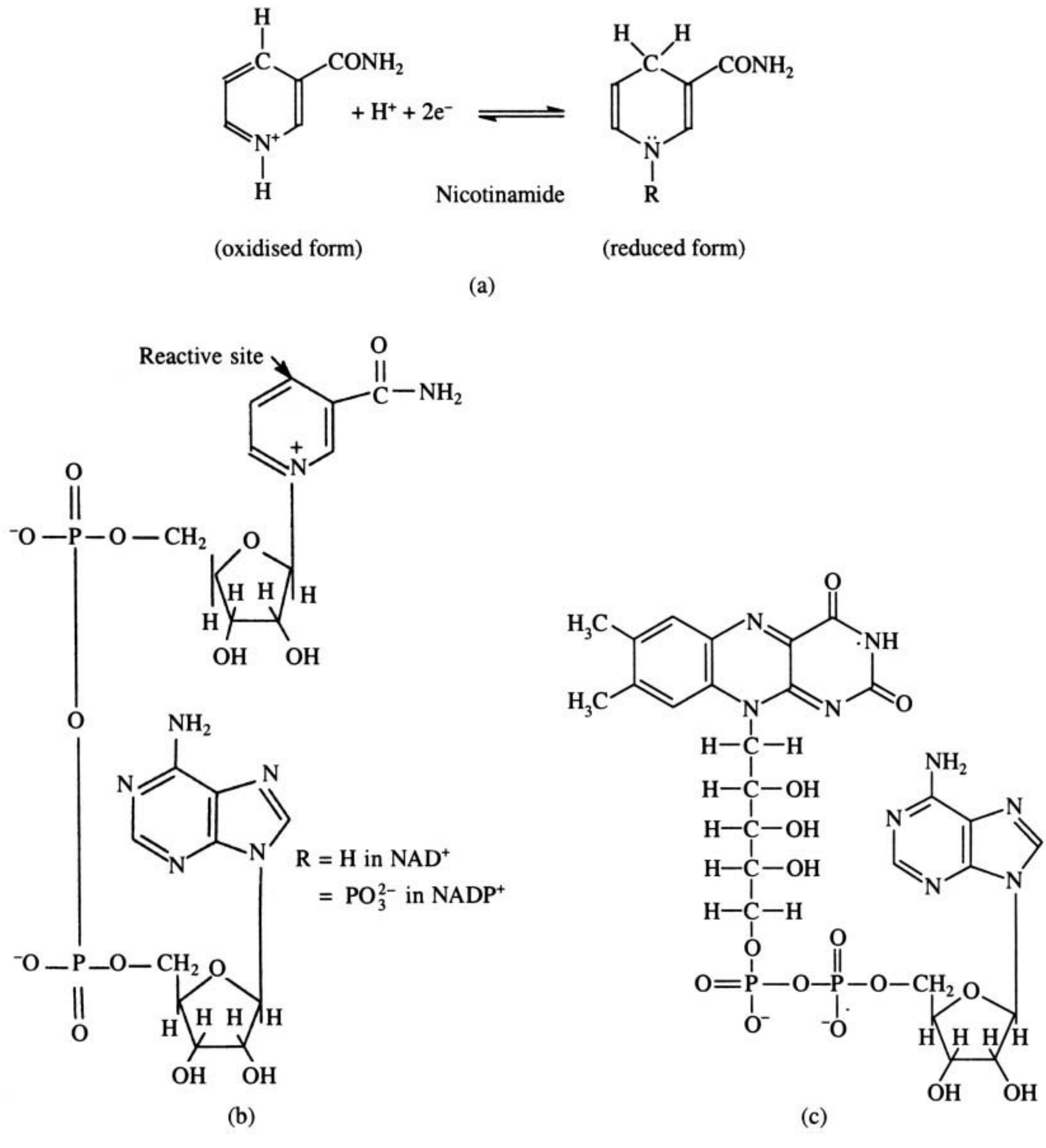
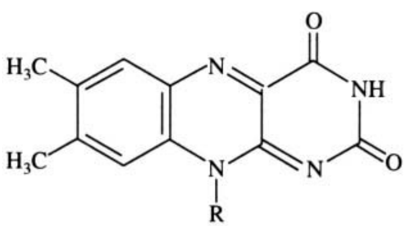


Fig. 11.4 Pyridine nucleotides. (a) Oxidised and reduced forms of nicotinamide. (b) NAD and NADP. (c) FAD

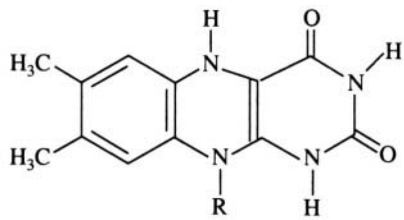
ADP and orthophosphate during this process, which is known as oxidative phosphorylation. **NAD$^+$** accepts a hydrogen ion and two electrons from the substrate in the oxidation of the substrate molecule. In the process, **NAD$^+$** is converted from the oxidised to the reduced state.

$$\mathrm{NAD^+}\ \mathrm{R_1{-}\overset{\displaystyle H}{\underset{\displaystyle OH}{C}}{-}R_2} \Leftrightarrow \mathrm{R_1{-}\overset{\displaystyle H}{\underset{\displaystyle \| \atop O}{C}}{-}R_2} + \mathrm{NADH} + \mathrm{H^+}$$

Similarly, the oxidised and reduced forms of FAD are shown in Figure11.5. NADP and NAD have structural similarity but their biological functions differ. NADP is used in reductive biosynthesis while NAD is used primarily in the energy metabolism. The cell has a mechanism by which NADPH can be converted to NADH and vice versa. Enzymes involved in this reaction are known as trans-hydrogenases. In the absence of catalysts, NADH, NADPH and FADH react very slowly with O_2. Similarly the hydrolysis of ATP in the absence of a catalyst is a slow process. This stability of these molecules is essential for their biological function as it provides a way for enzymes to control the reduction mechanism and hence the flow of free energy.



(a) FAD



(b) $FADH_2$

Fig. 11.5 Oxidised (a) and reduced (b) forms of FAD

11.6 Photosynthesis

Photosynthesis is the process by which the energy of sunlight is captured and used to sustain life on earth. Green plants, algae and some bacteria are capable of carrying out this process which takes place in the thylakoid membranes of the organelle called chloroplast (Figurel 1.6). Light is absorbed by the pigment molecules (carotenoids, chlorophylls and phycobillins) (Figurel 1.7) in these membranes and is transferred to the so-called photosynthetic reaction centres. The light energy is converted into chemical energy at these centres. The aggregate of the reaction centre and the photosynthetic pigments is called a photosynthetic unit. Each photosynthetic unit may consist of 50 to 400 chlorophyll molecules. The absorbed light excites the pigment aggregate and this in turn transfers the energy to the reaction centre. The photosynthetic reaction takes place in two phases (Figure 11.8). In the first phase a reductant is produced in the presence of light. In the second stage this reductant is utilised to form sugar in the absence of light. The excitation energy transfer from chloroplast to the next member in the chlorophyll cluster (light harvesting or antenna molecules) takes place through inductive resonance transfer till it reaches a specialised chlorophyll molecule at the reaction centre. The reaction centre is a protein-chlorophyll complex system. These reactions are described by the equations:

$$S + h\nu_a \rightarrow S^*$$

$$S^* + A \rightarrow S + A^*$$

$$A^* \rightarrow A + h\nu_e$$

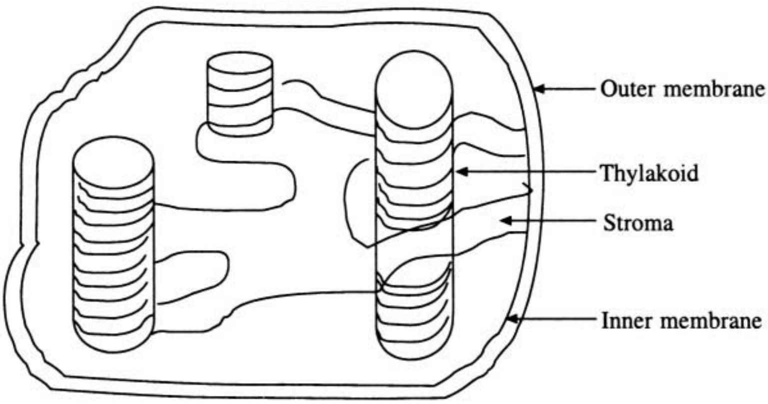


Fig. 11.6 A schematic sketch of a chloroplast showing some of the structures.

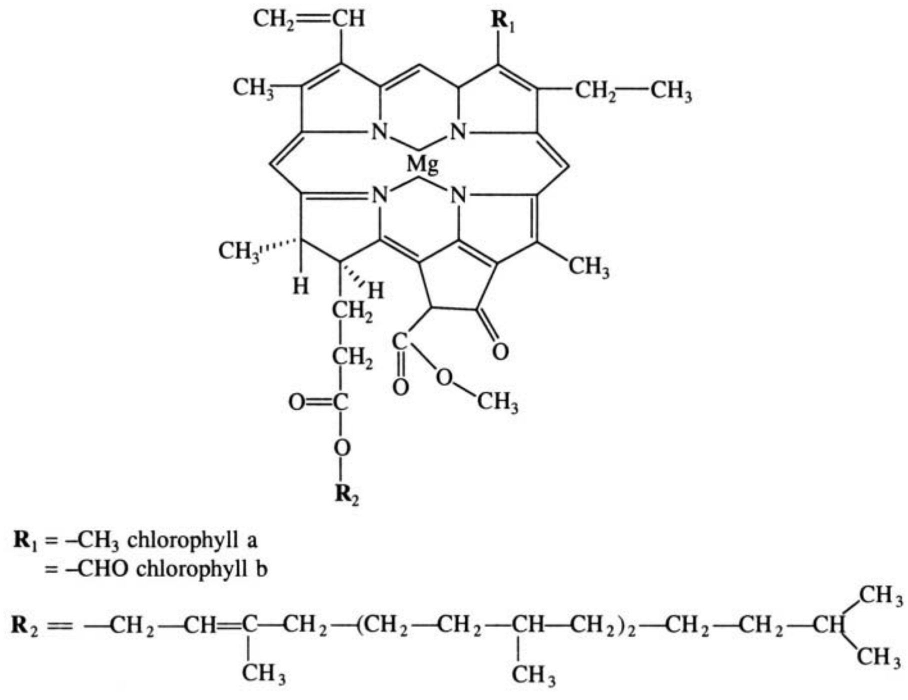


Fig. 11.7 Chemical structures of the pigment molecules in chloroplasts—chlorophyll a and b.

where $h\nu_a$ is the absorbed light, $h\nu_e$ the emitted light, S is the sensitiser molecule, i.e. the reductant, and A is the acceptor molecule, and S^* and A^* are the excited stages. The photosynthetic reaction can be written as

$$n\mathrm{H_2O} + n\mathrm{CO_2} \xrightarrow{\text{Light}} [\mathrm{CH_2O}]_n + n\mathrm{O_2}$$

However, photosynthetic bacteria do not produce oxygen but they give off elemental sulphur. The two phases of photosynthetic reaction described above are known as light reactions and dark reactions.

Two types of reactions involving light take place during photosynthesis and are called photosystem I and photosystem II respectively. In photosystem I, NADPH is produced. In photosystem II, O_2 is evolved and a reductant is generated. Photosystem I has a greater number of a particular type of chlorophyll molecule called chlorophyll *a* and is maximally excited at longer wavelengths. Photosystem II is maximally activated at wavelengths shorter than 680 nm and has more chlorophyll *b*. All photosynthetic cells of higher plants and cyanobacteria that evolve oxygen contain both photosystems I and II. Those bacteria that do not evolve the oxygen contain only photosystem I. Since photosystem I is maximally excited by light of wavelengths 700 nm, the reaction centre of this photosystem is called as P700. Similarly, the reaction centre of the photosystem II is called P680. Figure 11.9 shows the interactions between and within the two photosystems.

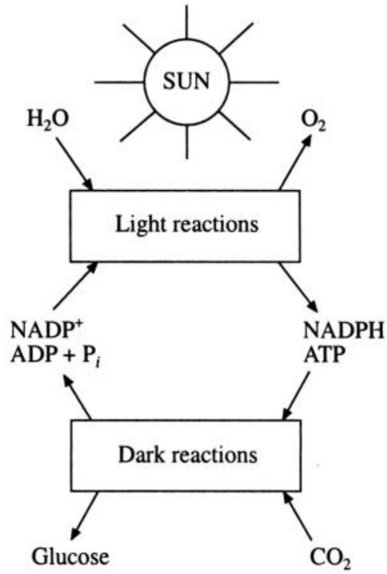


Fig. 11.8 Two stages of photosynthetic reaction

When a photon is absorbed by one of the chlorophyll molecules, the molecule goes from the ground state to an excited state. Generally when the molecule returns to the ground state it emits radiation known as fluorescence. There exists a coupled event wherein the de-excitation of one chlorophyll molecule is accompanied by the excitation of another molecule. This process is also

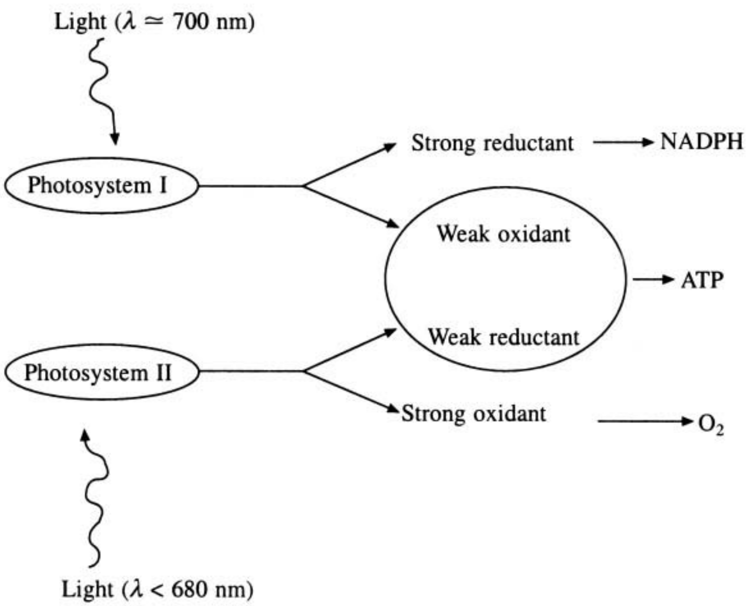


Fig. 11.9 Interactions between the two photosystems

known as sensitised fluorescence. As shown in Figure 11.10, the energy levels of the excited state of the reaction centre are lower than that of other chlorophylls. Hence it acts as an energy trap. This transfer of energy takes place in less than 10^{-10} sec.

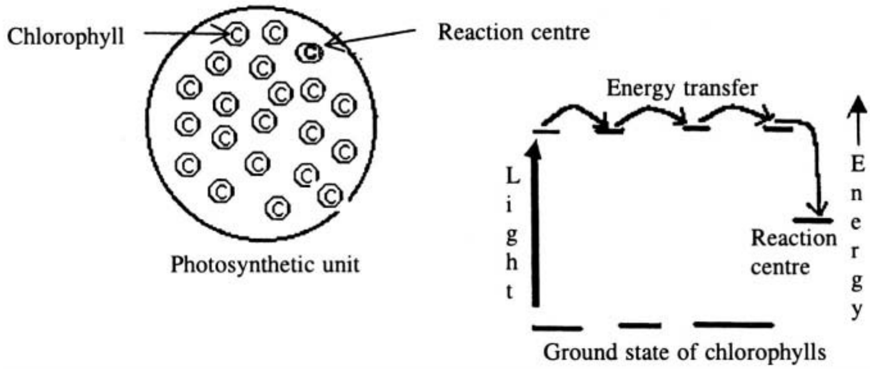


Fig. 11.10 Photosynthetic unit and energy level diagram of the antenna chlorophylls.

Recently, by the combined effort of X-ray crystallographers, spectroscopists and molecular geneticists, the mechanism of photosynthesis in the bacteria of Rhodopseudomonas family has been elucidated (Figure 11.11). Photosynthesis in these bacteria differs from that of higher plants in that no oxygen is given off during the process. However, the results obtained can be extrapolated to higher plants due to many other similarities. The study throws light on the mechanism of electron transport from one end of the reaction centre to the other end. The photosynthetic reaction centre primarily consists of a protein, embedded in the membrane of the photosynthetic vesicles. One end of the reaction centre is near the inner surface of the membrane and the other end is near the outer surface. Several small molecules known as prosthetic or helper molecules are also partially embedded in the protein complex. Photoelectrons flow through the prosthetic molecules during photosynthesis. The photosynthetic reaction centre has a two-fold symmetry axis running from one surface of the membrane to the other. The prosthetic groups are arranged in two spirals on either side of the central protein. When light falls on a special pair of chlorophyll molecules situated at the end of the reaction centre near the inner surface, an electron within the pair absorbs the energy and moves to the next prosthetic molecule viz. pheophytin. This step takes 2 picoseconds to complete. During this process the electron passes through another chlorophyll molecule but does not get attached to it. As an electron has left the special pair of chlorophyll molecules, they now have acquired a positive charge. Next the electron moves from pheophytin to a quinone molecule (Q_A) which is at the end of the spiral. The time taken for this process is about 4 picoseconds. Lastly, the electron moves through the central core of the protein to reach the quinone molecule of the other spiral (Q_B) through which electrons do not flow. This step is comparatively very slow and takes about 100 microseconds. In the meantime a cytochrome molecule donates an electron to the reaction centre and neutralises it. Thus the reaction centre is ready to receive another photon and the process repeats itself. When the quinone molecule has acquired two electrons it detaches itself from the reaction centre and participates in the later stages of the photosynthesis, which takes place on the outer surface of the membrane. Thus there will be two cytochrome molecules with positive charges at the inner surface and two electrons near the outer surface. The separation of the two charges represents stored energy. Later, the charge separation drives chemical reactions that are coupled to the bacterial metabolism. In most cases, the transfer of electrons can take place in both the directions and then the energy is lost as fluorescent radiation. But in the photosynthetic reaction centre the forward reaction is more than eight orders of magnitude faster than the backward reaction and hence the electric charges induced by light energy remain separated. The efficiency of the reaction is as high as 50% i.e. the energy stored in the separated charges is about half of the energy of the incident photons.

11.6.1 Photosystem I

As mentioned earlier, the reaction centre of photosystem I (P700) consists of chlorophyll *a* molecules. When P700 is excited by the absorption of light by the chlorophyll molecules, it transfers an electron to bound ferrodoxin, which is a Fe_4S_4 type iron-sulphur protein. The oxidation-reduction potential of

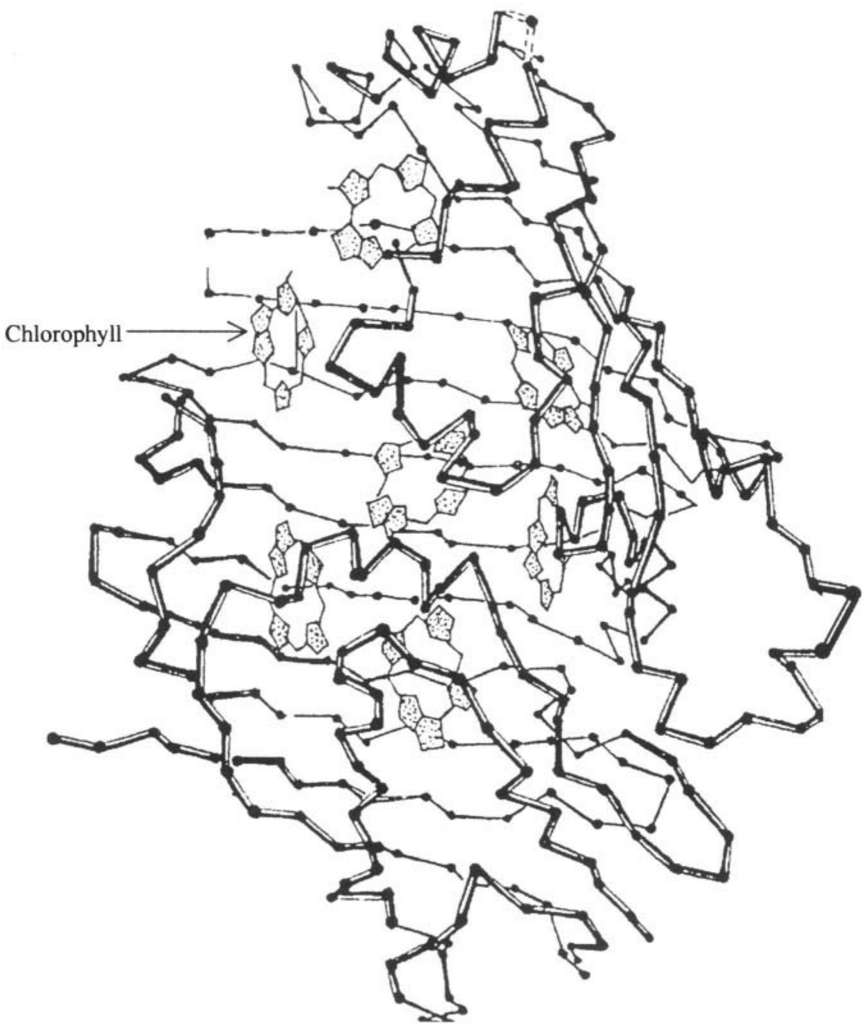


Fig. 11.11 (a) Structure of a bacteriochlorophyll-protein complex.

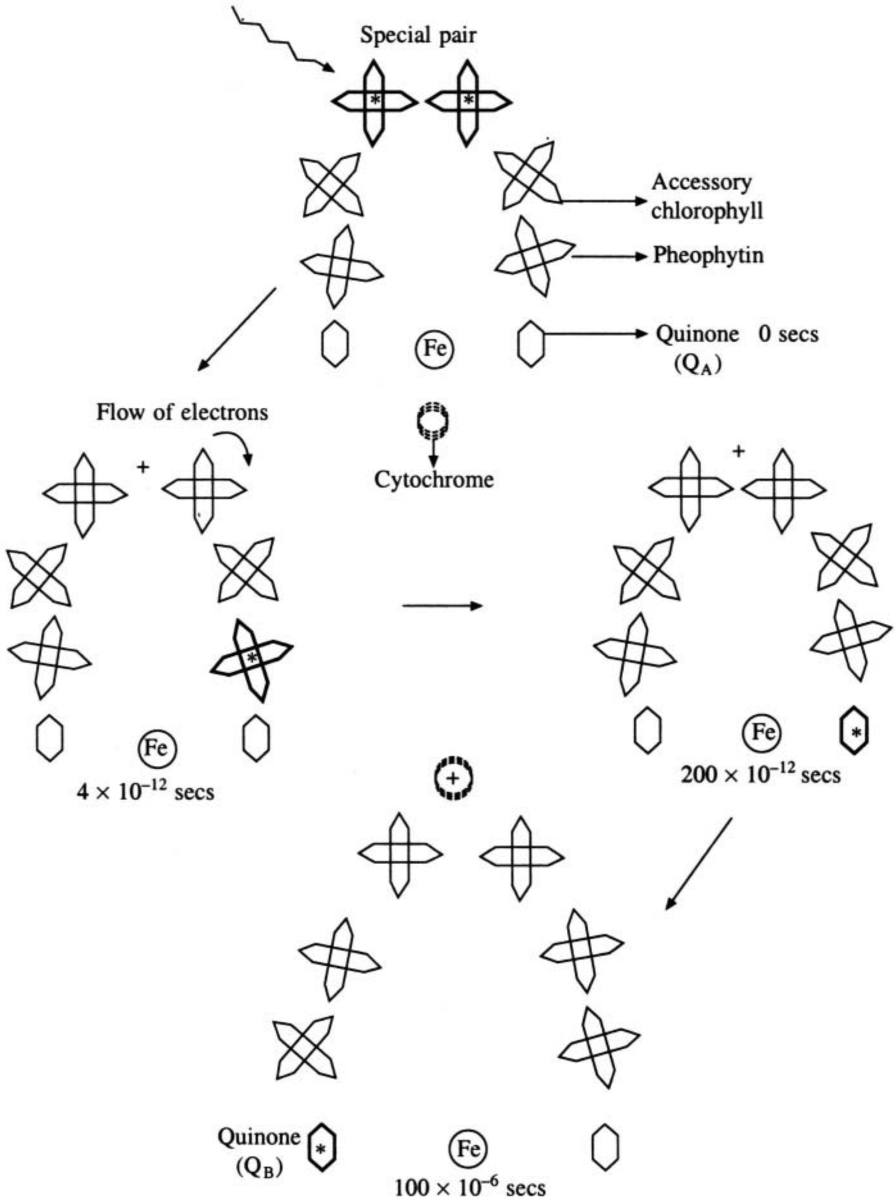
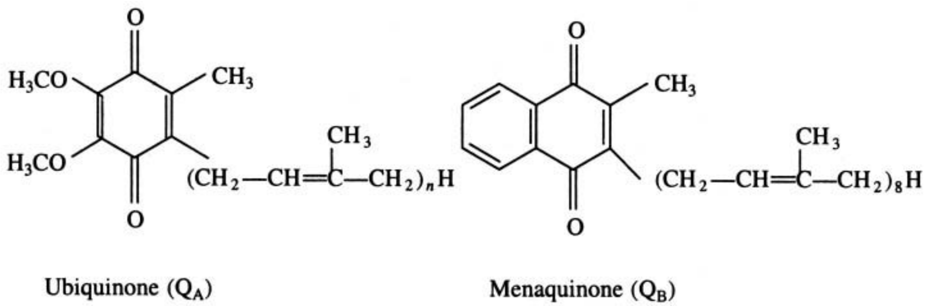


Fig. 11.11 (b) Mechanism of electron transport at photosynthetic reaction centre. Excited groups are shown in bold with a star.

P700 changes from +0.4 V to –0.6 V when it is excited by light. From the bound ferrodoxin the electron moves to soluble ferrodoxin and then to $NADP^+$ reducing it to NADPH. During this process the iron atom of ferrodoxin is alternatively oxidised and reduced. The reduction of $NADP^+$ to NADPH requires two electrons whereas ferrodoxin carries only one electron at a time. Hence electrons from two reduced ferrodoxins come together to convert $NADP^+$ to NADPH (Figure 11.12).



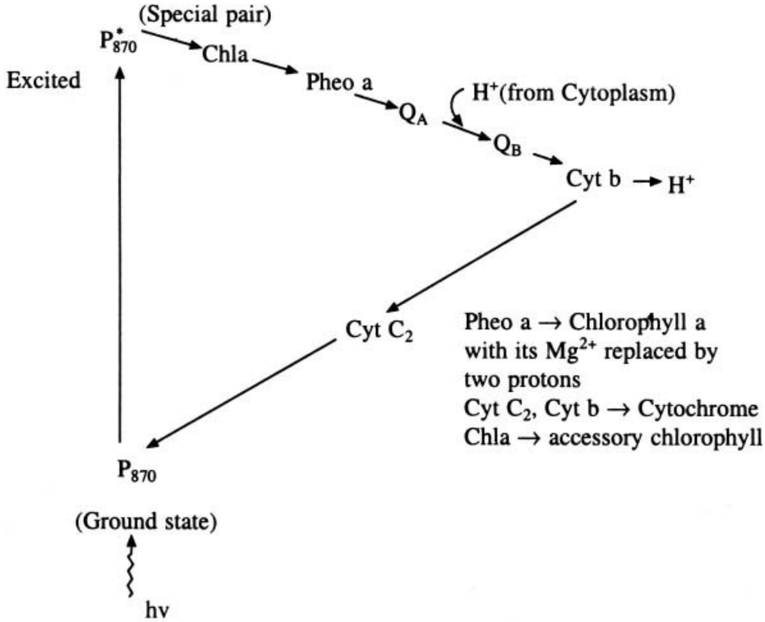


Fig. 11.11 (c) Photosynthetic reactions

11.6.2 Photosystem II

The photosystem II (P680) is not very well understood excepting for the fact that the redox potential of this reaction centre is +0.8 V. Photosystem II produces a strong oxidant **(A^+)** and a weak reductant **(B^-)** and transfers an electron to P700. A^+ attracts electrons from water and gets neutralised. In this process O is released.

$$4A^+ + 2H_2O \rightarrow 4A + 4H^+ + O$$

The electron from photosystem II goes through a bound molecule of plastoquinone, then through mobile plastoquinones, through cytochromes b_{559} and c_{552}, finally to plastocyanin, which transfers it to photosystem I. Thus photosystem I, which was in an oxidised state after absorption of light, is now reduced and is ready for the next cycle of photo reaction. A proton gradient is developed across the thylakoid membrane when electrons flow through the above carriers and, as described earlier, this gradient drives the formation of ATP. The electrons from P700 can also follow a cyclic path, if there is insufficient **$NADP^+$** to accept electrons from reduced ferrodoxin. In the cyclic path the electron gets transferred through cytochrome b_{563}, c_{552}, and plastocyanin, back to P700. Thus NADPH and ATP produced during light reactions are now available for subsequent dark reactions in which CO_2 is converted to carbohydrate.

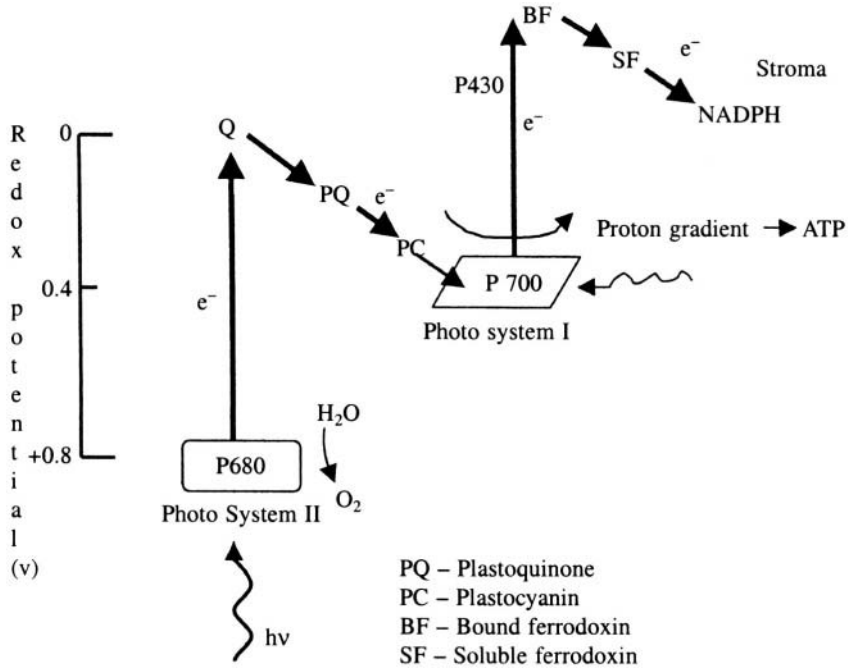


Fig. 11.12 Electron flow in photophosphorylation

11.6.3 Photophosphorylation and carbon fixation

In the second phase of photosynthesis, which occurs in the absence of light, CO_2 is fixed into carbohydrate through a series of reactions known as the Calvin cycle. CO_2 condenses with a five-carbon compound (ribulose-1,5-diphosphate) to form an unstable six-carbon compound which hydrolyses to two three-carbon compounds (3-phosphoglycerate). The reaction can be written as

$$C_5 \rightarrow C_6 \rightarrow C_3 + C_3$$

The cycle is shown in Figure 11.13. Three ATP molecules and two NADPH molecules are consumed in the process of CO_2 fixation as a sugar. As only one carbon is reduced in each cycle, a total of six cycles will be required to form hexose. The complete reaction is given by the equation

$$6CO_2 + 18ATP + 12NADPH + 12H_2O \rightarrow C_6H_{12}O_6 + 18ADP + 18P_i + 12NADP^+ + 6H^+$$

To calculate the efficiency of the system, consider the following: ΔG^0 for CO_2 reduction is 114 K cal/mole. $NADP^+$ reduction to NADPH in photosytem I requires two electrons and four photons. Photosystem II replenishes the lost electrons of photosystem I. It is known from a study of the photosystems that, for one electron to be transferred from H_2O to $NADP^+$, two photons must be absorbed by each of the two photosystems. Thus a total of eight photons will have to be absorbed for producing the required number (2) of NADPH molecules, per CO_2 fixed. The energy of the photons may vary from 72 K cal/mole at 400 nm to 41 K cal/mole at 700 nm. If the energy of the photon at 600 nm (47.6 K cal/mole) is taken as the average, the total energy absorbed from the 8 photons is 8 x 47.6 = 381 K cal/mole. The efficiency of the photosynthetic reaction at 600 nm wavelength is 114/381 = 30% under standard conditions.

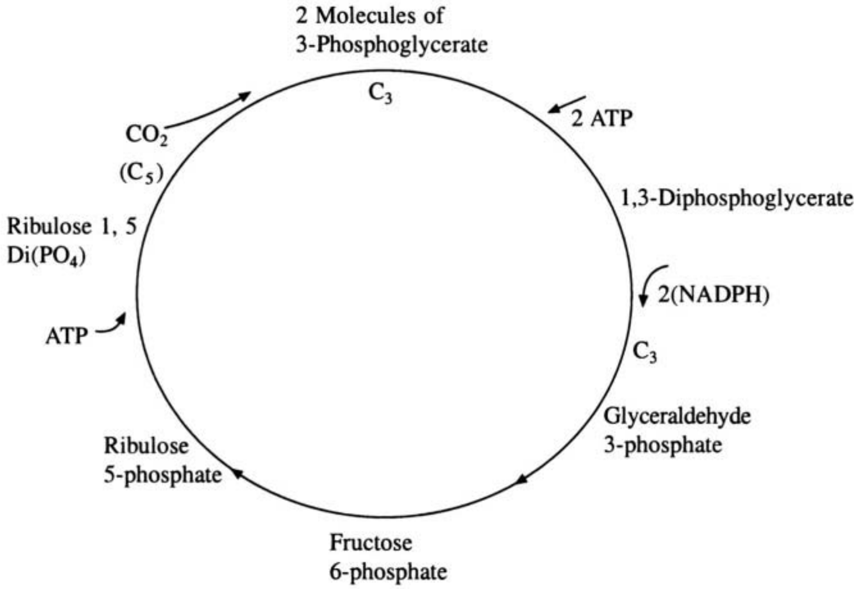


Fig. 11.13 Calvin cycle.

11.7 Energy Conversion Pathways

ATP and NADPH are continuously generated and degraded. These processes are regulated and the biosynthesis and the degradation have well-separated distinct pathways. When a high level of ATP is available then anaerobic reactions, which utilise ATP, are stimulated and ATP generating pathways are inhibited. The reverse occurs when the level of ATP is low.

11.7.1 Oxidation

Oxidation of food is always coupled with the formation of ATP and this process can take place even in the absence of oxygen. In some organisms anaerobic oxidation is the only available pathway for energy conversion whereas in others both anaerobic as well aerobic energy conversion can take place. (Figure 11.14)

11.7.2 Glycolysis

Glycolysis takes place in the cytosol (Figure 11.15). Glycolysis is a metabolic pathway in which glucose is converted into pyruvate with the concomitant production of ATP. This mechanism precedes a sequence of other oxidation reactions in higher organisms. Figure 11.16 gives in detail the role of NAD in the glycolytic pathways in anaerobic and aerobic metabolism respectively. When ethanol or lactate is produced from glucose in anaerobic organisms it is known as fermentation. The glycolytic pathway has a dual role in that it degrades glucose to form ATP and also provides building blocks for reactions like formation of long chain fatty acids. Glycolysis is controlled by essentially irreversible reactions, such as those catalysed by hexokinase, phosphofructokinase and pyruvate kinase. In these reactions, two molecules of ATP are generated during the conversion of glucose to pyruvate.

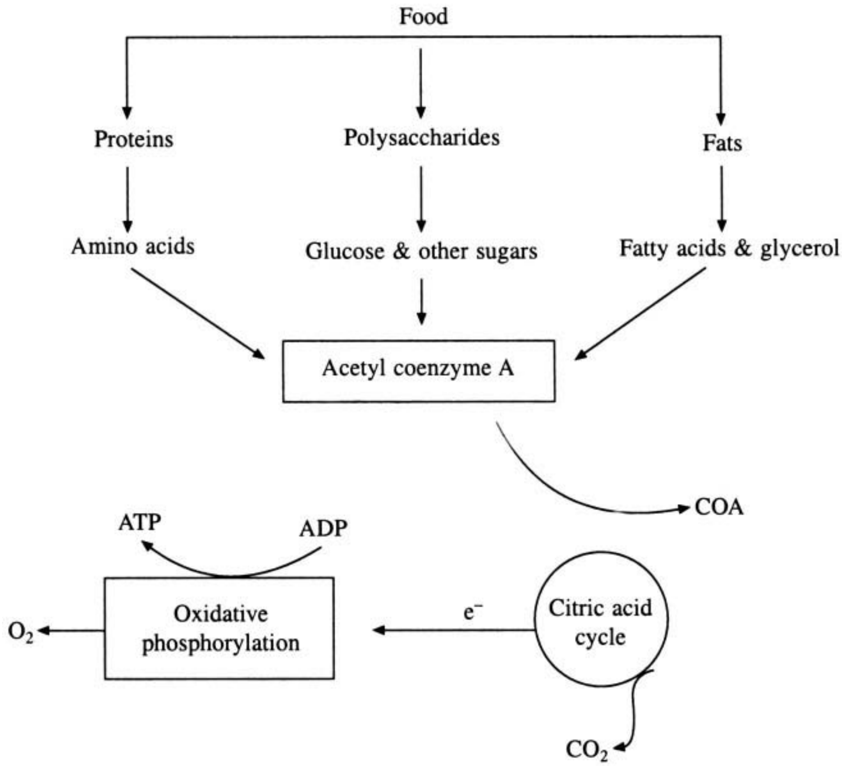


Fig. 11.14 Generation of energy from food

In aerobic organisms, glycolysis is followed by respiration, which takes place in the mitochondria (Figure 11.17). Mitochondria have two membranes. The outer membrane is permeable to small molecules. The inner membrane forms several inner folds thus increasing the total surface available. These folds are called cristae. They enclose an inner compartment known as the matrix. There are several enzymes in the membrane. The respiratory reactions take place in two steps, the first of which takes place in the matrix. This cycle is named after its discoverer Hans Krebs as Krebs cycle. The second step is part of the respiratory cycle and takes place in the cristae.

11.7.3 The Krebs cycle

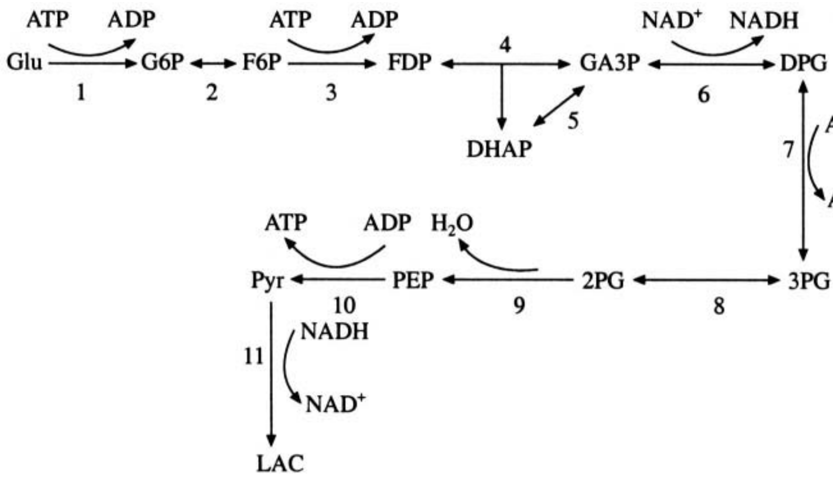
The Krebs cycle is also known as the citric acid cycle, or the tricarboxylic acid cycle (TCA). In addition to the oxidation-reduction reactions that lead to the generation of ATP, the citric acid cycle provides several intermediates, which are branch points for the synthesis of various compounds required by the cell, like amino acids. Most fuel molecules enter this cycle as acetyl coenzyme *A* (acetyl CoA). The reaction for the formation of acetyl CoA is as follows.

$$\textbf{Pyruvate + CoA + NAD}^+ \rightarrow \textbf{Acetyl CoA + CO}_2 \textbf{ + NADH}$$

The Krebs cycle can be written as

$$\textbf{Acetyl CoA + 3NAD}^+ \textbf{ + FAD + GDP + } P_i \textbf{ + 2HO} \rightarrow \textbf{2CO}_2 \textbf{ + 3NADH + FADH}_2 \textbf{ + GTP + 2H}^+ \textbf{ + CoA}$$

Two carbon atoms from acetyl CoA enter the cycle and condense with oxaloacetate (a four-carbon



Enzymes	Substrates	
1. Hexokinase	Glu	– Glucose
2. Phosphoglucose isomerase	G6P	– Glucose-6-phosphate
3. Phosphofructokinase	F6P	– Fructose-6-phosphate
4. Aldolase	FDP	– Fructose 1, 6-diphosphate
5. Triose phosphate isomerase	GA3P	– Glyceraldehyde-3-phosphate
6. Glyceraldehyde-3-phosphate dehydrogenase	DHAP	– Dihydroxy acetone phosphate
7. Phosphoglyceratekinase	DPG	– 1, 3 Diphosphoglycerate
8. Phosphoglyceromutase	3PG	– 3-Phosphoglycerate
9. Enolase	2PG	– 2-Phosphoglycerate
10. Pyruvate phosphokinase	PEP	– Phospho enol pyruvate
11. Lactate dehydrogenase	Pyr	– Pyruvate
	Lac	– Lactic acid

Fig. 11.15 Glycolysis

molecule) to form the six-carbon molecule citrate. Citrate is isomerised to isocitrate and oxidative decarboxylation of isocitrate gives **α-ketoglutarate** (a five-carbon molecule). One molecule of CO_2 is evolved during this reaction. A second molecule of CO_2 is given out when **α-ketogluterate** is oxidatively decarboxylated to succinyl CoA (again a four-carbon molecule). In the next stage of the reaction GTP is generated and succinate is formed. Succinate is then oxidised to fumarate (a four-carbon molecule) which is subsequently hydrated to malate. Finally oxaloacetate is regenerated from malate by oxidation. Figure 11.18 gives a schematic sketch of this cycle. In all there are four oxidation-reduction reactions and 3 pairs of electrons are transferred to NAD^+ and one pair to FAD. These reduced electron carriers are oxidised in the electron transport chain to form eleven molecules of ATP. One high-energy phosphate molecule, viz. GTP is produced during the cycle. Thus twelve high-energy phosphate molecules are produced in the oxidation of acetyl group into H_2O and CO_2. When NADH and FADH lose their electrons in the electron transport chain, NAD^+ and FAD are regenerated. As already mentioned intermediates for the biosynthesis of amino acids, fatty acids etc., are also formed during the Krebs cycle. The loss of intermediates of the cycle due to biosynthesis is constantly compensated, since many of the branching off reactions are reversible. In addition to this there are reactions known as anaplerotic (fill up) reactions. For example, if oxaloacetate is converted into amino acids then the supply of oxaloacetate for the Krebs cycle is replenished by the carboxylation of pyruvate.

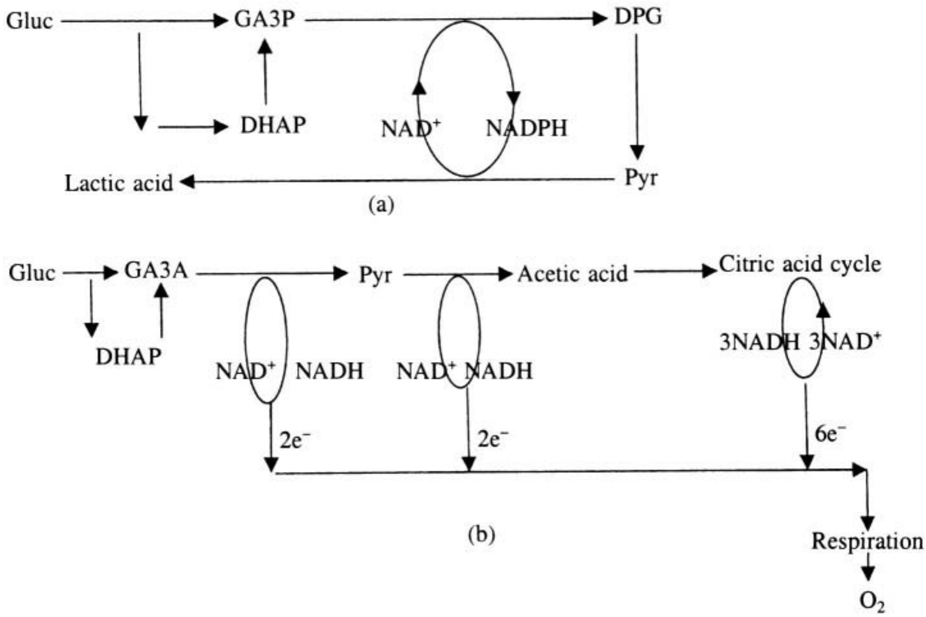


Fig. 11.16 Role of NAD in (a) anaerobic metabolism and (b) aerobic metabolism.

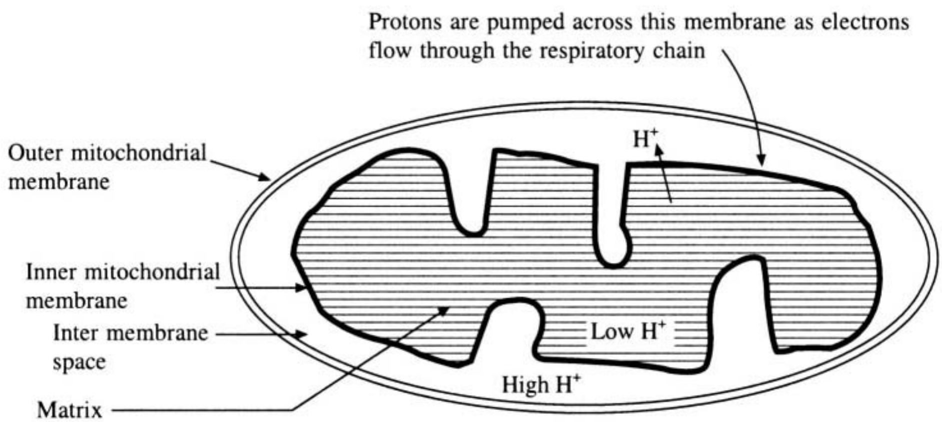


Fig. 11.17 Schematic diagram of mitochondria.

$$\text{Pyruvate} + CO_2 + \text{ATP} \rightarrow \text{Oxaloacetate} + \text{ADP} + P_i + 2H^+$$

The biosynthetic intermediates in the Kerb's cycle are shown in Figure 11.19.

11.7.4 The respiratory chain

The respiratory chain consists of a series of oxidation-reduction enzymes bound to the cristae membrane of mitochondria. The NADH and $FADH_2$ molecules formed in glycolysis, the citric acid cycle, etc. are energy rich molecules because each is capable of transferring two electrons to molecular oxygen, thus liberating a large amount of energy. This energy can be utilised to generate ATP. This is the major source of ATP molecules in aerobic organisms. The oxidation of NADH, $FADH_2$ and phosphorylation are coupled processes. Figure 11.20 shows the respiratory cycle. The proteins involved in the chain are

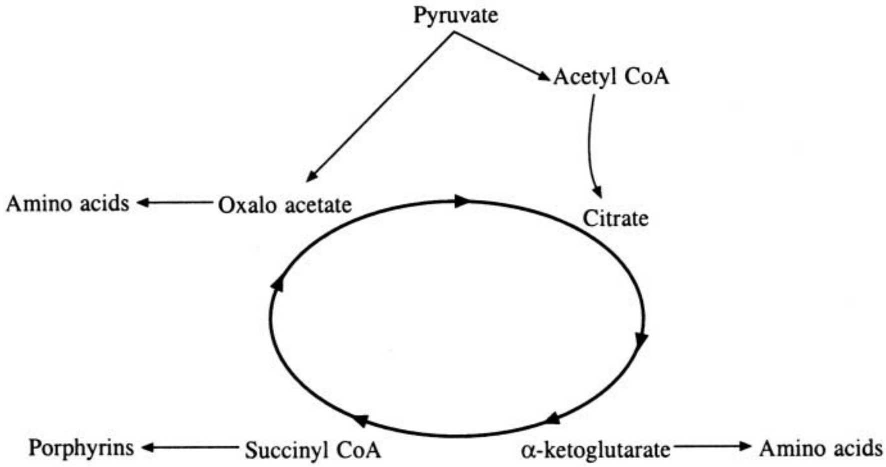
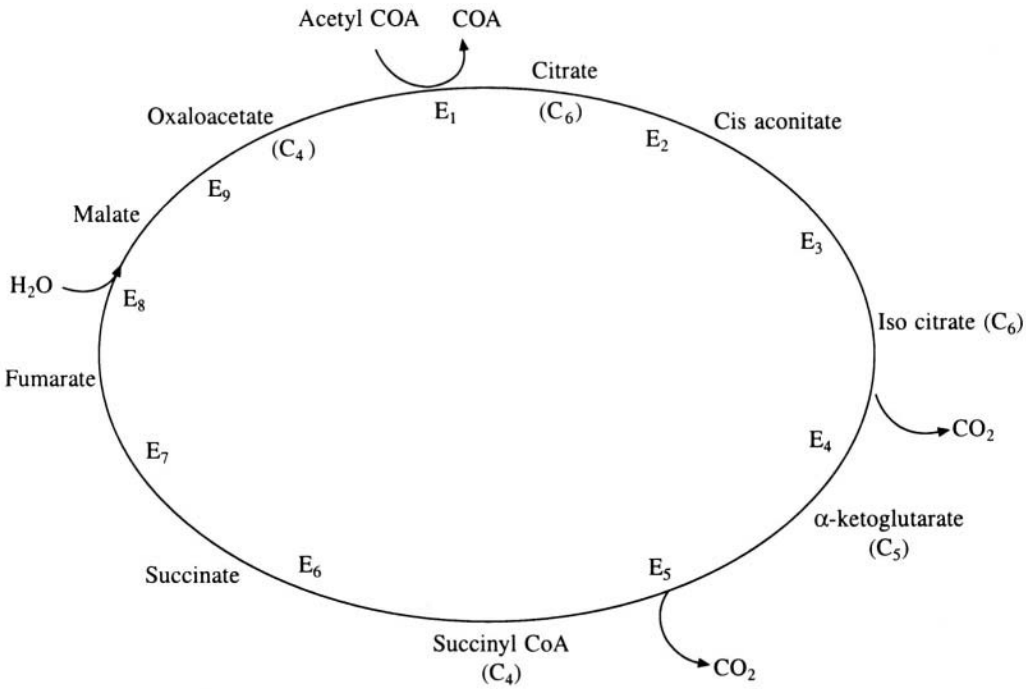


Fig. 11.18 Biosynthetic roles of Kreb's cycle

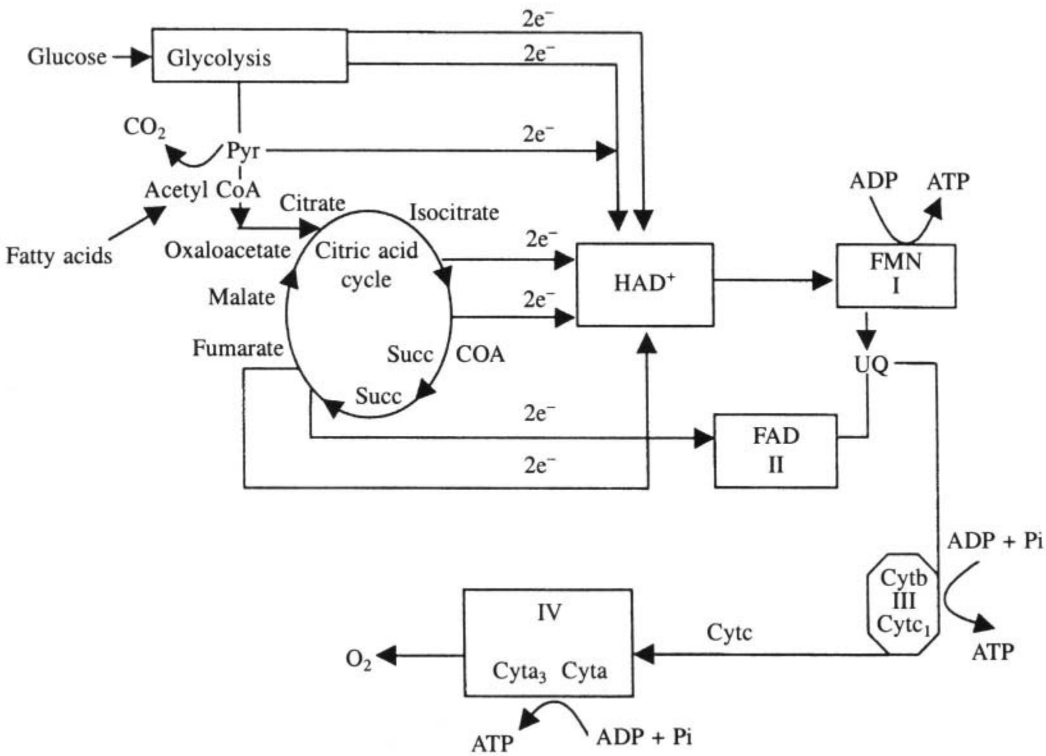


Enzymes (E)

1. Citrate synthase
2. Aconitase
3. Isocitrate dehydrogenase
4. Oxalo succinate decarboxylases
5. α-ketoglutarate dehydrogenase
6. Succinyl-COA synthase
7. Succinate dehydrogenase
8. Fumarase
9. Malate dehydrogenase

Fig. 11.19 Biosynthetic intermediates in the Kreb's cycle

FMN or NAD dehydrogenases, FAD or succinate dehydrogenases, cytochromes and iron-sulphur protein complexes. Electron transfer from NADH to O_2 takes place through a number of electron carriers like flavins, iron-sulphur complexes, quinones and hemes. It has been observed that when the mitochondrial (cristae) membrane is treated with specific detergents, the components separate into four structured complexes. Complex I contains NADH dehydrogenase and four Fe–S centres. Complex II consists of succinate dehydrogenase and its Fe–S centres. Complex III is made of cytochromes *b* and *c,* and a specific Fe–S centre. Cytochrome *a* and a_3 constitute complex IV. Ubiquinone is the connecting link between complex I, II and III while cytochrome *c* is the link between complex III and IV.



UQ	–	Ubiquonone	Succ CoA	–	Succinyl coenzyme
Cyt	–	Cytochromes	Succ	–	Succinate
FMN	–	Flavin mononucleotide	α-kglu	–	α-ketoglutarate
I, II, III, IV	–	Complexes	Pyr	–	Pyruvate
I, III, IV	–	Oxidative phosphorylation sites			

Fig. 11.20 Respiratory cycle

As discussed in Chapter *I*, chemical reactions in which electrons are transferred from one molecule to the other are termed as oxidation-reduction reactions. A redox (oxidation-reduction) reaction can be written as:

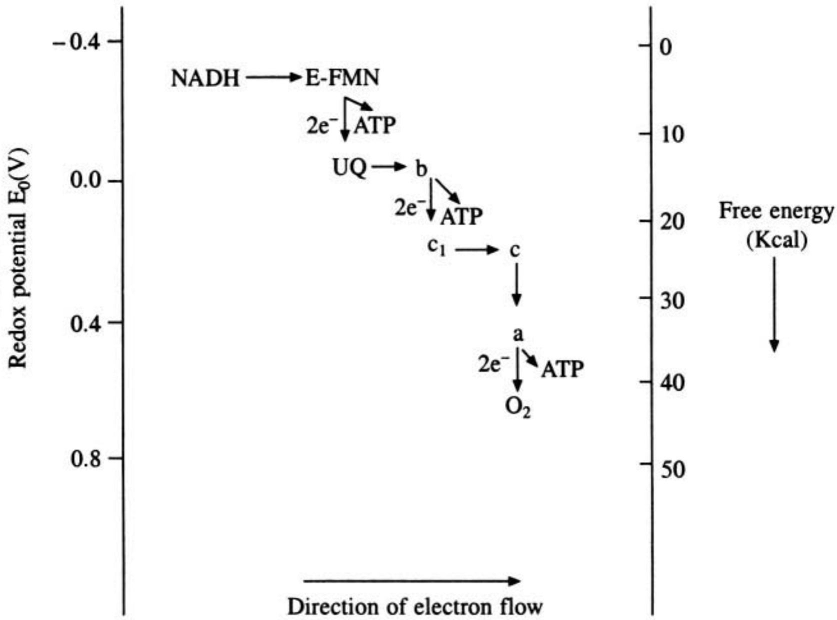
$$\text{Electron donor} \Leftrightarrow e^- + \text{Electron acceptor}$$

e.g.

$$Fe^{2+} \Leftrightarrow e^- + Fe^{3+}$$

Fe^{2+} and Fe^{3+} together form a conjugate redox pair. The standard redox potential (E_0') is defined as the electromotive force (emf) in volts measured by an electrode placed in a solution containing both electron donor and its conjugate acceptor in 1.0 M concentration at 25°C and pH = 7.0. Increasingly negative redox potential means an increasing tendency to loose electrons. Similarly increasingly positive potential indicates an increasing tendency to accept electrons. Thus electrons will tend to flow from a relatively electronegative conjugate pair like **NADH/NAD⁺** ($E_0' = -0.32$ V) to more electropositive redox pair like reduced cytochrome c/oxidised cytochrome ***c*** ($E_0' = 0.23$V) (Figure. 11.21). This electron flow leads to a loss of free energy, which is utilised for conversion of **ADP → ATP**. The complete equation for oxidative phosphorylation can be written as

$$NADH + H^+ + 2O_2 + 3Pi + 3ADP \rightarrow NAD^+ + 3ATP + 4H_2O$$



E-FMN → NADH Dehydrogenase
UQ → Ubiquonone
a, b, c_1, c → Cytochromes

Fig. 11.21 Direction of electron flow in the respiratory chain of mitochondria

ATP formation by oxidative phosphorylation is regulated by the **[ATP]/[ADP][P$_i$]** ratio. This ratio is known as the mass action ratio of the ATP system. The square brackets denote molar concentrations. Using absorption spectrophotometry one can follow oxidation-reduction reactions of the electron carriers. For example cytochromes, which absorb light in the visible region, undergo spectral changes when the enzyme changes its redox state, and these can be followed. Such coupled reactions can be inhibited by compounds like 2,4-dinitrophenol (DNP), certain classes of antibiotics etc. A protein called coupling factor is required for the coupled reaction to proceed. These coupling factors are located inside the matrix space and connected to the inner membrane by a narrow stalk (Figure 11.22). The coupling factor, which is referred to as F_1, catalyses the synthesis of ATP and is also known as ATP synthetase complex. Electron flow in the electron transport chain results in the pumping of protons across the inner mitochondrial membrane, i.e. from matrix to the cytoplasm (Figure 11.17). Thus the H^+ ion concentration is greater in the cytoplasm and a potential is generated across the membrane, with the cytoplasmic side positive. This proton motive force drives the synthesis of ATP by ATP synthetase complex. Oxidative phosphorylation generates 32 out of the 36 molecules of ATP produced when glucose is completely oxidised to CO_2 and H_2O. The electron flow from NADH $\rightarrow$ CO_2 is an exergonic process.

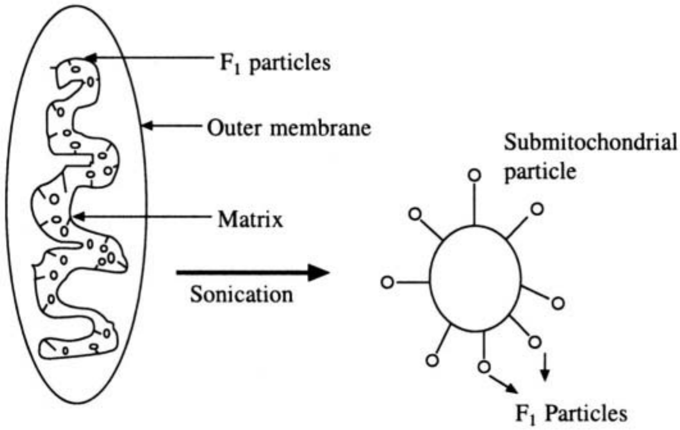


Fig. 11.22 Cross section of a mitochondrion

11.8 Membrane Transport

The inner mitochondrial membrane is impermeable to H^+, OH^-, K^+ and many other ionic solutes. The ATP formed inside the mitochondrial matrix will have to leave the matrix and reach the cytosol to become useful. Similarly any energy converting process requires a continuous supply of substrates and disposal of products and wastes. In several biological functions, products made at one part of the cell will have to be transported to another. Membrane transport, i.e. transport of molecules and ions across the membrane, is achieved in two general ways: (i) active transport and (ii) passive transport. For passive transport no energy input is required and the movement of the solutes is in the direction of the thermodynamic gradient. In active transport the movement is against the gradient and hence requires a source of energy. Membranes are made up of lipid bilayers, which divide the cell into compartments. They have a finite thickness ranging from 60 to 100 Å. They have selective permeability for certain ions and molecules. As the molecules will have to traverse through a hydrophobic lipid bilayer, there is good correlation between the lipid solubility of molecules and the penetration rate across membranes. Thus one would expect the rate of penetration of substances like ions, amino acids, etc. to be poor and this is indeed what is observed. However, this fact alone does not explain membrane transport completely, especially the high degree of selectivity observed.

A carrier is one of several mechanisms proposed for membrane transport (Figure 11.23). There are carrier molecules A in the membrane which have a high affinity for the species B to be transported. When A and B come together they form a complex $A + B$. Though B by itself has poor permeability, the complex $A + B$ can traverse the membrane. For simplicity, we assume that the complex is electrically neutral. Valinomycin is an example of a carrier molecule. It is a cyclic peptide antibiotic with a hydrophobic exterior and a hydrophilic interior and is highly specific to K^+ ions. Other antibiotics from micro-organisms, like gramicidin and alamethicin are known as channel formers. As their name suggests, these molecules form channels across the membrane. The specie to be transported binds at one end and travels through the channel to the other end. Channel former molecules do not move across the membrane to transport the ions. The interiors of the channel formers are hydrophilic and the exteriors are hydrophobic, just as in carrier molecules. Whether a particular mode of transport is active or passive is determined by the change in the free energy (ΔG) of transported species. If ΔG is positive it is active transport, and the mechanism requires a carrier molecule. If ΔG is negative it is passive transport and can occur spontaneously.

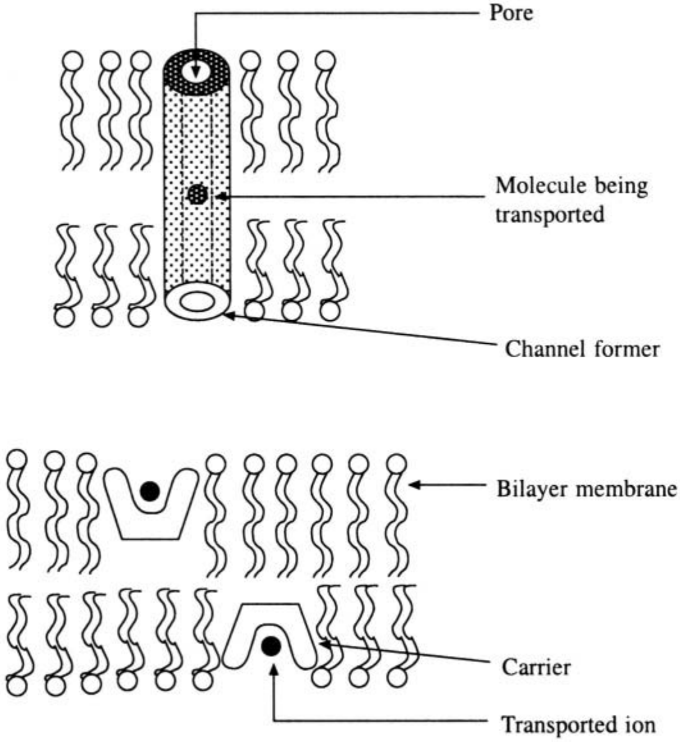


Fig. 11.23 Mechanisms of membrane transport

11.8.1 Active transport

In red blood cells, the cytoplasmic membrane is permeable to both Na^+ and K^+ ions but the K^+ ion concentration inside the membrane is greater than outside. For Na^+ the reverse is true. Thus an ionic gradient is maintained by a specific transport system known as the Na^+-K^+ pump. This ionic gradient is crucial for signal transmission through nerves. This difference in K^+ and Na^+ ion concentration can be maintained only by active transport and in fact the Na^+-K^+ pump maintains the gradient at the expense of hydrolysis of ATP. Thus active transport is coupled with the process that supplies energy. Under suitable conditions, this pump can be reversed to synthesise ATP from ADP and P_i. A model has been proposed for the Na^+-K^+ pump. The model takes into account the discovery, by Jens Skou, of an enzyme which hydrolyses ATP and which is present among the molecules that go to make up the pump mechanism. According to this model, hydrolysing enzyme in Na^+-K^+ pump must be in a position to adopt two conformations, and the affinity for transported species must be different for the two conformations. These are such that the cavity, or the site at which the species to be transported binds, is on the outside for one conformation and on inside for the other (Figure 11.24). From the figure we see that if E_1 has a higher affinity for Na^+ while E_2 has a higher affinity for K^+, the pump would function in the required manner. As stated earlier, the action of ATPase is reversible and hence transport of ions across the membrane in the direction of thermodynamic gradient must cause the synthesis of ATP from ADP and P_i. In fact the proton gradient across the inner membrane of the mitochondria is linked to the production of ATP. The proof of this comes from the fact that no ATP is synthesised when the membrane is not intact.

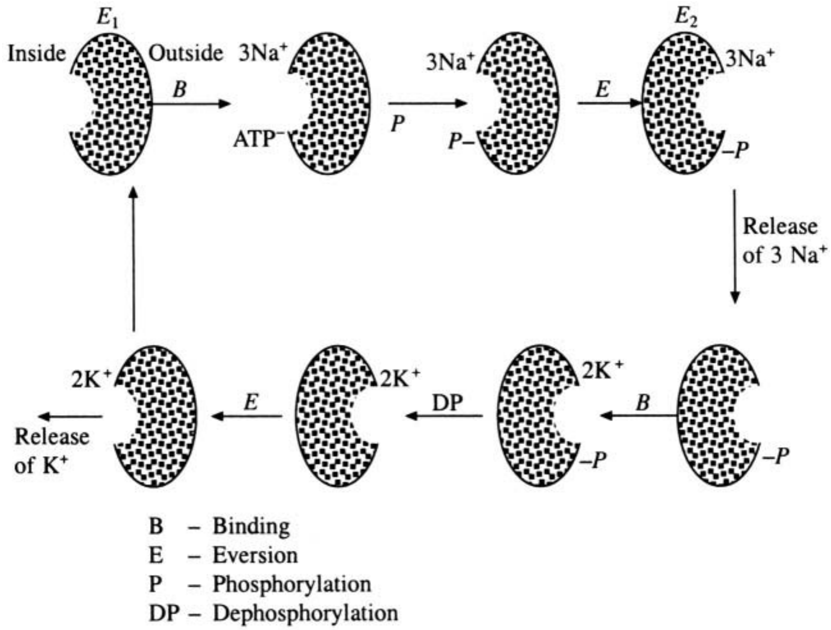


Fig. 11.24 Model of a Na$^+$-K$^+$ pump

11.8.2 Chemi-osmotic theory—Passive transport

As seen in a previous section, during electron transport in the inner membrane of the mitochondria, H^+ ions are ejected, generating a proton gradient across the membrane. According to the model proposed for this process (Figure 11.25), there are three H^+ conducting loops in the respiratory chain. Each loop translocates two H^+ ions from the inner matrix of the mitochondrion to the exterior through a carrier. The two electrons, which remain after the transport of H^+ ions are carried back to the matrix by a carrier. Thus for every pair of electrons passing from H_2 to oxygen, the loops carry $6H^+$ ions from the matrix to the external medium. Uncouplers like dinitrophenol (DNP) act by carrying protons across the membrane thus dissipating the proton gradient. This is necessary for ATP synthesis. In the case of chloroplasts, light causes water to loose electrons, which flow across the membrane and release protons in the inner medium. The reduced carrier on the other side of the membrane picks up protons from the outer medium and releases electrons to the second light reaction. The alternation of hydrogen and electron carriers on both sides of the membrane causes a net flow of protons across it. (Figure 11.25(b)). The thylakoid membrane and the mitochondrial membrane are permeable to water but not to hydrogen ions. Hence a concentration gradient of protons is formed and a proton motive force is built up. This proton gradient causes an electrochemical potential difference across the membrane which is given by

$$\Delta\mu = \mu_2 - \mu_1 = RT \log ([H^+]_2/[H^+]_1) + F(\psi_2 - \psi_1)$$

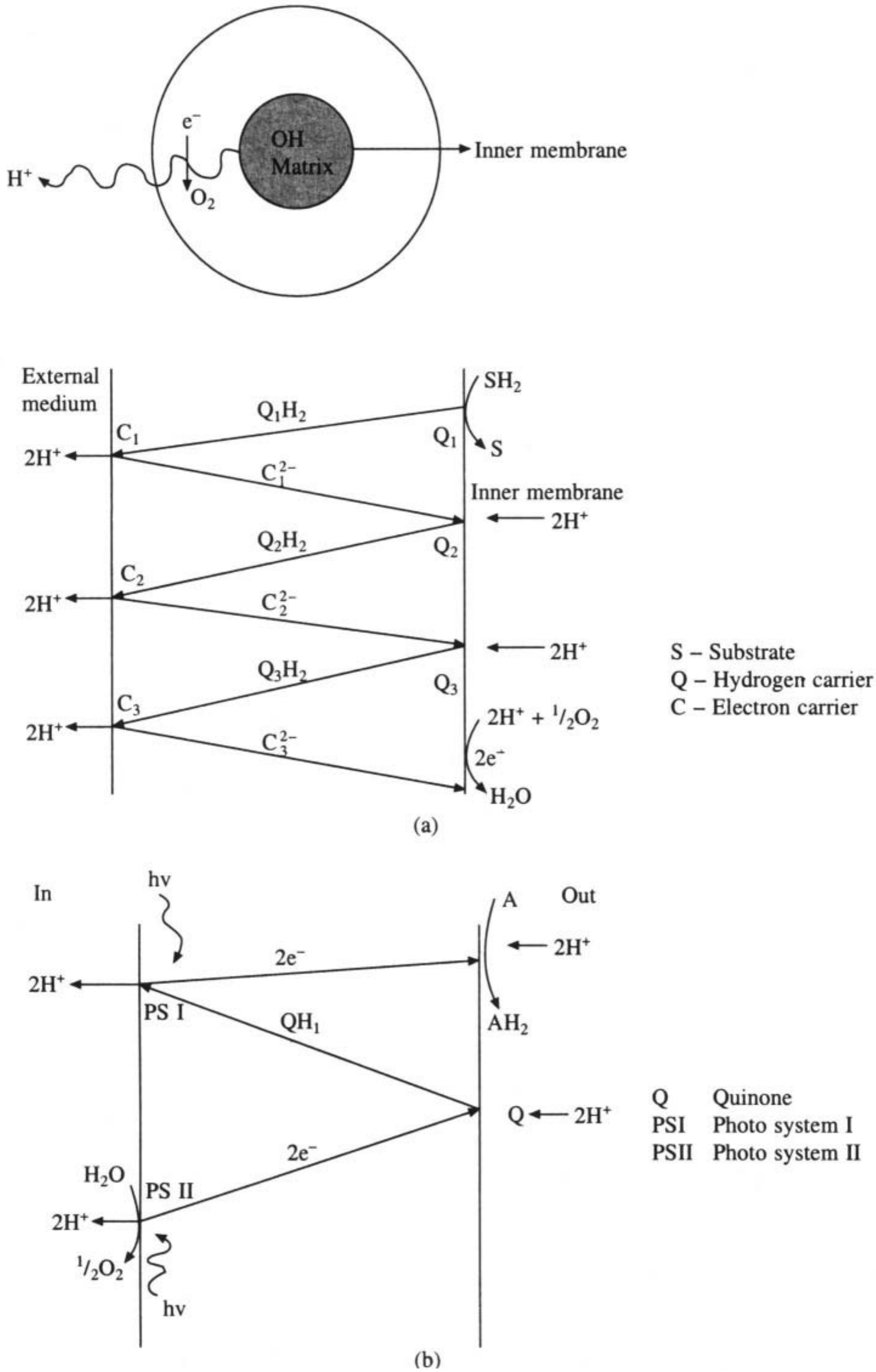


Fig. 11.25 Chemi-osmotic model for membrane transport in a: (a) mitochondrion and (b) chloroplast.

where F is the Faraday constant and $\Delta\psi = (\psi_2 - \psi_1)$ is the contribution from membrane electric potential. Since $pH = -\log [H^+]$, the above equation can be written as

$$\Delta\mu = 2.303RT(\Delta pH) + F\Delta\psi$$

where $\Delta pH = pH_1 - pH_2$ when hydrogen ions move from position 1 to position 2, and the factor 2.303 is due to conversion from natural logarithms to log base 10. Both ΔpH and $\Delta\psi$ are negative and that is the direction in which ions are pumped in. When the electrochemical potential is divided by the Faraday constant F we get the proton motive force or pmf. (analogous to the electron motive force, emf). In chloroplasts almost all the contribution to pmf is from the pH gradient. In mitochondria the contribution from the membrane potential is higher. This is due to the fact that the transfer of ions into the thylakoid space is accompanied by the transfer of either Cl^- in the same direction or Mg^{2+} in the opposite direction leading to neutrality. Hence no membrane potential is developed in chloroplasts.

12

Biomechanics

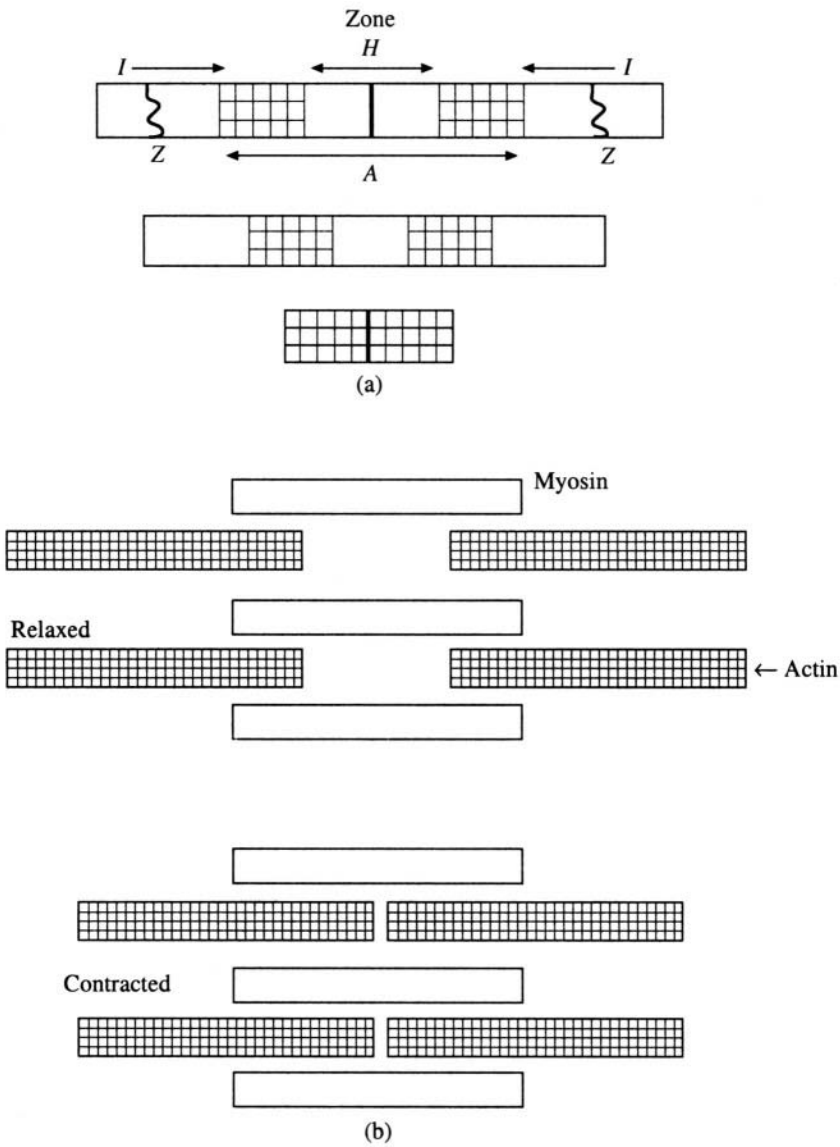
12.1 Introduction

In this chapter we will deal with the conversion of chemical energy into mechanical energy. Mechanical work is done by all living organisms. It gives the organism the ability to move away from a noxious environment or move towards a beneficial region. There are movements inside the cell such as pinocytosis, cell division, etc. However, the mechanism involved in these situations is not well understood. This chapter focuses on larger systems such as the entire organism or the limbs, etc. The best known example in this system of conversion of chemical energy into mechanical work is that of muscle action. The proteins involved in these processes are actin and myosin and the chemical substance providing energy is adenosine triphosphate (ATP). Whenever ATP is converted to ADP (adenosine diphosphate) or AMP (adenosine monophosphate) approximately 8 Kcal/mole of energy is released, and is used for mechanical work by the muscle proteins.

Muscles can be classified as smooth muscle, cardiac muscle and striated or skeletal muscle. The walls of some internal organs, like the bladder, the esophagus, intestines, etc. are made of smooth muscles. The contraction of smooth muscles is not under voluntary control but takes place in a slow, generalised manner. The cardiac muscles are fibrous and also slightly striated. They undergo periodic contraction and relaxation automatically. Striated muscles are under voluntary control. They are highly organised and consist of several elongated cells. These elongated cells or muscle fibres in turn contain the myofibrils. In a light microscope the muscle cells show a characteristic repeating structure which gives it a striated appearance.

12.2 Striated Muscles

Striated muscle cells are multi-nucleated and are enclosed in an electrically excitable membrane called the sarcolemma. The intracellular fluid surrounding the myofibrils is sarcoplasm. It contains glycogen, ATP, phosphocreatine and glycocyte enzymes. When a longitudinal section of myofibrils is examined under an electron microscope one can see the repeating unit, sarcomere, every **2.3 μm** along the fibre axis. A sarcomere has a banded appearance due to regions with different optical properties. A dark *A* band and a light *I* band alternate regularly (Figure 12.1). The middle of the A band is known as the *H* zone and a dark line is found in the middle of the *H* zone. The Z lines bisect *I* bands. There are two kinds of filaments in the sarcomere. The thick filaments make up the *H* zone of the *A* band and only thin filaments are found in the *I* band. These filaments are protein filaments of diameters 150 Å and 70Å respectively. The thick filament consists mainly of the protein myosin while the thin filament consists of actin, tropomyosin and troponin. Each thick filament is surrounded by six thin filaments in the region of overlap, and each thin filament has three neighbouring thick filaments. The filaments interact with each other through what are known as cross bridges. Electron



microscopy and optical measurements show that the *I* zone shrinks when the muscle contracts. Based on this, a 'sliding filament' model of muscle action has been proposed. According to this model, the lengths of the filaments do not change but the length of the sarcomere decreases as the filaments slide past each other during contraction. Thus, though the lengths of the thick and thin filaments do not change, the sizes of the *H* zone and the *I* band decrease. The cross bridges change angle during the sliding motion of the filament. When at rest the thin and thick filaments overlap only to an extent of about one third of their length. At maximal contraction, the thick and thin filaments fully overlap and shortening is about 32 Å (Figure 12.1(b)).

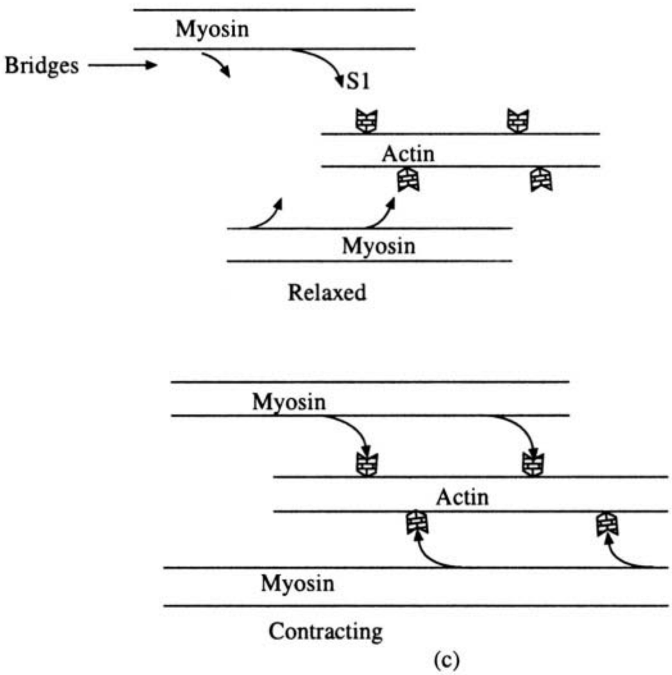


Fig. 12.1 A schematic representation of the sliding model of muscle action. (a) As seen under a microscope. (b) Thick and thin filaments in a contracting muscle. (c) Formation of coss bridges

12.2.1 Contractile proteins

The major components of myofibrils are the protein myosin (50 to 55% of the total protein) and actin (20 to 25%). Actin is the major constituent of thin filaments. When extracted from muscle, a globular protein known as *G*-actin emerges. In the presence of Mg^{2+} and ATP *G*-actin polymerises to form the fibrous *F*-actin. *F*-actin looks like two strings of beads wound round each other when looked under electron microscope. The other two proteins that are closely associated with actin in thin filaments are tropomyosin and troponin. Tropomyosin is a double stranded α-helical rod arranged parallel to the long axis of the thin filament. Each tropomyosin molecule covers about seven *G*-actin monomers. Troponin is a complex protein with three subunits viz. TnC, TnI and TnT having the molecular weight of 18, 24 and 37 kDa respectively. Troponin is attached to the tropomyosin filament, one on each molecule. TnC binds calcium ions, TnI binds to actin and TnT binds to tropomyosin. The troponin-tropomyosin complex plays an important role in the regulation of contraction, which is effected by Ca^{2+} ions (Figure 12.2).

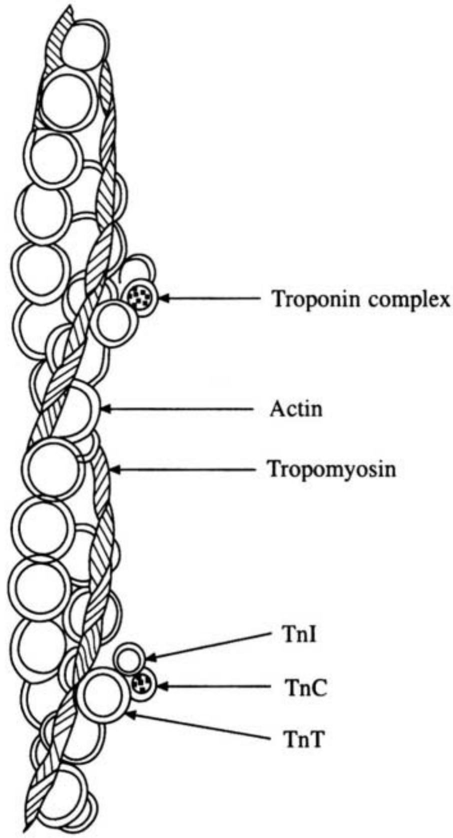


Fig. 12.2 Model for a thin filament in resting stage

The thick filament consists mainly of myosin molecule whose molecular weight is about 500 KDa. Electron microscopy shows that myosin has a double headed globular protein attached to a long rod. The rod is a two-stranded **α-helical** coiled coil. When myosin molecule is cleaved using the enzyme trypsin, it splits into two fragments, known as light meromyosin (LMM) and heavy meromyosin. The heavy meromyosin molecule, which contains the globular heads, has ATPase activity while the light meromyosin lacks this activity. The heavy meromyosin (HMM) does not form filaments (Figure 12.3). The globular heads also have a strong affinity for actin suggesting that this part of the myosin molecule must be involved in forming the cross bridges with actin in the myofibril. The formation of this complex leads to a large increase in viscosity, but this is reversed on addition of ATP. Figure 12.3(a) is a schematic diagram of the myosin molecule showing the six-polypeptide chains that form this macromolecular assembly. Myosin has a molecular weight of about 500 KDa and is made up of six polypeptide chains, viz. two heavy chains (= 400 KDa) and two pairs of light chains (each with a molecular weight of about 20 KDa). Each one of the myosin heads is globular in nature and has about 850 amino acids contributed by the respective heavy chain and two light chains. The remaining portions of the two heavy chains assemble in an extended coiled coil form, which is about 1500 Å long.

12.3 Mechanical Properties of Muscles

Muscles are transducers capable of converting chemical energy into mechanical energy. In a typical muscle function a load is moved through a distance due to shortening of the muscle fibres. The contractile unit of a muscle fibre is known as a tension generator. A muscle is elastic and develops tension when it is stretched. When a muscle is greatly stretched, the tension increases to a point beyond which muscle fibres can break. The elastic system of a skeletal muscle basically consists of three components, a contractile component and two elastic components. Sarcoplasm and sliding filament constitute the contractile component and tendons, disc, connective tissues and sarcolemma form the elastic components. Experimental observations of muscle contraction can be done using thermal, mechanical or electrical stimulus. However, electrical stimulus is preferred as the muscle is not damaged in the process and the intensity of the stimulus can be accurately measured using a galvanometer. A muscle twitch, which is a simple response to the threshold stimulus, is useful in the study of properties of muscles. A muscle switch goes through three stages: (1) latent period, (2) contraction phase and (3) relaxation phase. A muscle fibre can continue to be in tension if a series

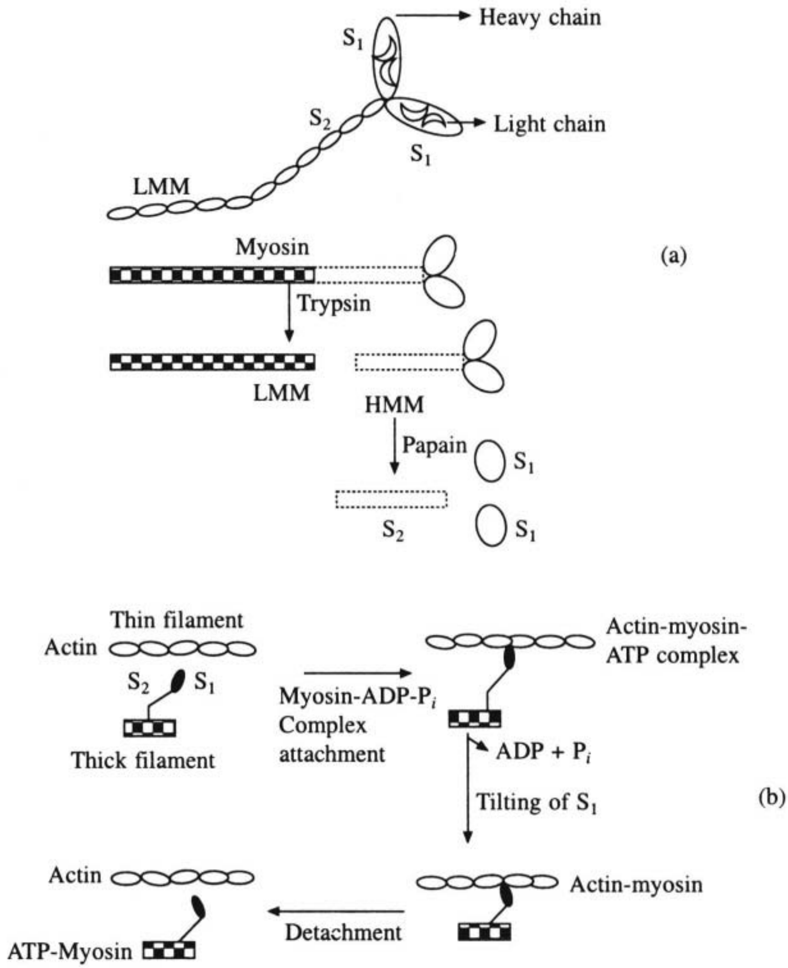


Fig. 12.3 (a) Schematic diagram of myosin molecule. (b) Proposed mechanism of muscle contraction

of stimuli are applied at high frequency without allowing the muscle to relax. This is known as tetanus. A fibre capable of maintaining tetanus for a considerable time can stop contraction due to. fatigue if the tetanus is prolonged too long.

12.3.1 Contraction mechanism

The decrease in viscosity when ATP is added to the troponin-tropomyosin (or actomyosin) complex undergoes slow recovery when ATP is hydrolysed. This is because ATP is required for breaking the cross bridges between actin and myosin rather than for making it. Thus the steps involved in muscle contraction are cyclic and can be designated as the following.

1. Binding of ATP to the myosin head (also known as $\mathbf{S_1}$ head).
2. Activation of myosin ATP complex.
3. Binding of the activated myosin-ATP complex to actin in thin filaments in the presence of $\mathbf{Ca^{2+}}$ ions.

4. ATP hydrolysis followed by the change of orientation of S_1 heads with respect to the filament.
5. Binding of ATP to myosin heads leading to the dissociation of myosin-ATP complex from actin filament.
6. The cycle repeats from step 2.

Thus the above model proposes two forms for the actomyosin complex: (1) The activated form. (2) The inactivated form. The binding of myosin-ATP complex to thin filaments in the presence of Ca^{2+} ions is the activated form. The part that is left after hydrolysis of ATP is the inactivated form. Figure 12.3(b) shows the details of the proposed mechanism of muscle contraction. The contractile force is generated by the making and breaking of the complexes between *S* heads of myosin and actin. In the resting stage, S_1 heads are detached from thin filaments, and myosin contains tightly bound ADP and P_i. When muscle is stimulated, S_1 heads move away from thick filaments, and get attached to the thin filaments in a perpendicular orientation. The tightly bound ADP and P_i are released and the orientation of S_1 head changes to make an angle of 45° with the thin filament axis. This tilt of the S_1 head is the power stroke of muscle contraction. The tilt is then transmitted to the thick filament through the S_2 unit (rod region) of the myosin molecule. This results in pulling the thick filament by about 75 Å with respect to the thin filament. In the final step, the S_1 head is detached from the thin filament by the binding of ATP and reverts back to its perpendicular orientation. Finally, hydrolysis of ATP by S_1 completes the cycle. Two types of hinges are invoked in the contractile mechanism—one between the S_1 and S_2 units, and the other between S_2 and light meromyosin.

12.3.2 Role of Ca^{2+} ions

It has been shown that the onset of contraction is mediated by Ca^{2+} ions. The contraction of muscle is initiated by the arrival of a nerve impulse at the end of the muscle fibre and the consequent release of acetylcholine. The muscle membrane (the sarcolemma) has a resting potential of 70mV. The response of the membrane is a change in membrane permeability, with Na^+ ions flowing into the muscle and K^+ ions out of it, and the consequent change in membrane potential. Within a few milliseconds the cell begins to contract as the local response propagates over the membrane. The nerve excitation triggers the release of Ca^{2+} ions by the sarcoplasmic reticulum. The released Ca^{2+} ions bind to TnC component of troponin and causes conformational changes, which are transmitted to tropomyosin and actin. Consequently tropomyosin moves towards the helical groove of the thin filament which enables S_1 heads of myosin molecules to interact with actin unit of the thin filament. Contraction takes place with concomitant hydrolysis of ATP till Ca^{2+} ions are removed and this causes the S_1 head to be blocked by tropomyosin again. Thus the binding of Ca^{2+} to troponin causes the activated myosin-ATP binding site on actin to be exposed. This is otherwise blocked by troponin and tropomyosin (Figure 12.2). The flow of information can be summarised as

$$\mathbf{Ca^{2+} \rightarrow Troponin \rightarrow Tropomyosin \rightarrow Actin \rightarrow Myosin}$$

Though it is known that ATP is the source of energy for muscle contraction, the supply of ATP is through a phosphocreatinine which has higher phosphate group transfer potential than ATP.

$$\mathbf{Phosphocreatinine + ADP \Leftrightarrow ATP + Creatine}$$

However phosphocreatine itself cannot make muscles contract.

The experiments on muscle contraction are performed by applying the potential directly to the tissue. Single fibres respond only when the potential exceeds a threshold and hence are not preferred. Two types of experiments on muscles can be carried out. In one case, both the ends of the muscle are fixed so that contraction occurs without shortening. This condition is known as isometric. In the second case only one end of the fibre is fixed and hence the muscle contracts and shortens and this condition is known as isotonic. A.V. Hill carried out fundamental studies on muscle behaviour and he showed that the maximum tension exerted by a muscle is about **5 Kg-m/cm²** of the muscle cross section. Figure 12.4 shows how the mechanical advantage and efficiency of muscle contraction may be calculated.

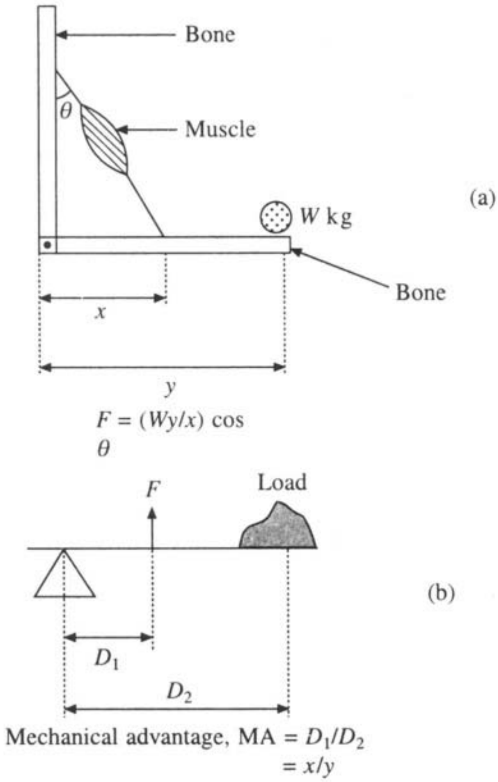


Fig. 12.4 (a) Force exerted by a muscle. (b) Mechanical advantage of the arm

12.4 Biomechanics of the Cardiovascular System

The blood circulatory system is a transport system that transports gases like O_2 and CO_2, nutrients, wastes, hormones and immunologically active substances. Blood flow can be considered as a more or less closed circulatory system, which is pumped by the heart. (It is not a completely closed system since some fluid, called lymph, leaves the capillaries and flows into the tissue spaces). Blood is a suspension of different types of single cells in a viscous medium containing proteins, organic salts etc. Kidneys, heart and lungs are the major organs that interact with blood. The vessels into which blood is pumped by heart are known as arteries. They branch into smaller and smaller arteries and finally form capillaries. Most of the exchanges between blood and the surrounding tissues take place at the capillary level. The capillaries then join together to form venules, leading to larger and larger veins which bring back the blood to the heart. The kidney is the most crucial organ. It removes waste products and excess water from the blood, thus maintaining the blood volume, ionic concentrations, arterial pressure etc. Thus the circulatory system is in a dynamic state, losing water to kidneys, lungs, exocrine organs (e.g. salivary glands), etc., and re-absorbing it from the gut and also that formed during metabolic processes.

12.4.1 Blood pressure

The fundamental properties of any fluid are pressure, velocity and density. Pressure is force per unit area, and is generally measured in terms of the height of a column of liquid that can be supported. The most frequent system of units used to measure blood pressure is millimetres of mercury, being the height of the column of the liquid metal supported by blood pressure. The maximum arterial pressure is called the systolic pressure and the minimum arterial pressure is called the diastolic pressure. The pressure falls when the blood reaches the capillaries. At the venous system it is still lower. The arteries and veins have very similar inner diameter (0.5 to 12.5 mm) and flow rates, but the pressures are different. This is due to the fact that arteries have thick, elastic walls while veins have thin walls. The larger pressures in arteries prevent any reverse flow of blood from veins. The pulsations in the arteries are due to the stretching of its elastic walls from the force of the heartbeat.

The mammalian heart is shown in Figure 12.5. It consists of four chambers, two auricles and two ventricles. Blood from the entire body, excepting from the lungs, enter the heart at the right auricle. From there it is pumped into the right ventricle, then into the lungs and finally back into the left auricle. From the left auricle the blood goes to the left ventricle and from there through the aorta (which is the largest blood vessel leading away from the heart) to the arteries and is then distributed to the entire body, excepting the lungs. This system of circulating blood is very efficient in supplying oxygen and removing carbon dioxide. Four valves viz. the tricuspid valve and mitral valve separating the auricles and ventricles, and the pulmonary valve and semilunar valve prevent the back flow of the blood from the ventricles to the arteries or from the main arteries into the respective ventricles (Figure 12.6).

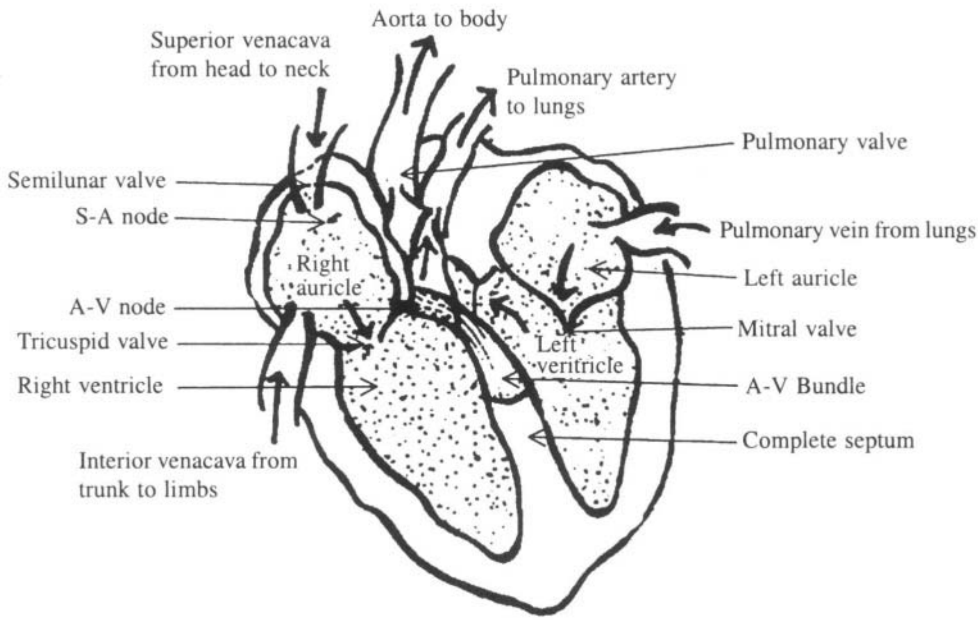


Fig. 12.5 Mammalian heart. Thick arrows show direction of blood flow.

The mechanical action of the heart is due to the heart muscle (myocardium) and is analogous to those of the skeletal muscle. As in other striated muscles the cardiac muscle fibre membranes are also polarised, i.e. they have 90 mV negative potential between the inside of the membrane and the outside. Just as in nerve fibres, the polarity reverses for a short period of time, when an impulse is received by the membrane, just before contraction occurs. The action potential is around 120 mV, i.e. the outside will be 30 mV more negative than the inside at the peak of the spike. The large collection of cardiac muscle fibres acts like a group of electric cells and leads to measurable potential changes on the body surface. These potential differences can be measured using an electrocardiogram. The heartbeat is initiated by a sino-auricular (*s-a*) node and the node acts like an electronic multivibrator letting out one electrical pulse per heart cycle. The heart pulses are periodic and rhythmic. During systole, which is the active part of the heart cycle and when the arterial pressure is at a maximum, the blood from ventricles is forced into aorta and pulmonary artery respectively. In the beginning of this cycle all the four valves, are closed. As the pressure increases, and the ventricular pressure just exceeds the pressure at the roots of the aorta or the pulmonary artery, the semilunar valve opens. The ventricular pressure reaches a maximum and gradually decreases and this causes the semilunar valve to close. The second part of the heart cycle known as diastole then begins with the relaxation of the ventricular wall and a drop in the ventricular pressure with respect to the corresponding attrium. Hence the atrio-ventricular valves (*a-v*) open and the blood enters the ventricles. At the end of the diastolic cycle the atria contract and the pressure inside the ventricles increases. Immediately after this the ventricular valves close and the systol and diastole cycle is repeated.

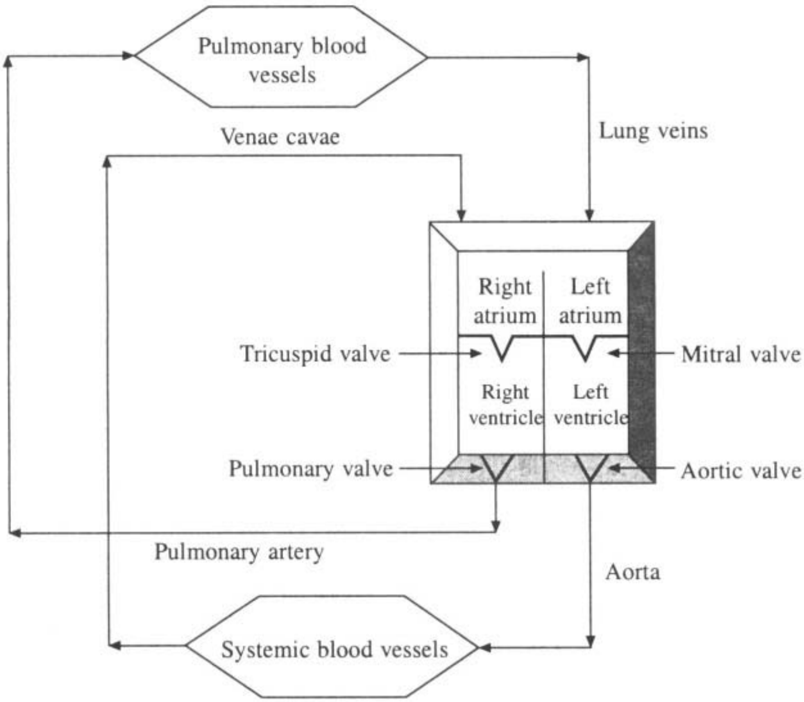


Fig. 12.6 Circulation of blood

12.4.2 Electrical activity during the heartbeat

The heart pulses generated by the *s-a* node spread over the surface of the auricle and make the muscle fibres to contract. In addition, the electrochemical pulses stimulate the *a-v* node. After a time delay of 0.1 seconds or less, this node lets out a new electrical pulse that is carried by special fibres known as *a-v* bundle to the septum. These fibres terminate at different points in the septum and on the outer ventricular wall. Thus the fast conduction of the impulses by *a-v* bundle synchronises with the ventricular contraction. The (*a-v*) node can take over the control of heart rate when the (*s-v*) node has failed. Under these circumstances auricular contraction will not be properly synchronised with ventricular action. If the *a-v* node also fails, the auricular and ventricular walls take over the control of heart rate. These conditions are not fatal.

The difference between cardiac muscle fibres and nerve fibres lies in the recovery process. The recovery of resting potential in nerve axons takes about a fraction of millisecond to 5 milliseconds. On the contrary, the cardiac muscle fibres take as long as 200 milliseconds to recover their resting potential. Similarly, there is a subtle difference in the permeability of $\mathbf{K^+}$ ions during and after the action potential in the case of cardiac muscles as compared to nerve axons. This was seen by experiments using squid axons. The permeability of K^+ ions increases suddenly and then drops when the resting potential of a voltage-clamped squid axon is suddenly decreased. In the case of the cardiac muscles, the original low or zero permeability to potassium ions is recovered only after the membrane potential returns to its original value.

12.4.3 Electrocardiography

As mentioned above, electrical potential changes occur during heartbeats and this is spread over the surface of the body. Electrodes at any two points of the surface of the body will be able to measure this potential difference and it can be recorded using an electrocardiogram (ECG). As the heart is not the only organelle capable of generating such potentials at the body surface, adequate precautions will have to be taken while recording an ECG. Even a movement of a skeletal muscle can give rise to body surface potential difference and hence the patient is made to lie down during recording of the ECG. The electrodes are placed generally at the right arm and left leg. Sometimes three wires are attached, one to each arm and the third to left leg. The ECG is then recorded between each pair of electrodes. The maximum potential difference is about 1 mV and this is adjusted to show 1 cm deflection on the ECG. The electrical signals corresponding to one heart cycle, also known as the waves of the signal, are shown in Figure 12.7. The *P* wave occurs just before the auricular contraction and the QRS complex occurs at the start of the ventricular contraction. The end of ventricular contraction gives rise to *T* waves. The ECG cycle is described in Table 12.1

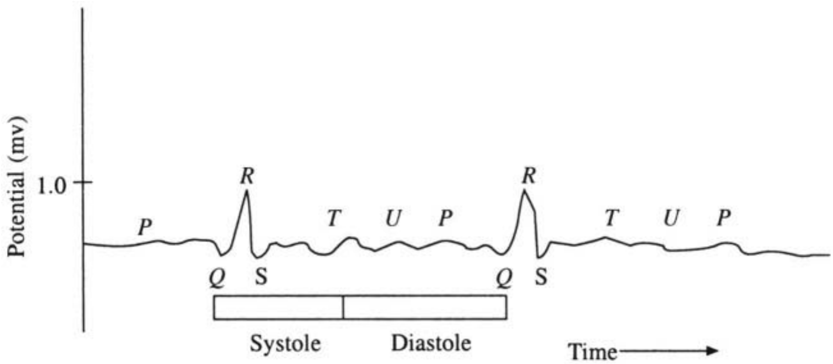


Fig. 12.7 Electrocardiogram. P wave precedes auricular contraction, QRS is due to ventricular contraction

Table 12.1. The ECG cycle of signals and the corresponding cardiac functions

ECG waves	Duration (milliseconds)	Corresponding heart cycle
P	8	Precedes auricular contraction
P–Q	150–200	*a-v* delay time
Q	40–80	Precedes ventricular contraction
R		
S		
S–T	100–250	Ventricular ejection
T	100	Follows ventricular ejection
T–P	300	Diastole

The potential difference between two points can be written as

$$V_1 = V_{LA} - V_{RA}$$

$$V_2 = V_F - V_{RA}$$

$$V_3 = V_F - V_{LA}$$

(The subscripts LA, RA and F refer to the left arm, the right arm and foot respectively. Instead of the foot, the chest or the back can also be used for potential measurements.) Therefore

$$V_2 = V_3 + V_1$$

This is known as Einthoven's Law. Thus if any two of the three potential differences (also known as leads in ECG terminology) are measured the third can be calculated. Further if $\mathbf{V_1}$, $\mathbf{V_2}$ and $\mathbf{V_3}$ are taken to be vectors, they are chosen such that

$$\mathbf{V_{LA} + V_{RA} + V_F = 0}$$

taking into account the signs of the potentials (Figure 12.8). The triangle thus constructed is known as Einthoven's triangle. Einthoven further showed that the three vectors could be considered as projections of a single hypothetical heart vector **H** on this triangle. Though this relation is true for any three points, the **H** vector will be meaningful only when it is related to the electrical axis of the heart. The three points chosen are such that they are equidistant from the heart so that the triangle is an equilateral one.

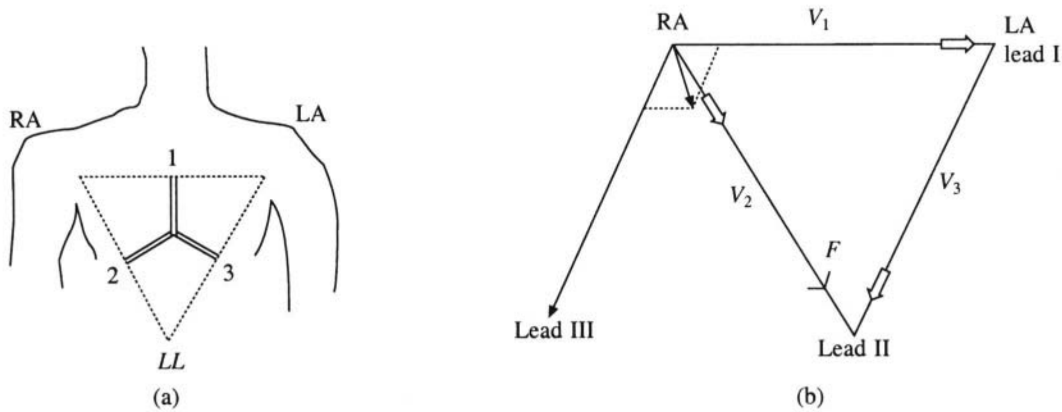


Fig. 12.8 Einthoven's triangle

A realistic model for the electrical events of the heart cycle is based on the assumption that the heart acts as a single dipole, represented by the **H** vector, and the torso is a homogeneous conductor. The recording of the magnitude and spatial orientation of the equivalent heart dipole as a function of time is known as vector cardiography. However, in practice it is not possible to locate a single equivalent dipole and to monitor its magnitude and orientation with respect to time. One of the ways of improving the performance of Vector ECG is to increase the number of electrodes used in order to get a refined value of the dipole. Clinical ECG tests may therefore involve use of 12 or more leads to aid in diagnosis In spite of its great utility, many of the abnormalities in the functioning of the heart may not be obvious in vector cardiography. A more appropriate method would be to represent heart as a collection of dipoles. By the use of such techniques and of computer based automated diagnosis, vector cardiograms have proved clinically very useful.

13

Neurobiophysics

13.1 Introduction

The external world is perceived by organisms through stimuli received by them. These can be mechanical, thermal or chemical. They are transduced by the sensory receptors of the organism into electrical signals and passed on to the central nervous system. The internal communications within the organism are taken care of by the nervous system. This transmission of information relating to the environment and the ability of multi-cellular organisms to respond appropriately to them is essential for survival. This chapter deals with the nerve transmission and other forms of information transport.

As far as the human beings are concerned vision, hearing and smell play vital roles in communications. Hence, the mechanism of hearing and vision is discussed in some detail here. Chemical signals, like the odours emanating from flowers, are described in the later part of the chapter.

13.2 The Nervous System

The nervous system consists of neurons, which transmit information from one part of the organism to the other. This is known as the neuronal response. The nerve cells (neurons and Swann cells) have a cell body from which small projections known as dendrites and large projections called axons protrude. The dendrites carry impulses towards the central cell body (Figure 13.1). The cell body is the thick region of the neuron containing the nucleus and most of the cytoplasm. The axons carry impulses away from the cell body. A fatty covering known as the myelin sheath covers most of the larger axons. The myelin sheath is interrupted somewhat periodically at sites called the nodes of Ranvier. Several axons are grouped together in each nerve. Similarly nerve cell bodies form clusters known as ganglia. Some axons are very long and can extent up to 3 or 4 metres. For example the nerve controlling the muscles in the human finger extend up to the spinal cord where they have their cell bodies. The diameter of the axons is of the order of 1 to $100\ \mu m$. Connections between neurons are called synapses and these occur at the branched ends of axons, dendrites and collateral branches of different neurons. Axons can also end in a synapse at the receptors such as the hair cells of the organ of the corti in the ear.

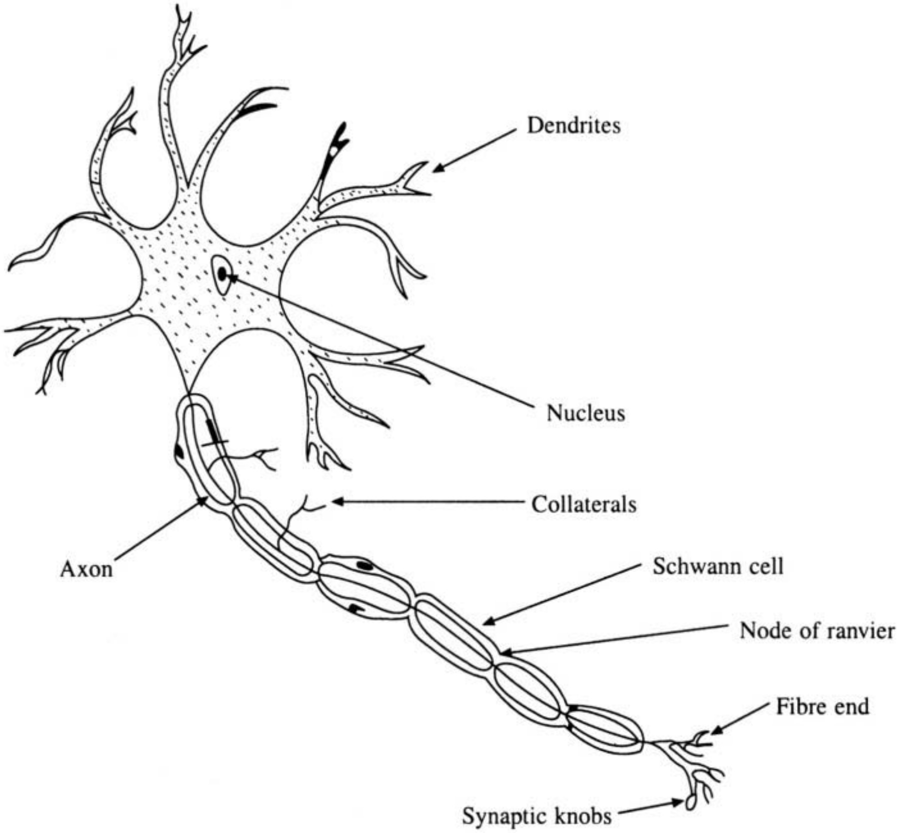


Fig. 13.1 Nerve cell

The nervous system of vertebrates is organised into the central nervous system and the peripheral nervous system. The central nervous system is in the skull and the spinal cord while the peripheral nervous system consists of nerves and ganglia. Generally the peripheral nervous system contains both sensory axons and motor axons. The sensory axons conduct signals toward the central nervous system while motor axons conduct them away from it. Though the neurons within the central nervous system look similar to those outside it, there are several differences. For example when a nerve fibre within the central nervous system is injred, the entire neuron degenerates. On the contrary if a peripheral nerve is cut, the fibres will re-grow out of the old nerve trunk.

A single axon can be removed from an organism and be examined in the laboratory. The maximum diameter of a vertebrate axon is about $20\ \mu m$ while some of the invertebrate axons can be as large as $200\ \mu m$ in diameter. Squid axons are easy to work with, due to their size, and are used routinely for laboratory studies. It is now possible to insert a microelectrode inside a nerve axon and to measure the electric potential relative to the surrounding medium (i.e. the interstitial fluid outside the fibre). This potential normally ranges from -50 to 100 mV. Axons can be stimulated by electrical pulses, heat, chemical changes and also mechanical pressures. When the axon is stimulated its surface potential changes in a specific way, producing a signal called the action or spike potential. The spike potential thus formed is an all-or-none response and is transmitted down the axon in both the directions. However, due to the nature of the synapse present in an intact nerve, only one of these directions is usually effective. Each spike generally starts with a sudden change from its normal negative resting potential to a positive membrane potential, and then abruptly falls back to the resting potential. The total duration is about one millisecond (Figure 13.2).

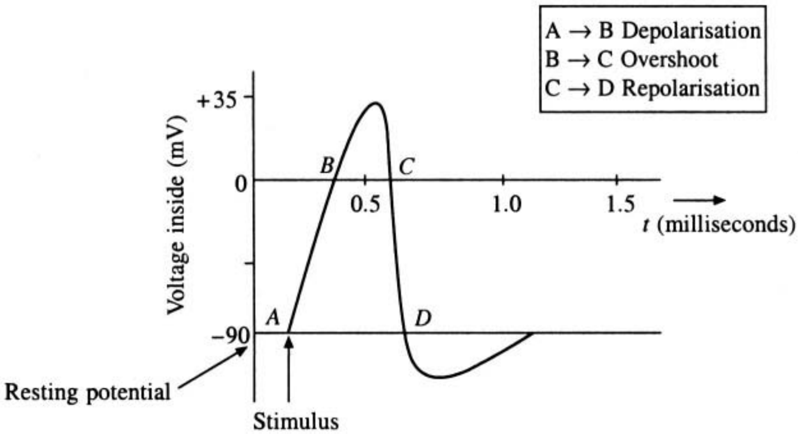


Fig. 13.2 A typical action potential

13.2.1 Synapse

In the nervous system the message is transmitted from one point to another. This means that a two-way passage of the message along the axons has to be prevented by some means. This is achieved by the synapse. The point at which an axon and a dendrite associate is called a synapse (Figure 13.3). When an action potential arrives at the presynaptic fibre, transmitter molecules are released from the vesicles and these diffuse through the synaptic gap and reach the sensitive surface of the postsynaptic fibre. The transmitter molecules alter the permeability to Na^+ and other monovalent ions in the membrane of the sensitive surface, thus creating an action potential. The transmitter molecules are

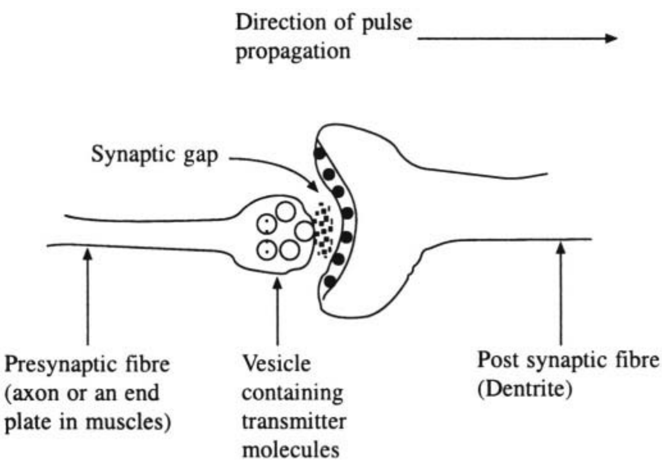


Fig. 13.3 A synapse

destroyed immediately after this so that the synapse could be ready to receive the next signal. Acetylcholine and noradrenaline have been identified as synaptic transmitters and a very small quantity **(10^{-18} M)** of the substance is enough to evoke an action potential. This amount is known as threshold quantity. The destruction of acetylcholine is achieved by the enzyme acetylcholine esterase, which is situated in the receptor surfaces of the postsynaptic fibres. Synaptic conduction can also occur through charge transfer without any chemical intermediates. In this case the impulses travel from one axon to another electrically with negligible time delay. One spike potential at the presynaptic fibre can induce one spike potential at the postsynaptic fibre. In some cases, when the response due to spike is below the threshold value, then sub-threshold responses from several synapses can be added together to form a spike potential at the post synaptic fibre. Thus the neurons can act like a digital computer and can add, subtract or do other arithmetical operations.

13.3 Physics of Membrane Potentials

There are two ways by which membrane potentials can develop: (1) Diffusion of ions through the membrane causing an imbalance of negative and positive charges on the two sides of the membranes (Figure 13.4). (2) Active transport across the membrane again leading to imbalance of the charges on either side of the membrane. Measuring the membrane potential of a nerve fibre is a difficult task due to the small size of the fibres. A microelectrode with a diameter of 1 micron is inserted through the membrane. The other electrode, called the indifferent electrode, is kept in the interstitial fluid (Fig 13.5). The potential difference is measured using a sophisticated voltmeter, which is capable of measuring even very small voltages despite extremely high resistance to electrical flow through the tip of the microelectrode.

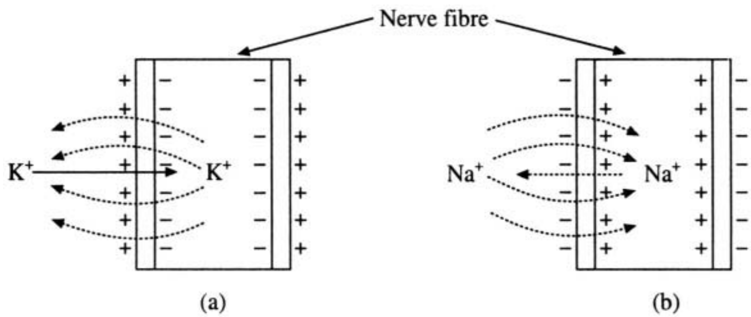


Fig. 13.4 Diffusion potential across cell membranes. Nernst potential for (a) K^+ ions and (b) Na^+ ions.

13.3.1 Membrane potential due to diffusion

Generally the K^+ ion concentration is greater on the inner side than on the outer side of the membrane. The reverse is true for Na^+ ion concentration. When there is no active transport, and if we assume that the membrane is permeable only to K^+ ions, then these ions will diffuse from inside to outside the membrane (Figure 13.4(a)), since the K^+ ion concentration inside is greater. The K^+ ions carry the positive charge to the outside and hence a state of electropositivity outside and electronegativity inside is developed. Thus a potential difference across the membrane is developed and this prevents further K^+ ions moving outward. The potential across the membrane is now known as the Nernst potential for potassium ions. Figure 13.4(b) shows the same effect for Na^+ ions, where we now assume that the membrane is highly permeable to Na^+ ions and impermeable to all other ions. As the

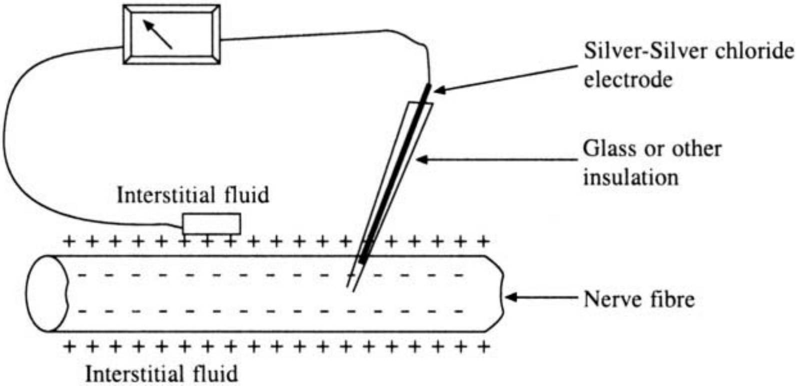


Fig. 13.5. Measuring membrane potential

Na^+ ion concentration outside is more, sodium ions move inward and a membrane potential with reverse polarity is generated. When the membrane potential rises high enough to prevent further diffusion of Na^+ ions, it is known as the Nernst potential for sodium ions. The magnitude of the potential is given by

$$\text{EMF(in mV)} = -\ 61 \times \log \left[\frac{\text{Concentration of positive ions (inside)/}}{\text{Concentration of negative ions (outside)}} \right]$$

When the concentration of positive ions inside is ten times that of negative ions outside, then log (10) is 1, and the EMF = -61 mV. The above equation is called Nernst equation. For sodium ions, the Nerst potential is $\cong +61$ millivolts. For potassium ions, the Nernst potential $\cong -94$ millivolts. Under resting conditions the membrane potential averages to –90 millivolts which is near the Nernst potential for potassium. This is because at resting stage the membrane is more permeable to K^+ and only slightly permeable to Na^+ ions.

We can now see more details of how the action potential, mentioned above, develops across the membrane. There are three stages.

(a) *Resting potential:* This is the stage before the action potential actually occurs. The membrane potential is negative and is said to be polarised during this stage.

(b) *Depolarisation stage:* At this stage, the membrane becomes very permeable to sodium ions and a tremendous number of sodium ions rush into the interior of the axon. The membrane potential rises in the positive direction and sometimes overshoots the zero value.

(c) *Re-polarisation stage:* Sodium channels close immediately after this and potassium ions flow to the exterior re-establishing the resting stage negative potential (Figure 13.2). Voltage gated channels and Na^+–K^+ pumps play a vital role in both depolarisation and re-polarisation of the nerve membrane during the action potential. These are membrane proteins or peptides or molecular complexes, which open and shut appropriately to allow or prevent the flow of the ions. As the names suggest, the gates play a passive role, while the pumps are proactive. The sodium channel is activated when the membrane potential rises from its resting value of –90 mV, in the positive direction, to somewhere between –70 and –50 mV. When the sodium channel opens, it changes its conformation (Figure 13.6). After a few 10,000ths of a second, it again changes its conformation and closes, and the Na^+ ions cannot enter the nerve axon. The conformational changes of the sodium channel during depolarisation state are rapid, while those during the closing of the channel are delayed. The membrane potential starts recovering back to its resting stage as soon as the sodium channel is closed. The inactivated gate will not reopen till the membrane potential reverses back to the resting negative potential. The potassium channel is closed at the resting stage preventing potassium ions flowing to the exterior. When the membrane potential moves in the positive direction, i.e. from –90 mV to zero then the potassium ion channel opens up and K^+ ions diffuse outward. As this process is relatively slow, it takes place when the sodium channel is inactivated. Thus the closing of the sodium channel and simultaneous opening of the potassium channel speeds up the repolarisation process.

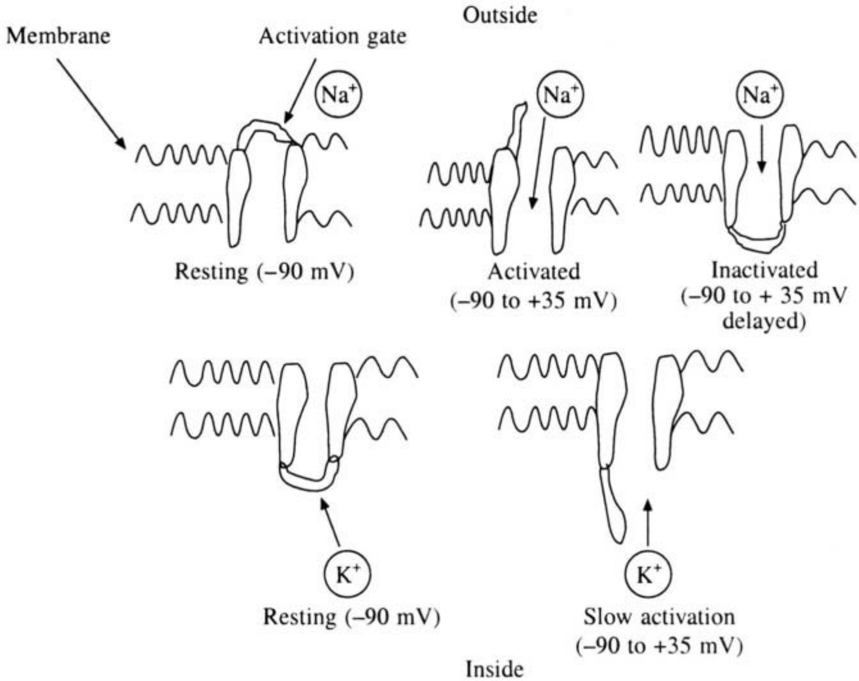


Fig. 13.6 Voltage gated sodium and potassium channels

13.3.2 Voltage clamp

Membrane potentials change very rapidly and it is possible to study the kinetics of these processes only when the membrane potential is held constant at a specified value, and if the measurements are carried out for a specified time. This is achieved by the voltage clamp technique. It was first successfully employed by Hodgkin and Huxley to measure the flow of ions through different channels. The voltage clamp has two electrodes inserted into the nerve fibre. One of these is for measuring the voltage and the other is to conduct electric current inside or outside the fibre (Figure 13.7). The investigator chooses a voltage to be established inside the nerve fibre, which is greater than the threshold value, and the voltage is clamped at the chosen value by a feedback system. The membrane potential (E_m) is measured by an intracellular electrode, and this is compared with the chosen voltage in a control amplifier. The output of this amplifier is connected to a second electrode, which passes a current into the cell, which is proportional to the deviation of the membrane potential from the desired voltage. Thus the deviation is compensated. It is to be noted that the compensating clamp current has the same amplitude but opposite polarity as the membrane current at the chosen potential. The current flow is measured by connecting the current electrode to an oscilloscope. For example, when the membrane potential is suddenly increased from –90 mV to zero by this method, then

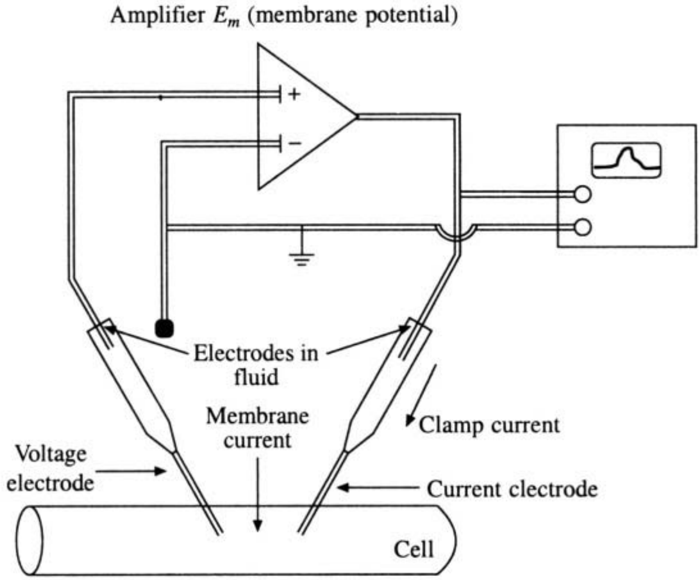


Fig. 13.7 Voltage clamp

sodium and potassium channels open up and the ions begin to flow through the channels. Electric current is injected through the current electrode to compensate for the net current flow through the channels and to maintain the intracellular voltage at zero (or a pre-determined value). The conductance, which varies with time, is measured. By selectively inhibiting one of the channels, the flow of ions through individual channels can be studied. Toxins like tetrodotoxin are used for blocking sodium channels while terra ethyl ammonium ion is used for blocking potassium channels. Figure 13.8 is an example of the results obtained from such measurements. Changes in conductance when the membrane

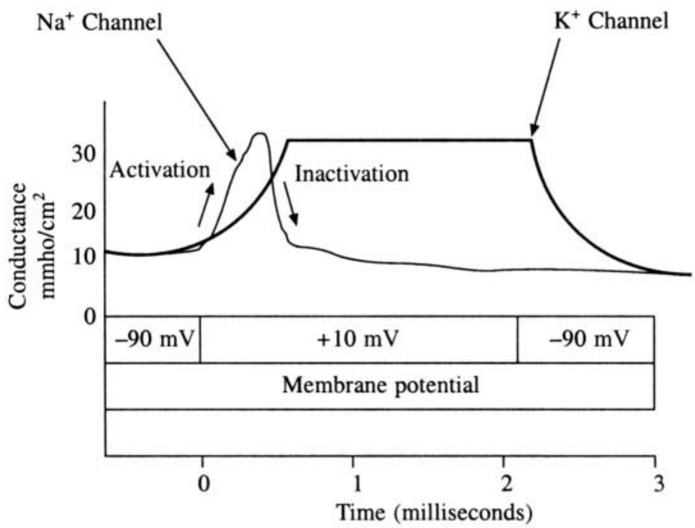


Fig. 13.8 Changes in conductance during membrane transport

potential is increased from its normal resting value of –90 mV to +10 mV for 2 milliseconds are shown. The sodium channels are activated and inactivated within this time while potassium channels are only activated during this time.

The behaviour of the nerve fibre in terms of changes in permeability of the membrane to Na and K ions has been formulated as a set of equations, known as the Hodgkin-Huxley equations, as follows:

$$g_K = n^4 \overline{g_K}$$

$$dn/dt = \alpha_n(1 - n) - \beta_n n$$

for K^+ conductance, where $\bar{g}_K$ is the constant maximum conductance. The state α is that in which the membrane allows K^+ to pass, while β is the state in which the membrane is impermeable to K^+ ions. n is the proportion of molecules in α state while $(1 - n)$ is the proportion of molecules in state. β_n is the reaction rate of the change from state α to state β and α_n is that of the change from β to α. At the resting stage β_n is large and α_n is small. Consequently most of the regulating molecules are in state β. On depolarisation α_n increases and β_n decreases depending on the membrane potential and this leads to a rise in n and hence in g_K. After about 10 milliseconds, α_n is larger than β_n and an equilibrium is reached where most of the molecules are in α state. n and g_K attain maximum values and this will last as long as the depolarisation state lasts. For sodium conductance, the Hodgkin-Huxley equations are as:

$$g_{Na} = m^3 \bar{g}_{Na}$$

$$dm/dt = \alpha_m(1 - m) - \beta_m m$$

$$dh/dt = \alpha_h(1 - h) - \beta_h h$$

$\bar{g}_{Na}$ is the constant maximum sodium conductance m and h denote the state of the two molecules which regulate sodium conductance. α_m, β_m, α_h and β_h are reaction rate constants that depend only on membrane potential but not on time. Though the conductance of sodium ions is similar to that of potassium, the equations will have to accommodate complications like inactivation of the sodium channel after about 1 millisecond. The variable '*h*' has been introduced to take care of the inactivation process.

13.4 Sensory Mechanisms—The Eye

An organism is continuously bombarded by different signals from the environment. AH the sense organs that are provided to the organism are transducers of energy; i.e. they convert energy in one form into another. Unlike neurons the response in these cases is not an 'all or none' process. The intensity of the response by the sensory organs will vary with the extent of the stimulation. Sensory detectors are classified into teleceptors, chemical receptors, somatic receptors and visceral receptors. Organs like eyes and ears are teleceptor as they receive information from distant objects. Sensors of smell and taste belong to the class of chemical receptors as they are excited by chemical stimulation. Touch, pressure, pain and temperature are experienced through the somatic receptors, while the information regarding body conditions like thirst, hunger etc are conveyed to the brain by the visceral receptors. When stimulated, each of these receptors will initiate nerve discharges in the form of action potentials. However, the way in which sensory information, such as the image on the retina or the pitch of a sound, is converted into the binary code of action potential is unknown. The action potentials generated by sensory nerves are usually evoked by synaptic stimulation and are therefore called generator potentials. Generator potential magnitudes vary with the intensity of the stimulation of the receptors and this is achieved by modulating the frequency of a sequence of action potentials in such a way that it reflects the intensity of the stimulus. Let us look at the basic question of how a stimulus like light, sound or temperature is converted into electrical polarisation of the membrane in the case of visual receptors and auditory receptors.

13.4.1 The visual receptor

There are different kinds light sensitive receptors of which vary from the most primitive system like photokinesis (light induced motion) in plants to most elaborate ones like the vertebrate eye. A schematic diagram of the vertebrate eye is given in Figure 13.9. The lens system of the eye, which consists of cornea, aqueous humour, ciliary muscle, and vitreous humour of the eyeball helps in focussing the image of the object onto a layer containing visual receptors. The layer on which image is formed is called the retina. The iris is an important structure and it controls the amount of light entering the eye. A small iris opening increases the depth of focus in addition to restricting the distortions such as spherical aberration, field curvature etc. For example in the night, sensitivity is more important than depth of focus or acuity. Hence the iris diaphragm will be open to the maximum extent.

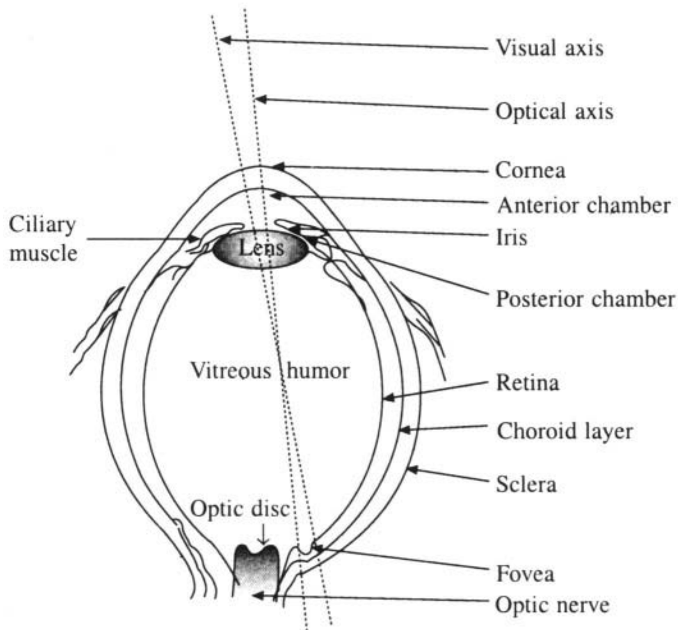


Fig. 13.9 Schematic diagram of the vertebrate eye

The crystallin lens or eye lens is the most important structure in the eye. The eye lens is a cellular structure, which is more curved at the rear than in front. The focus of the lens is varied by changing the curvature of the front face by the action of ciliary muscles. The space between the retina and the lens is filled with a jelly-like substance called vitreous humour. Vitreous humour is optically similar to the aqueous humour filling the cornea and the eye lens. Light enters the eye through the cornea and passes through the aqueous humour and the crystalline lens into the vitreous humour. The light is received by the retina and an image is formed. Figure 13.10(a) shows the geometrical optics of the eye. Vision itself is made up of several processes. The photons (light) which fall on the retina trigger a series of events in the light sensitive photoreceptor cells. As a result the energy carried by the photons is transduced into chemical energy which in turn is transformed into an electrical signal. This signal is carried by the optic nerve to the brain for processing. The transduction of light into chemical energy is fairly well understood but the subsequent conversion into electrical signals is not completely understood. The image on the retina is formed by refraction of light by the cornea and lens and can be described by the well-known Snell's law

$$\sin \theta_2/\sin \theta_1 = n_1/n_2$$

where θ_1 is the angle made by the incident light with the normal and θ_2 the angle made by the refracted ray with the normal and n_1 and, n_2 are the refractive indices of the two media, respectively (Figure 13.10(b)). Before reaching the retina, light is refracted at three places. (1) At the surface of the cornea, going from the air into the cornea. (2) At the surface of the lens, going from aqueous humour into the lens. (3) At the posterior surface of the lens, in passing from lens to the vitreous humour, the refractive indices being $n_{air} = 1.0$, $n_{humour} = 1.33$, $n_{lens} = 1.413$. Thus the greatest refraction takes place at the posterior surface of the cornea. In order to calculate the image size we can replace the cornea-lens system by a single equivalent lens. In that case, we know that

Object size/Image size = Object distance/Image distance

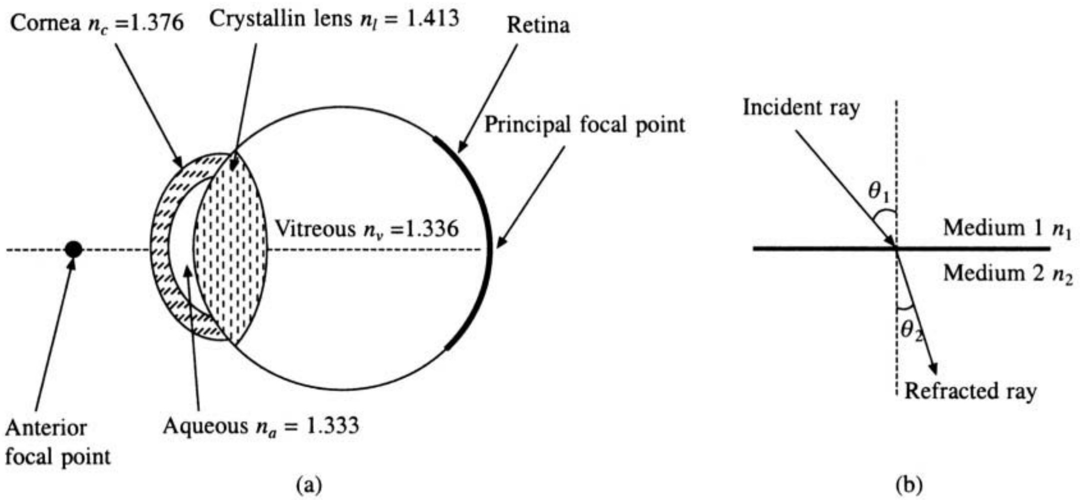


Fig. 13.10 (a) Optical properties of the eye. (b) Refraction of light

This expression can be used to calculate the image size. For example if the object distance is 1 km and the object size is 10 m then the image size = 0.15 mm, where the average image distance is taken to be 1.5 cm. The process of vision is dynamic, since the eye lens is not a rigid object with fixed focal length. The focal length can be adjusted by changing the curvature of the lens and this process is called accommodation. The standard relation between the object and image distances (p and q) is given by

$$1/p + 1/q = (n - 1)(1/R_1 + 1/R_2) = 1/f_L$$

where R_1 and R_2 are the front and rear radii of curvature of the lens, respectively, f_L is the focal length of the lens, and n is the refractive index of the lens. If the focal length of the lens is fixed, then objects near the lens will be focussed behind the retina and vice versa. As the lenses have a flexible focal length the images are always focussed on the retina. The change of focal length is achieved by a change in curvature of the lens, which is made possible by the action of muscles attached to the lens. As age progresses the power of accommodation reduces and external lenses are used to correct this defect.

The image formed on the retina is received by photoreceptor which are called rods and cones (Figure 13.11). Cones are more concentrated at the central part of the retina while rods are more abundant at the periphery. Colour vision is due to cones while rods are responsible for vision in weak light. As rods are insensitive to colours, three types of cones have been implicated in colour vision. Though Newton's experiments demonstrated that a spectrum of white light consists of a continuum of colours, Maxwell showed that almost all colours could be obtained by mixing three primary colours. This suggests that at least three different receptors are necessary for colour vision. However, the perception of colour is achieved only by a great deal of processing of the signals by the brain. The fovea, which is the central part of the retina, has a high degree of acuity of vision and this is due to the exclusive presence of cones in this region. The optic disk lies very close to the fovea. This disk is also known as the blind spot. Objects focussed on this region cannot be seen as there are no cones or rods in the optic disk. The photoreceptor cells contain visual pigment molecules that are light sensitive. The light sensitive molecule in the vertebrate eye is rhodopsin and has a molecular weight

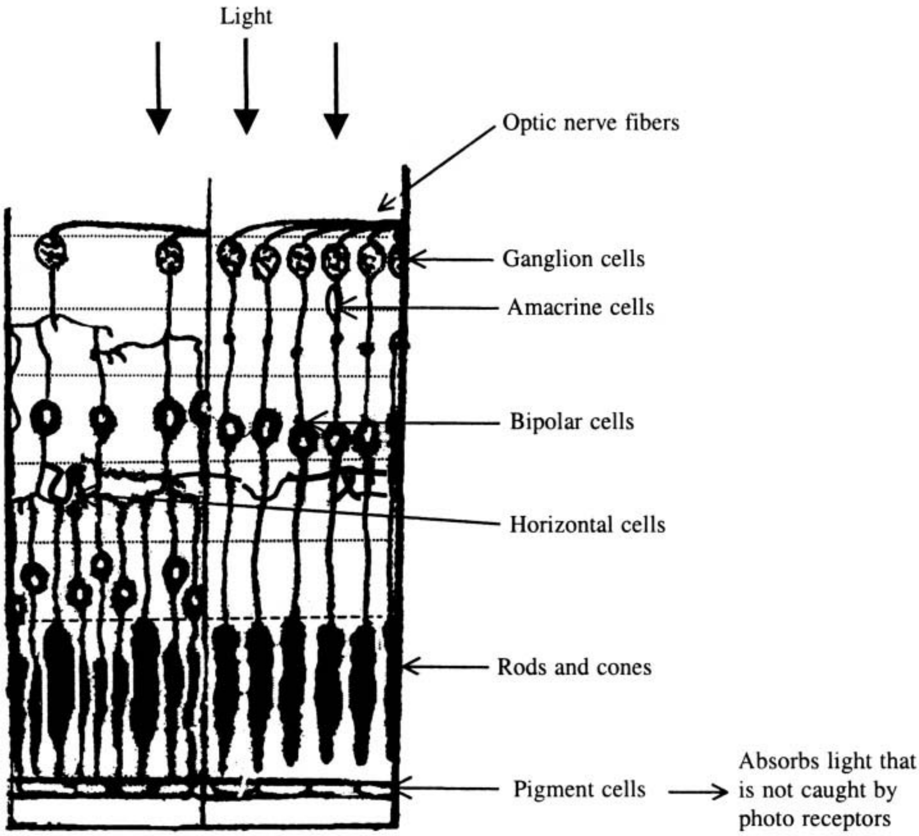


Fig. 13.11(a) Structure of retina

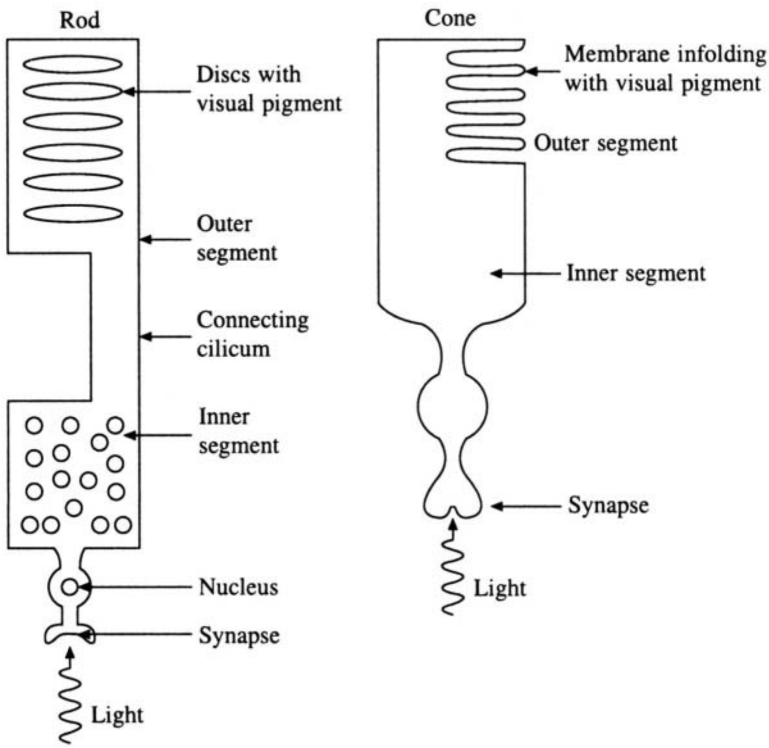


Fig. 13.11(b) Schematic diagram of rods and cones

of about 35 to 40 KDa. This consists of a single-polypeptide protein, opsin, to which other molecules are covalently linked, viz. a chromophore and two oligosaccharide chains. The chromophore is retinal and is responsible for absorption of light in the visible region. Retinal is a derivative of vitamin A and is obtained by oxidation of vitamin A by **NAD^+**. Figure 13.12 shows the two isomeric forms of retinal. Of the two oligosaccharides, at least one is composed of three *N*-acetyl glucosamine and three mannose units. Rhodopsin is a membrane protein and is insoluble in water. Figure 13.13 shows the structural arrangement of seven **α-helical** segments of rhodopsin. The absorption spectrum of rhodopsin is shown in Figure 13.14. The absorption maximum is at approximately 500 nm. This is also the most effective wavelength for human vision in a dark-adapted eye that has been exposed to weak light. In strong light the peak is shifted to 550 nm which is close to the absorption maximum of iodopsin. In strong light when the rod vision changes to cone vision, iodopsin is the visual pigment involved.

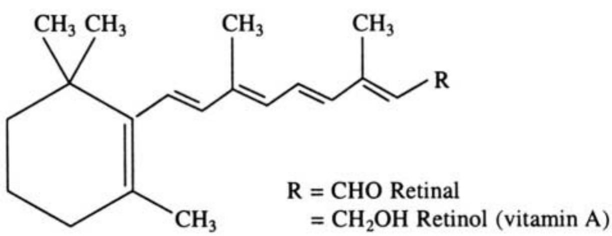


Fig. 13.12 Structure of retinal and retinol

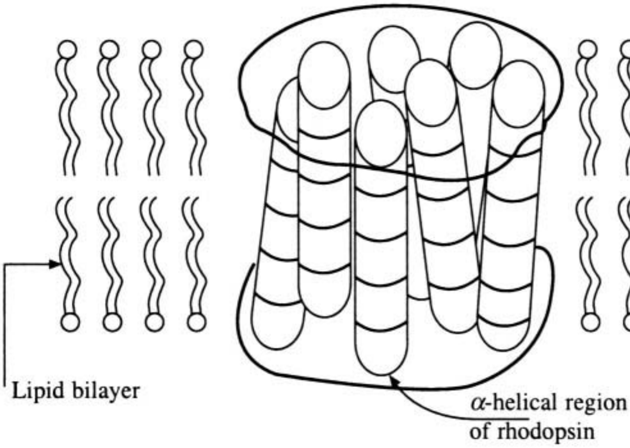


Fig. 13.13 Arrangement of the α-helical segments of rhodopsin in the lipid bilayer

The first step in the energy conversion procedure is the absorption of photons by chromophores. This causes a change in conformation of retinal from '11-cis' to 'all-trans' leading to detachment of the chromophore from opsin. This is known as the bleaching reaction. The recovery process involves reduction of retinal to 'all-trans' vitamin A, diffusion of vitamin A into the epithelium and re-conversion of vitamin A by light or enzymatically into the '11-cis' form. The '11-cis' vitamin A is then oxidised to '11-cis' retinal and linked to the protein opsin. The protein probably goes through a number of conformational changes before the 'all-trans' retinal comes off. Thus detachment and reattachment of the chromophore is the crucial part of the photochemistry of vision. An enzyme in the retina maintains an equilibrium between retinal and retinol (vitamin A). Generally a higher concentration of the alcohol (retinol) is maintained. However if the aldehyde (retinal) concentration is low then it

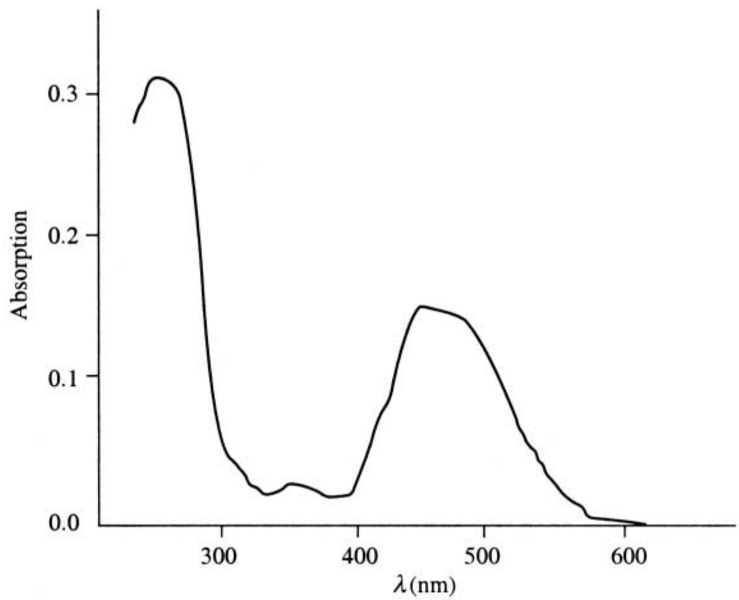


Fig. 13.14 Absorption spectrum of rhodopsin

is produced from the alcohol by the action of an oxidoreductase enzyme and NADPH. The vitamin produced by oxidoreductase enters the blood through the rod membrane and is maintained at a constant concentration by the liver (Figure 13.15).

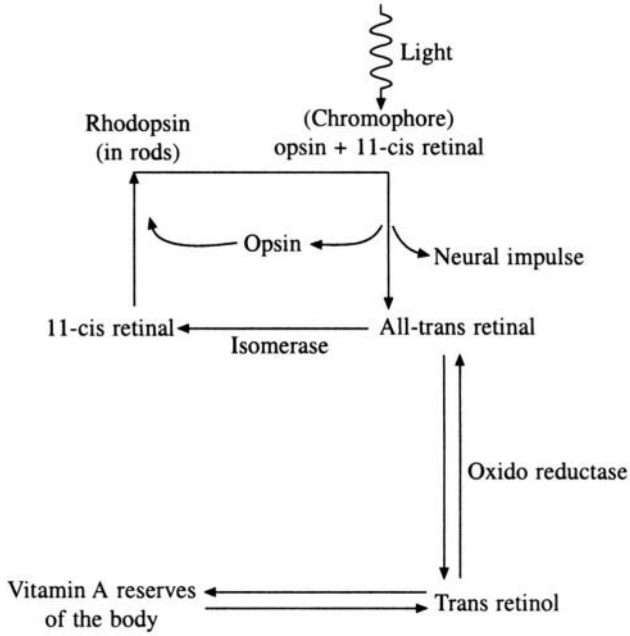


Fig. 13.15 Visual cycle undergone by rhodopsin

13.4.2 Electrical activity and visual generator potentials

When the rhodopsin molecule absorbs a photon, it undergoes a change in its electrical state, which is passed on as a membrane potential. This electrical signal, which is known as receptor potential, is proportional to the intensity of the light stimulus. The receptor potential is processed at the synapse and translated as series of nerve impulses, which are carried to the brain for visual perception. Information about the firing of action potentials in optic nerve fibres has been obtained by implanting microelectrodes in selected places. In vertebrates the ionic permeability decreases while in invertebrates the reverse occurs. The measurements show that the $\mathbf{K^+}$ ion concentration is about 20 times more inside the cell than outside while the sodium ion concentration is more on the outside than in the inside. This is the same as the situation seen in nerve cells. The association of electrical impulses with the visual process is different from that of transmission of electrical impulse in the axon. The receptor cells behave like photocells. The membrane potential of the vertebrate photoreceptor is also mainly a diffusion potential. The potential difference across the cell membrane in the inner segment is larger than in the outer segment and this causes a permanent ion current in darkness. This current flows from the inner segment to the outer segment through the extracellular medium. In the dark, the external membrane of the rod outer segment is more permeable to $\mathbf{Na^+}$ ions while the membrane of the inner segment is preferentially permeable to potassium ions. On illumination, the permeability for the sodium ions across the outer segment cell membrane drops and this drop is greater if the intensity is higher. This transient light induced reduction of conductance causes a reduction of the membrane current and a transient increase in the membrane potential (i.e. the membrane is more polarised) and this is communicated to the horizontal cells. The signal is actually produced in the amacrine and ganglion cells, and is transmitted to the brain by the optic nerve.

13.4.3 Optical defects of the eye

A person with normal vision can see objects as close as 250 mm. In order to focus objects from 250 mm to infinity the crystalline lens curvature has to be changed by about 20%. The strength of the human eye is about 48 diopters. If the eye is stronger than this, then images of distant object will be focussed in front of the retina. Thus distant vision will be not be clear as compared to near vision and the person is said to be myopic or suffering from myopia. This can be corrected by using a diverging lens in front of the eye (Figure 13.16). If the strength of the eye is too weak, then the images of near objects will be formed behind the retina. This condition is known as far-sightedness or hypermeteropia, and requires a convergent lens for correction. The adaptability of the ciliary muscles drastically reduces, as the age increases. This also leads to long sight and a person who is myopic would require bifocals for correcting the near and far vision independently. Astigmatism is another eye defect due to the deviation of cornea from sphericity. When astigmatism is present, a point object does not form a point image on the retina. This is because the eye lens has different focal lengths for different directions. This defect can be corrected by using cylindrical lens.

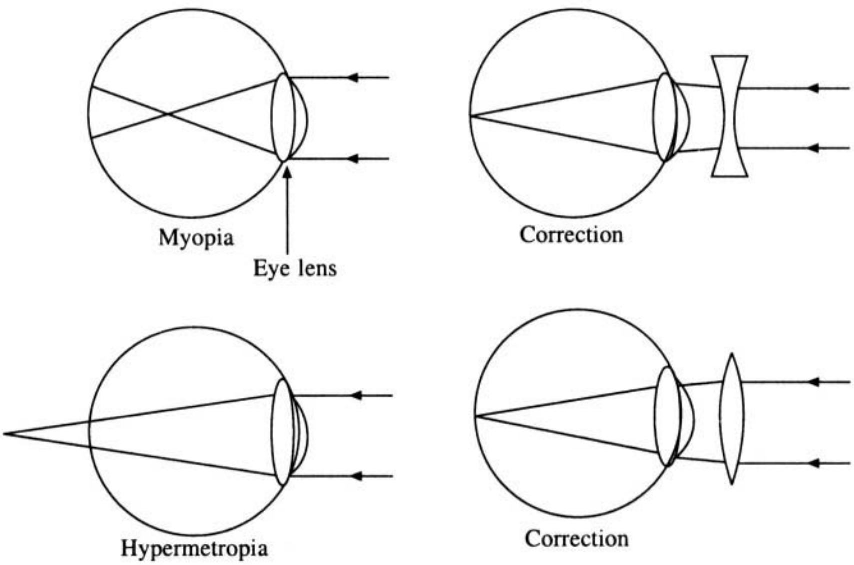


Fig. 13.16 Defects in vision and their corrections

The response of the eye extends beyond the visible spectrum. For example the cornea absorbs ultraviolet light which can damage it. Hence dark glasses are recommended in strong sunlight.

13.4.4 Neural aspects of vision

The retina acts as a photo-neural transducer i.e. it converts light energy into a neural response. This response, which is in the form of a spike potential, travels along the optic nerve and reaches the central nervous system (CNS). From the central nervous system, the neural response reaches specific areas in the cerebral cortex. The information is analysed both in the retina and in the CNS to generate the sensations of shape, colour, brightness, etc. The responses from either eye for an object are projected on to the occipital lobe of the cerebral cortex opposite to the half of the visual field containing the object. There are three layers of cells which are important in the function, viz. the retinal cells, intermediate cell and a third layer of cells. These layers are interconnected and the information flows from retinal cells to the brain through these layers of cells and optic nerve. The information coming out of the optic nerve is divided into two bundles before reaching the brain. The visual cortex in the brain is responsible for the analysis of these impulses and subsequent vision. There are a few fibres that go to the midbrain rather than to visual cortex directly. Information such as the intensity of light, clarity of vision, etc., are processed in the midbrain and a feedback is sent to the appropriate regions of the eye. Figure 13.17 shows the neural connections between the eye and the visual cortex. From this figure one can see that the left side of the brain receives information from the right side of the visual field and the right side of the brain receives signals from left side of the visual field. They run through the optic chiasma and the information from both the eyes are put together to achieve binocular vision. Every point in the retina seems to have a corresponding point in

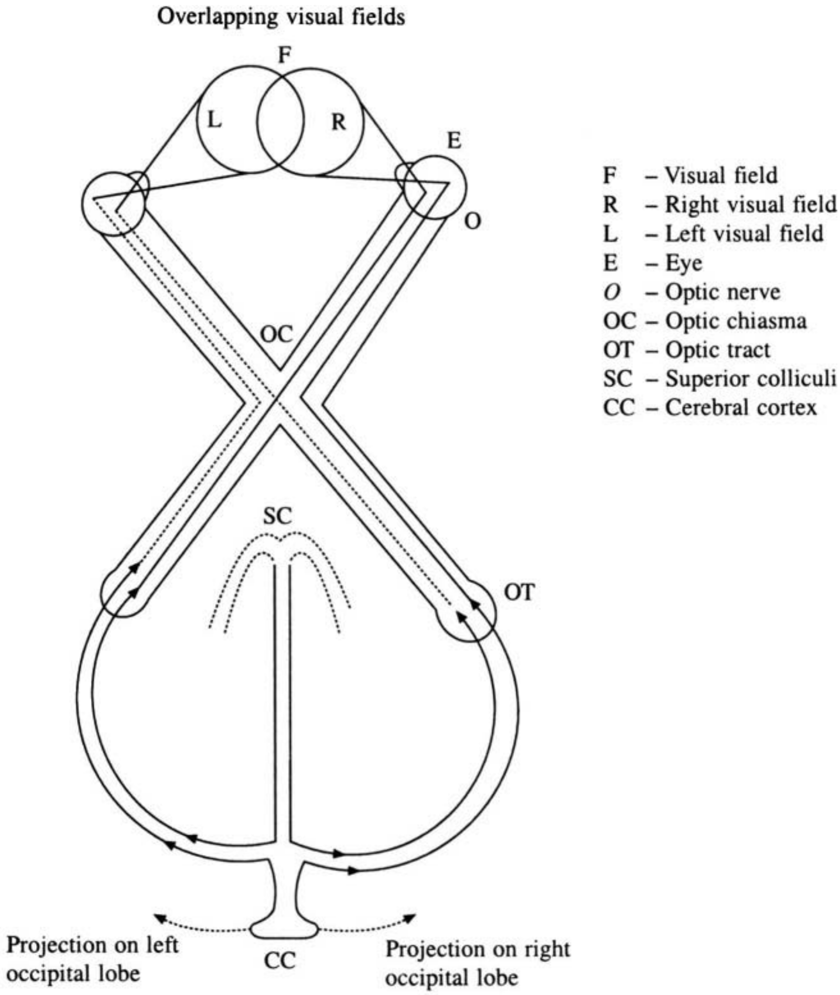


Fig. 13.17 Pathways of vision in the brain

the visual cortex. The points that are close together on the surface of the retina are also close together in the visual cortex. However, the central field of vision, which occupies a very small region on the retina, is expanded and is represented by several cells in the visual cortex.

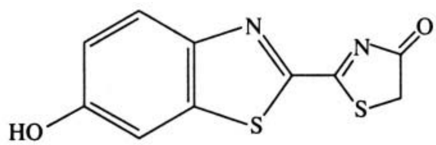
13.4.5 Visual communications, bioluminescence

Several insect families communicate with others in their species by producing a light, without accompanying heat. The visual signals can vary in intensity, colour and periodicity. The production of light without heat by living organisms is called luminescence. Most bioluminescent signals are not continuous and are mostly produced in the night or in special habitats like caves, ravines etc. A diverse variety of organisms like bacteria, fungi, sponges, corals, and fish can produce luminescence. Amphibians, reptiles, birds and mammals do not possess this property of producing light. The bioluminescent reaction in fireflies has been well characterised. The two substances responsible for this reaction, viz. luciferin and luciferase, have been isolated and crystallised, and their structures have been established. Figure 13.18 gives the chemical structure of luciferin. The reaction involving these two chemicals is a luciferase-catalysed oxidation using molecular oxygen and a substrate, viz. D(–)luciferin. A large amount of energy is liberated in this reaction and the intermediate or product is left in the excited state. When the excited product returns to the ground state, light is emitted. ATP is hydrolysed in this reaction, which is written as follows:

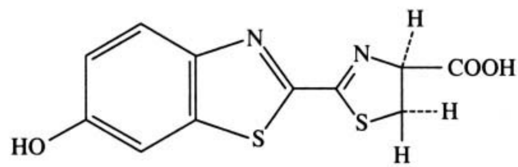
$$\textbf{Enzyme} + \textbf{D(–)luciferin}(\textbf{LH}_2) + \textbf{ATP} \overset{\textbf{Mg}}{\Leftrightarrow} / \textbf{Enzyme–LH}_2\textbf{–AMP} + \textbf{PP}_i$$

$$\textbf{Enzyme–LH}_2\textbf{–AMP} + \textbf{O}_2 \rightarrow \textbf{Oxyluciferin} + \textbf{AMP} + \textbf{CO}_2 + \textbf{Light}$$

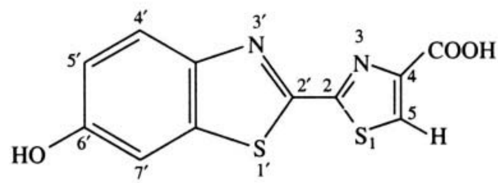
where **enzyme–LH$_2$–AMP** is the enzyme bound-luciferyl adenylate.



Oxyluciferin



D(-) Luciferin



Dehydroluciferin

Fig. 13.18 Chemical structure of luciferin

13.5 Physical Aspects of Hearing

Hearing, like vision, is an important sensory mechanism that enables many animals, including humans, to perceive their environment. Though the biophysics of hearing has been one of the earliest to be studied, information on the transduction of sound is meagre. This is due to the fact that the sensory organ, viz. the ear, will have to be intact for it to function and it is very difficult to perform experiments and measurements on the inner ear, where transduction takes place. However the structure of the ear is known in very great detail from anatomical and histological studies, and this has helped to reconstruct the hearing process.

Hearing is actually the perception of pressure waves that are generated by a vibrating medium. The ear, which is the hearing apparatus, is homologous in most species though their sensitivity varies. The frequency range of the normal human ear is 20 to 20,000 Hz.

13.5.1 The ear

The ear is divided into three parts - the outer, the middle and the inner ear. Figure 13.19 shows the human ear. The outer and middle ear are rilled with air and their primary function is to conduct the sound waves to the inner ear, where the acoustic energy is converted to neural spikes, which are analysed by the brain. The outer ear consists of an external auricle, a narrow tube called the ear canal or external auditory meatus and the ear drum or tympanic membrane. The ear canal is similar to a closed end organ pipe with the tympanum serving as the end closing membrane. The vibration of air molecules produced by an external sound is passed on to the small bones in the middle ear. The middle ear has three small bones or ossicles called malleus, incus and stapes. The ossicles amplify the acoustic pressure of vibrations transmitted via the tympanum. They also restrict the amount of energy that is transmitted at high sound levels. Thus when a high sound reaches the system then the amplification at the middle ear is decreased and this acts as a volume control. The ossicles also discriminate against

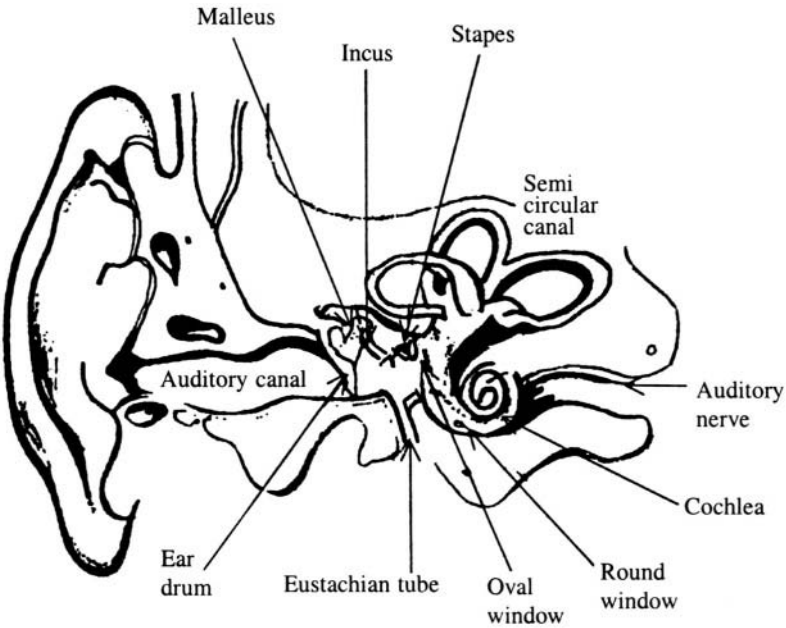


Fig. 13.19 Human ear

the vibrations reaching them through the skull. The outermost ossicle or the malleus presses against the tympanic membrane while the innermost one, the stapes pushes against a membrane called the oval window. The oval window separates the air-filled middle ear from the liquid-filled channels of the inner ear. There is a second membrane known as the round window, which separates another channel of the inner ear from the middle ear. The maximum pressure amplification achieved by the outer and middle ear together is about 35 dB. The middle ear is connected to the pharynx through a tube called the eustachian tube. The eustachian tube prevents distortion in the shape and position of the tympanic membrane when a change in pressure takes place due to atmospheric conditions or changes in the altitude. The mammalian inner ear consists of several portions of which only the cochlear portion is involved in hearing. The cochlea has two and half turns and has three parallel fluid-filled tubes. These tubes are called tympanic, vestibular and cochlear ducts. The individual ducts or canals are separated by membranes. Reisner's membrane separates the cochlear duct and the vestibular duct while the basilar membrane separates the tympanic duct from cochlear duct (Figure 13.20). The organ of corti, which is situated in the basilar membrane, is considered as a neuro-mechanical transducer. This organ is a complicated system of cells. It includes hair cells. The bending of these hair cells excites the nerve endings located in the organ of corti. These excitations are transmitted to the brain where it is perceived as sound. In order to understand the physical aspects of hearing a basic knowledge in acoustics is required.

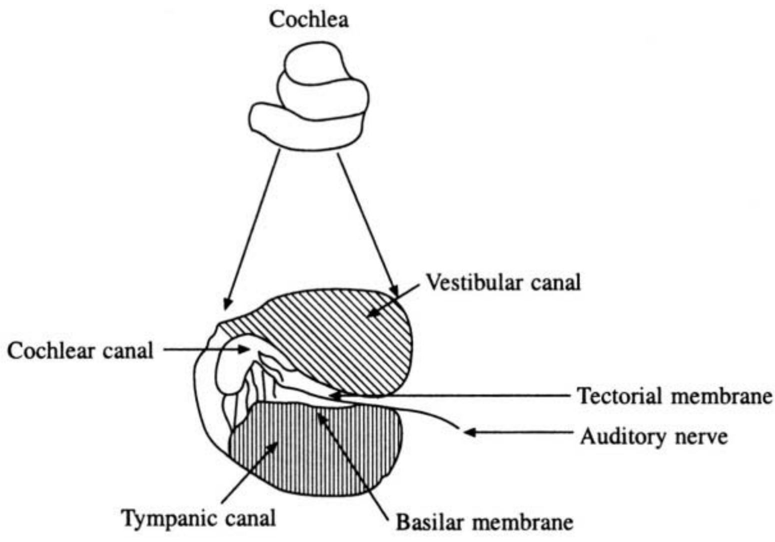


Fig. 13.20 Cochela and its cross section

13.5.2 Elementary acoustics

Sound is actually an elastic disturbance in a continuous medium. Supposing a particle in the medium is disturbed from its equilibrium position by a sound wave. If v is the frequency of the displacement, and ψ is the amplitude, then we can write the expression describing this effect as

$$\psi = A \sin(2\pi v t) + B \cos(2\pi v t) \text{ or}$$

$$\psi = Ce^{2\pi i v t}$$

where the amplitude varies as a function of time t. Most sound waves are mixtures of frequencies. If the velocity of sound waves is c then $\lambda = c/\nu =$ **wavelength.** In a gas the pressure variation produced by the passage of sound wave is very rapid and hence the process is assumed to be adiabatic, i.e. occurring at a constant temperature. The process of hearing can be considered as the conversion of mechanical energy in the form of pressure variations in the air to electrical energy in the form of nerve pulses, which are transmitted to the brain for processing by auditory nerves. The conversion starts at the tympanic membrane in response to the pressure variations in the atmosphere and causes the three bones in middle ear to deflect. The three small bones in the middle ear have a mechanical linkage, and cause compressions in the cochlear fluid. The inward motion of the stapes causes downward motion of the basilar membrane, which triggers the nerve signal. The movement of the basilar membrane is a travelling wave, with its maximum amplitude at some small distance from the oval window. The responses of basilar membrane to sound have been measured using microelectrodes placed in the cochlear space and in the fibres of auditory nerves. Some typical responses are and that the displacements are inversely proportional to the frequency of the exciting sound wave, are shown

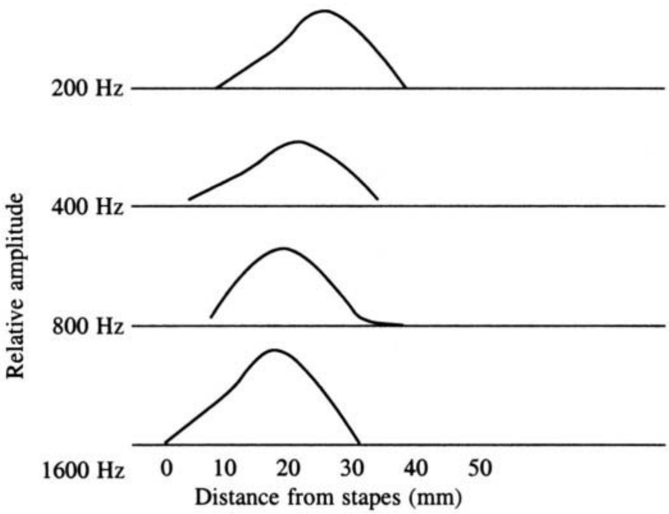


Fig. 13.21 Amplitude of the basilar membrane displacements

in Figure 13.21. The magnitude of amplification is estimated as follows. The force on the stapes is

$$F_{\text{stapes}} = 1.3 \times 52 \times P_{\text{tympanum}}$$

where 1.3 is a constant signifying the mechanical advantage of ossicles, $52\ \text{mm}^2$ is the area of contact between the membrane and the hammer, and P is the pressure. The area of contact between stapes and round window is $3.2\ \text{mm}^2$. Therefore the pressure at the window is

$$P_{\text{window}} = F_{\text{stapes}}/3.2 \text{ (since Pressure = Force/Unit area)}$$

Therefore

$$P_{\text{window}}/P_{\text{tympanum}} = 68/3.2 \cong 21$$

This is roughly the maximum amplification observed. In humans, the average value of magnification is about 17 and is less than the calculated value because of friction, which has not been taken into account in the above calculations.

13.5.3 Theories of hearing

The different theories that have been formulated to explain hearing are known as the spatial and temporal theories. The spatial theory proposed by Helmholtz suggests that a particular point on the basilar membrane responds to a particular audible frequency. Anatomical investigations have shown the basilar membrane to have a fibrous structure and individual fibres are expected to resonate at a particular frequency. The Helmholtz theory met with a few difficulties. Experiments conducted with resonators showed that the fibres respond to a narrow band of frequencies rather than a single frequency. Secondly, any resonating system produces an 'echo' which continues even after the excitation of resonance has stopped. However, no such effect is observed in the hearing process. Nevertheless, hearing loss due to damage to particular regions of the membrane clearly supports the view that the loss is due to the lack of response by the fibres expected to resonate at those frequencies. Helmholtz compared the resonance of the fibres to that of a stretched string. The resonating frequency of such a stretched string is given by

$$\nu_{res} = {}^{1}/_{2}\, l\sqrt{(T_0/\rho)}$$

where l is the length of the string, T_0 is the tension on the string, and ρ is its density. Using the above equation we can calculate the frequencies over which the ear is sensitive. When this is done we find that the audible frequency range is reduced by five octaves from the experimentally known values. This shows that Helmholtz's theory does not completely represent the hearing process.

Bekesey showed that in addition to the variable compressions one should take into account the hydrodynamic waves that are produced in the cochlear fluid in explaining the hearing process. He could successfully predict the point of maximum response in the basilar membrane as a function of frequency. The maximum displacement position is dependent on the initial frequency. The mathematical theory dealing with hydrodynamic waves is quite complex and a simplified explanation is given here. We have seen earlier that the vibration of the stapes produces a wave in the cochlear fluid, which spreads from the oval window throughout the cochlea in about 25 microseconds. This leads to vibrations in the basilar membrane, but the membrane being less elastic at the base, the displacement moves towards the apex. Thus the fluid and the membrane form a coupled system and hence individual parts cannot resonate separately. As mentioned earlier, the position of the maximum displacement depends on the initial frequency; for example when the frequency is low the entire membrane moves at the tip but if the frequency is high then the oscillations are produced near the oval window. This explains why damage to certain regions of the membrane leads to hearing loss at certain frequencies. Thus, the high frequency response depends on the functional response at the base while low frequency depends on the condition of the apex. The organ of corti acts as a transducer and the motion of the basilar membrane is converted into an electrical signal and propagated along the nerve fibres to the auditory cortex of the brain. The threshold potential required to fire a cell depends on the frequency of stimulation. The processing at auditory cortex of the brain is still an unsolved problem.

13.6 Signal Transduction

Biologically significant information is generally transmitted through chemical signals. The message carriers are special chemicals. Messages are also transmitted by intracellular substances, like mRNA, cAMP (cyclic adenosine monophosphate), etc. Communications between cells and organs are carried out through substances like hormones. Neural communication, as seen earlier, is through neurotransmitters. Chemical signals like odours emanating from bugs, skunks, flowers etc., are useful in communication between organisms of the same species or different species. The antigens that stimulate the production of antibodies can also be considered as biological signals.

In any signal mechanism there is a sender, a receiver and mode of transport of signal. In biological systems the receiver of chemical signals are chemosensitive organs, chemoreceptor cells and other receptor molecules. For the recognition of the signal, the receiver molecule must have chemical specificity. The nature of the response depends on the concentration and nature of the signal substance. In most cases the chemical specificity is due to macromolecules which are the receptors. The molecules on the cell surface that enable cell-cell recognition (e.g. glycoproteins, glycolipids) can also be considered as chemical signals. The three stages in molecular recognition are: (a) capturing of the signal substance by the receptor molecules, (b) causing of the physiological effects and (c) removal of the signal substance by decomposition or by removal from the receptor.

13.6.1 Mode of transport

Transport of signals could be due to diffusion or convection. The diffusion process is effective only when the distance between receiver and sender is small (of the order of a few tens of nanometres) and the concentration of signal substance is high. For example, in nerve transmission the neurotransmitters travel only a distance of about 20 nm. In the case of hormones, which may travel from one organ to the other, the signal substance is first adsorbed on the cell membrane, and then diffused through the membrane for action. If the affinity for the signal substance to the receptor molecules on the membrane is high, then these molecules could accumulate specifically on this cell, and help in transporting the signal. Thus the strong affinity of the lipophilic surface of the olfactory organ for odour-producing substances helps insects to respond to even low concentrations of the signal substances in the air.

Convection is another method of transport of signal molecules and is especially useful when the distance between the sender and the receiver is large. The time taken for transport by diffusion is

$$t = x^2/D$$

where x is the distance to be travelled, and D diffusion coefficient. Hence the time taken for transport by diffusion over large distances is large. Convection plays an important role under these circumstances. For example odorous substances are known to move across distances of even several kilometres and yet be recognised at the destination.

13.6.2 Signal transduction in the cell

The chemical signals are recognised by receptor proteins. The binding of the signal molecules to the active site of the receptor is reversible, due to its non-covalent nature. The strength of the binding is also dependent on the extent of complementarity between the signal molecule and the active site of the receptor. The receptor active site could be, for example, the opening of an ion channel in the membrane of the muscle end plate when the neurotransmitter acetylcholine binds to the receptor molecule. Another such example is the binding of antigen molecule to the hypervariable region of antibodies. In the case of insect olfactory cells, even a single odour molecule is sufficient to trigger a nerve impulse. This is made possible by opening of a single ion channel in the sensory cell membrane.

If the signal substance is continuously produced, then accumulation of the signal substance in the transporting medium can take place. This has to be avoided by removal of the signal substance by decomposition, inactivation or by dilution due to diffusion. For example, in the antigen-antibody complex, irreversible chemical binding inactivates the signal substance. In the case of neurotransmitters the acetylcholine molecule is broken down very rapidly by acetylcholine esterase. The receptor surface of the olfactory cells is cleaned by being continually rinsed by convection and by the flow of fresh mucosa over the cells.

14

Origin and Evolution of Life

14.1 Introduction

Life can be broadly defined as something that can replicate and sustain itself. No other precise or exact definition of life is available. Biochemists and molecular biologists generally describe the properties of living organisms as the following.

1. Systems that are separated from the environment by an envelope (membrane).
2. Systems that have basic structures and mechanisms common to all. For example, the genetic code or the mechanism of translation of the genetic messages into polypeptides is essentially the same in all.
3. Systems that sustain themselves by an internally controlled metabolism, which is essentially the exchange of matter with the environment and the synthesis of energy rich compounds.

It is possible to find living systems that do not possess all three properties listed above. But certainly every system that can be called 'Life' will possess at least one or two of them.

Evolution can be defined as change or adaptation by the organism, in response to environmental factors. Evolution can be classified into chemical evolution, molecular evolution and Darwinian evolution. Chemical evolution is generally used to denote the prebiotic (before biology) formation of organic molecules in the period between the formation of earth and the first appearance of living cell. Molecular evolution refers to the changes in living systems that have occurred thereafter, leading eventually to the formation of unicellular and multicellular organisms. Darwinian evolution refers to the events thereafter, which have lead to the species seen today on earth. Figure 14.1 provides the evolutionary sequence schematically.

The origin of life is linked to the environmental conditions prevailing at that time and also to the elements available. Any attempts to propose a theory for origin of life should involve a chain of logical steps that are consistent with known laws of physics and chemistry. The interesting possibilities

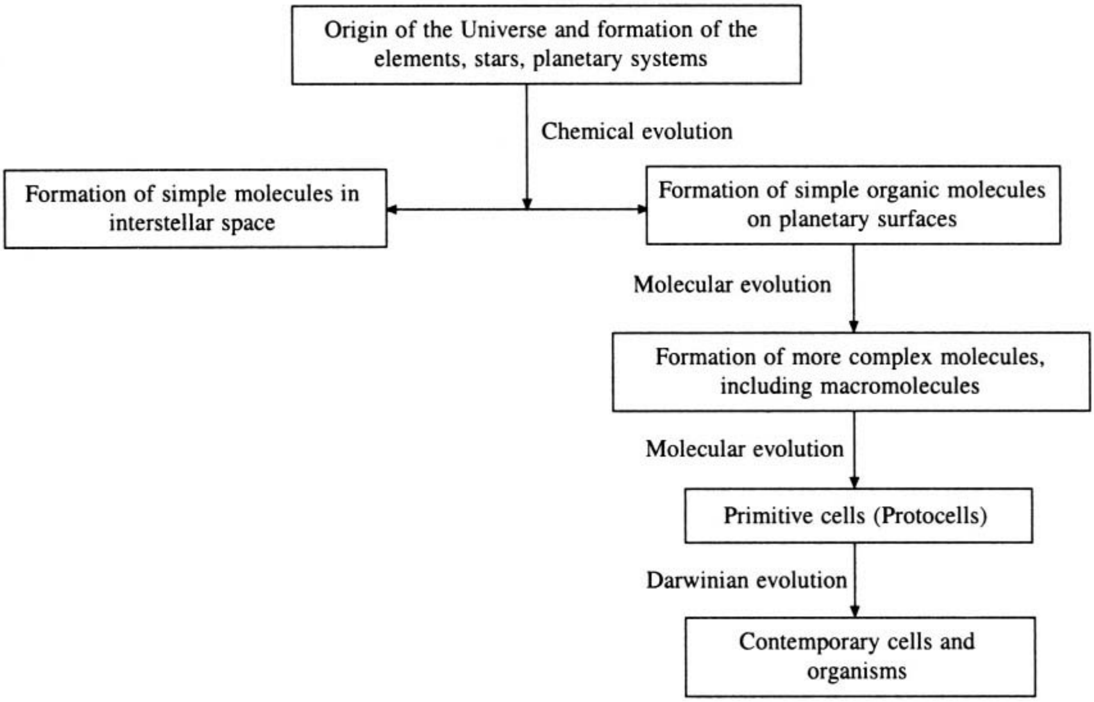


Fig. 14.1 Scheme of evolution

that arise in the discussion of origin of life include the following: (1) Life evolved elsewhere and was introduced on the earth. (2) Life as we know it today, originated fully developed, all-at-once, spontaneously by chance. (3) Life originated on earth first in a primitive form and then evolved gradually to the present-day forms. It is difficult to prove that life originated elsewhere and was later implanted on the earth. Similarly it is hard to expect the present day system to have originated spontaneously, i.e. to expect very high order to emerge from total disorder by chance. Thus we are left with the third option. To explore this option further, we have to consider how primitive molecules could have arisen on earth in the environment which prevailed at those early times, soon after the cloud of gases condensed into a solid planetary mass with continents and oceans. It becomes necessary to know the conditions prevailing at that time on earth.

14.2 Prebiotic Earth

The earth is believed to be 4500 to 4750 million years old and it has been suggested from fossil records that life in the form of unicellular organisms appeared about 3000 million years ago. The atomosphere of the primitive earth was devoid of molecular oxygen, which made it more suitable for chemical evolution. UV light could reach the earth undiminished due to the absence of an ozone layer. Other forms of energy that were available include ionising radiation, electric discharges (thunderstorms), heat etc. The most important organic elements like carbon, hydrogen, nitrogen and oxygen were also the most abundant elements on earth, besides helium, neon and silicon. The synthesis of carbon based compounds would have proceeded at fairly high rates in such an atmosphere. This is borne out by the fact that most of the interstellar compounds are carbon based, like CO, CH_4,

CH_3OH, cyanoacetylene etc. Also carbon compounds are more stable to the action of water, oxygen, UV light and heat.

It is very difficult to know exactly how life originated on eath. We can reconstruct the earliest stages of evolution only by conducting experiements in the laboratory under the conditions knowns to be prevalent at that time or by looking for evidence in fossils. Fossils are the remains of living matter after their slow mineralisation into stone. These fossils can only give information about the gross structure of the cell, as most of the organic matter would have been destroyed by repeated heating and cooling of the rocks due to change in the environment. However, molecular fossils, which is the name given to those structures or functions in the cell that are shared by diverse present-day organisms, may yield clues to build a coherent picture of primordial events.

14.3 Theories of Origin and Evolution of Life

The evolution of life after the formation of the first cell can be considered to have been controlled by fairly well established Darwinian principles. However, prebiotic evolution is an area of intense debate as it depends on the interpretation of fossil remains or on extrapolations from existing life forms. There are no fossils of the early period of evolution i.e. 4 billion years ago. Whenever experimental evidence is lacking, extrapolations have been made or theories have been proposed. Not surprisingly, these contain several contradictions.

Eigen proposed a theory for the self-organisation of macromolecules and the emergence from chaos of ordered structure, which in turn created information. He said that this is possible only when we have an autocatalytic open system, which is far away from equilibrium. Accidental ordering from chaos has a very low probability. For example, given that a polynucleotide chain of 100 nucleotides is to be formed from four nucleotides, the number of possible sequences is equal to 4^{100} or 10^{60}. For any particular primary structure to evolve, the probability is close to zero. Eigen proposed a model in which he assumed that macromolecules do not make an exact copy of themselves, (i.e. do not self-reproduce) but replicate by synthesising complementary strands. In a more complex system of reactions, there could be a cycle of molecules and reactions, each supporting the next in line. Such a cycle of reactions is called a hypercycle (Figure 14.2). Eigen's hypercycles have the following properties: (1) All species (molecules) linked through a cycle have a stable and controlled coexistence. For example, a hypercycle can be formed by a sequence of reactions in which the product of one reaction may become the reactant of the next reaction and so on. The citric acid cycle is one such set of reactions. (2) A hypercycle can increase or decrease in size if these changes have a selective advantage. (3) Once established, a hypercycle has a distinct advantage over a new, yet-to-be-established species (of hypercycle). To explain how these hypercycles come into existence in the first place, Eigen proposed that the first self-reproducing system with enough information content should have been a short tRNA-type molecule. This lead to the proposal of a so-called 'RNA world', in which the catalytic molecules as well as the information storage and transfer molecules are RNA. Eigen and Schuster have extended this theory with models for the origin of the genetic code and the translation system.

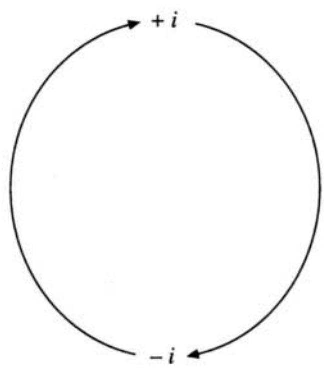


Fig. 14.2 Eigen's hypercycle

Kuhn's model proposes the formation of short polynucleotide chain like pre-tRNA molecules that have secondary and tertiary structure. The periodic changes in the environment act as a selection factor. Molecules are assumed to have been synthesised in the pores of clay minerals. This theory also proposes steps related to self-assembly of tRNA molecules, synthesis of proteins, formation of membranes, the genetic code etc.

Ebeling and Feistel's theory assumes that compartmentation of the reactants and products started in the early stages of chemical evolution. This compartmentation, also known as coacervate formation, gave rise to micro-reactors, which played the role of primitive cells during early evolution.

14.4 Laboratory Experiments on the Formation of Small Molecules

At the molecular level all forms of life appear to be similar and hence it is not inappropriate to assume that life on earth descended from single ancestral life form. Chemists have shown that essential building blocks of life can be synthesised from elements that were present on the primitive earth. This is also known as prebiotic synthesis. Miller's experiment showed for the first time that simple amino acids could be assembled spontaneously from the constituents of primitive earth.

14.4.1 Prebiotic synthesis of amino acids—Miller and Urey's experiment

The experiment attempted to simulate the pre-biotic atmosphere of the early earth. Air was pumped out of a small flask and then a mixture of ammonia, methane and hydrogen was added. Water vapour was introduced by adding water and boiling it. This also helped to mix and circulate the gases into a larger flask that was connected to the smaller one. An electric spark was generated across a spark gap set into the arrangement (Figure 14.3). The product of the discharge were condensed by a condenser and washed through a U-tube into the small flask. The non-volatile products remained in the small flask, but the volatile products re-circulated past the spark. The reduced forms of carbon, nitrogen and oxygen in the gas phase represented the prebiotic atmosphere; the liquid phase represented the ocean. When the products were analysed, it was seen that many different organic molecules had been synthesised. The results of Miller and Urey's experiment were unexpected in two respects. A small number of relatively simple compounds accounted for a large proportion of the products. Further the major products themselves were not a random selection of organic compounds but included a surprising number of substances that occur in living organisms (Table 14.1).

Table 14.1 Compounds that were synthesised in the Miller-Urey experiment

Compound	Yield (%)
Glycine	2.1
Sarcosine	0.25
α-Alanine	1.7
α-amino isobutyric acid	0.007
Aspartic acid	0.024
Formic acid	4.0
Urea	0.034
β-Alanine	0.76

α-Alanine was produced in greater quantity than **β-alanine.** When the experiment was performed over a longer period, large amounts of cyanide and aldehydes were formed during the first 125 hours

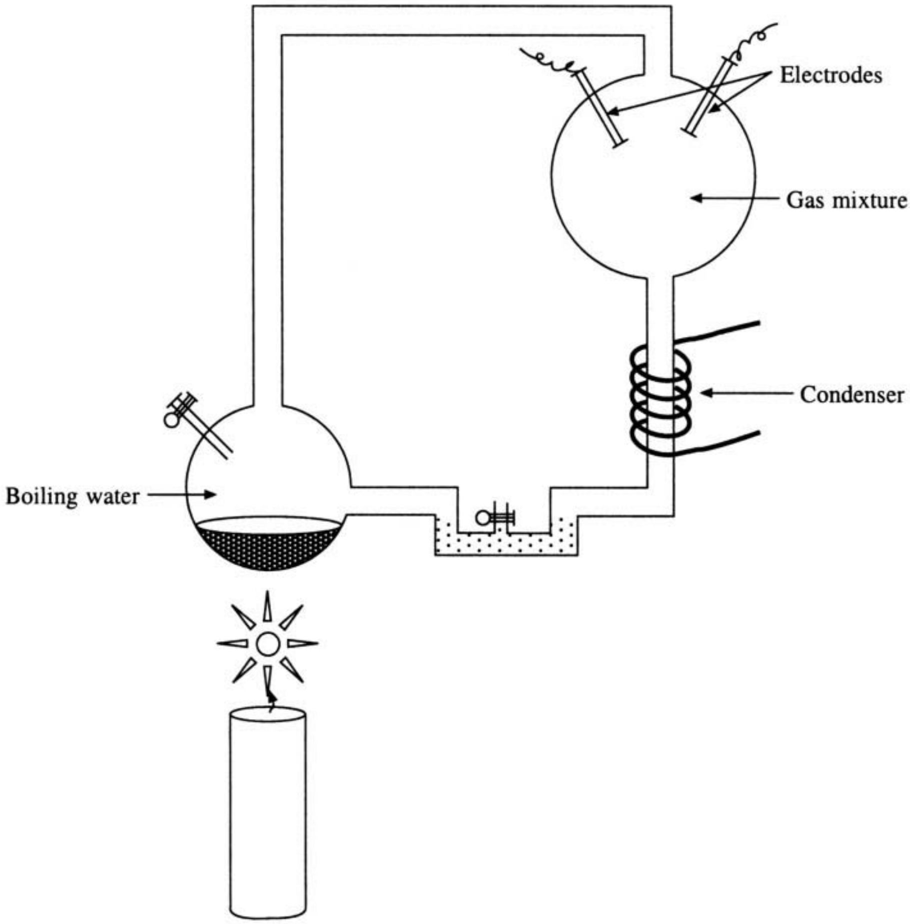


Fig. 14.3 Miller and Urey's experimental setup

of sparking and then the rate of their synthesis fell off, whereas the amino acids were formed at a more or less constant rate throughout the run. The mechanism proposed for the formation of glycine is as follows:

$$CH_4 + H_2O \rightarrow H_2CO + 2H_2$$

$$CH_4 + NH_3 \rightarrow HCN + 3H_2$$

$$H_2CO + HCN + NH_3 \rightarrow CH_2(NH_2){-}C \equiv N + H_2O$$

$$CH_2(NH_2){-}C \equiv N + 2H_2O \rightarrow CH_2(NH_2){-}COOH + NH_3$$

The general reaction can be written as

$$RCHO + HCN + NH_3 \rightarrow \underset{\displaystyle NH_2}{\underset{|}{RCH}}{-}C{\equiv}N + H_2O$$

$$\underset{\displaystyle NH_2}{\underset{|}{RCH}}{-}C{\equiv}N + 2H_2O \rightarrow \underset{\displaystyle NH_2}{\underset{|}{RCH}}{-}COOH + NH_3$$

The above reactions depend on the formation of hydrogen cyanide.

14.4.2 Prebiotic synthesis of purines, pyrimidines and nucleosides

The most important purines are adenine, guanine and, to a lesser extent hypoxanthine (Figure 14.4). If concentrated ammonia solution containing 1-10 M cyanide is refluxed, adenine is formed in yields up to 0.5 % along with large amounts of the usual hydrogen cyanide polymer. This reaction does not easily explain the synthesis of purines under prebiotic conditions since useful yields of adenine cannot be obtained except in the presence of 1.0 M or stronger ammonia. The highest reasonable concentration of ammonia or ammonium ion that can be postulated in oceans and lakes on the primitive earth is about 0.01 M. In order to accommodate this fact, an alternative mechanism has been proposed. This mechanism makes use of an unusual photochemical rearrangement, which does not involve additional reagents as shown in Figure 14.5.

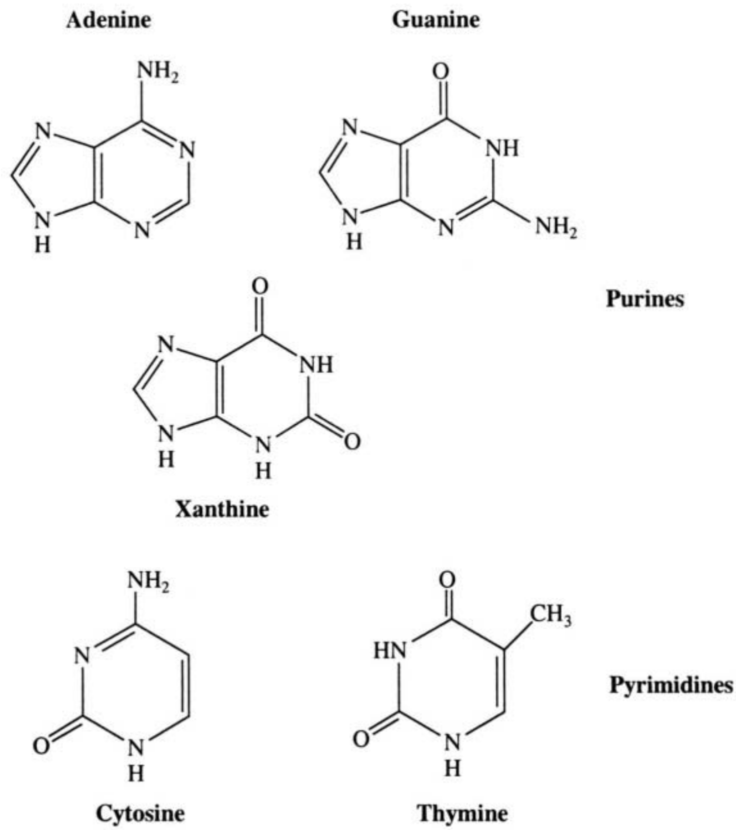


Fig. 14.4 Purines and pyrimidines

Synthesis of pyrimidines, viz. C, U and T has not been studied extensively. Action of an electric discharge on a mixture of methane and nitrogen led to the formation of cyanoacetylene. Cyanoacetylene reacts with aqueous cyanate to give cytosine in quite good yield (20%). Uracil may be obtained by hydrolysis of cytosine.

14.4.3 Sugars

Formaldehyde is one of the simplest of organic molecules that is readily formed under pre-biotic

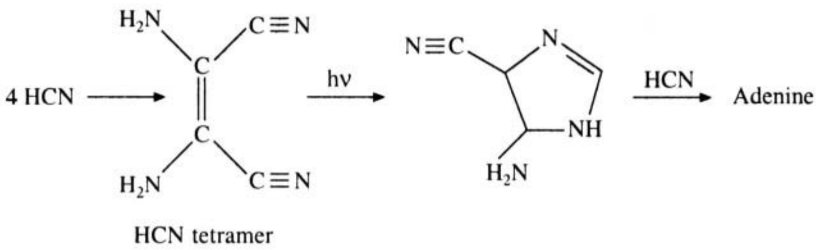


Fig. 14.5 Photochemical rearrangement

conditions, as attested by its presence in intersteller clouds. Sugars are formed by treating formaldehyde with certain alkaline catalysts like sodium hydroxide.

$$nH_2C = O \xleftrightarrow[\text{Inorganic ashes as catalysts}]{} \text{triose, tetrose, pentose, hexose}$$

While the suggested mode of synthesis is attractive, it is not without difficulties, such as the following. (a) Sugars are unstable in aqueous solution. (b) The mixtures of sugars obtained usually contain a very small percentage of ribose, which is the constituent of nucleic acids. (c) The reaction requires a minimum concentration of 0.01 M formaldehyde.

Groth and Suess also produced organic compounds by irradiating a gaseous mixture of H_2O and CO_2 during their work related to the origin of life. Similarly Garissian and coworkers bombarded an aqueous solution of carbon dioxide with some Fe^{2+} and formed formaldehyde, formic acid and succinic acid.

14.4.4 Nucleotide synthesis

Ribose exists in two different ring forms, viz. furanose and pyranose (Figure 14.6). The natural nucleosides are all furanosides, but most organic syntheses give a mixture of isomers. Further, heterocyclic bases can form bonds to a sugar at more than one position. Thus, when purines are heated along with sugars in the presence of certain inorganic salts, a mixture of nucleosides is obtained in reasonable yield. A good percentage of the yield consists of natural nucleosides. The mixture of salts obtained by evaporating seawater is one of the most effective catalysts tested so far. Two alternative approaches to the synthesis of nucleosides are possible. Either a sugar can be attached to the preformed base or a base can be constructed on the preformed sugar. The later model has been

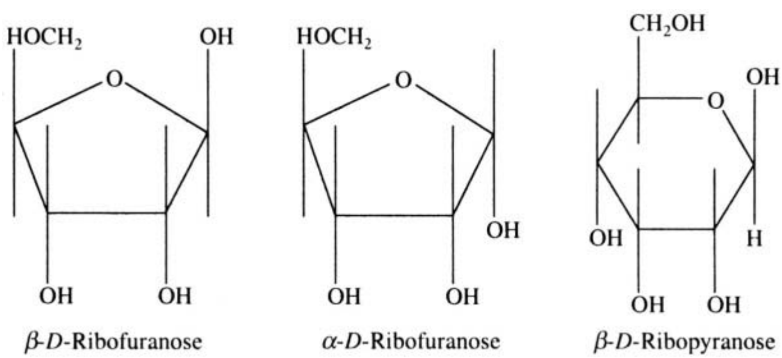


Fig. 14.6 Furanose and pyranose

accepted as the correct one in the case of prebiotic synthesis. Phosphorylation of nucleosides can be achieved by heating them with acid phospates such as NaH_2PO_4 and $Ca(H_2PO4)_2$. The reaction occurs quite rapidly at 130°C. This yields the required nucleotides. Thus, there is substantial evidence to prove that a variety of small organic molecules relevant to living systems could be synthesised under prebiotic conditions.

14.4.5 Optical activity

Most biological molecules exhibit optical activity i.e. they rotate the plane of polarisation of polarised light. In an organic synthesis reaction, optically inactive starting materials will give rise to optically inactive products. In this case, the product molecules may either be individually optically inactive or the product may contain L and D molecules in equal quantities so that the product solution is optically inactive. There is no obvious reason why molecules of one chirality should be produced more than the other in a symmetric environment. In biomolecules only L-amino acids and D-ribose (Figure 14.7) occur, even though molecules with opposite chirality have an equal chance of forming a polymer. This asymmetry needs to be explained. Among the many plausible explanations that have been offered are the following. There are several asymmetric phenomena in nature that could contribute to an asymmetric environment. For example sky light is circularly polarised to a small extent. Radioactive β-decay exhibits handedness or chirality. Similarly the rotation of earth, the magnetic field of earth, etc. may also contribute to the asymmetric environment and this could lead to the preference for one handedness of the synthesised molecules over the other.

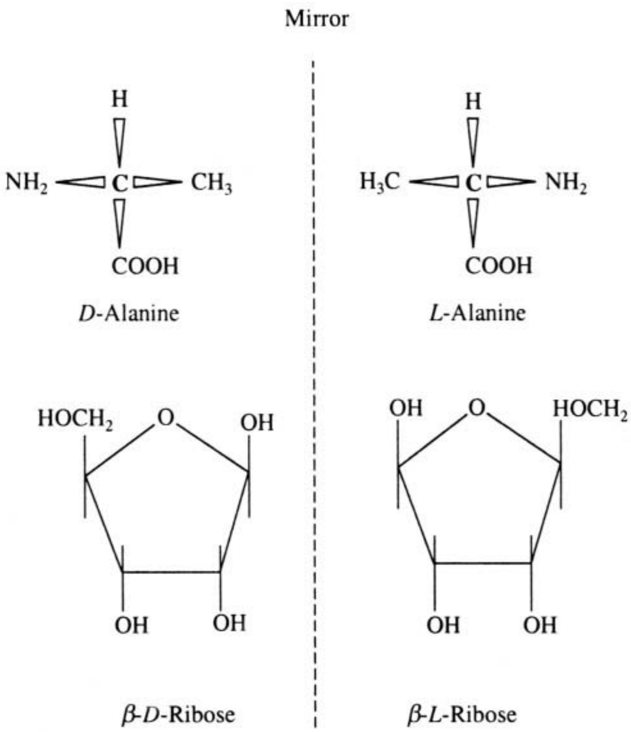


Fig. 14.7 L and D amino acids and riboses

14.4.6 Polymerisation

Once the relevant monomers have been synthesised, the second step in prebiotic chemical evolution is the aggregation or self-organisation of molecules. The following experiment demonstrates how this could have occurred for proteins. When a pure amino acid solution is heated to about 180°C, varying quantities of diketopiperazine (a cyclic dipeptide) and small amounts of polypeptides are formed. If a mixture of 20 natural amino acids is used in this experiment, all of them are incorporated in the polypeptide like materials. These materials are called "proteinoids". Most of the bonds between the different residues in proteinoids are the normal HN–CO peptide bonds. But those formed from mixtures which include aspartic acid and glutamic acid also contain α-β and α-γ amide linkage (Figure 14.8). A surprising feature is that many of the polypeptides thus synthesised begin with glutamic acid. If the proteinoids are boiled in water, a dispersion of microspheres of size 2 microns is obtained. Some microspheres are surrounded by structures that look like bilayer membranes. Some of them look much like simple bacteria under a microscope and they can be made to divide into two, or to form buds, under appropriate conditions. These proteinoids, also known as thermal polypetides, display a catalytic activity in some reactions. However, there are some facts, which argue against thermal synthesis of polypeptides as a mechanism for prebiotic polypeptide synthesis. Temperatures of the order of 150 to 180° C are required. In the prebiotic atmosphere at the time when the monomers are available, such temperature would be found only several kilometres below the surface of the earth

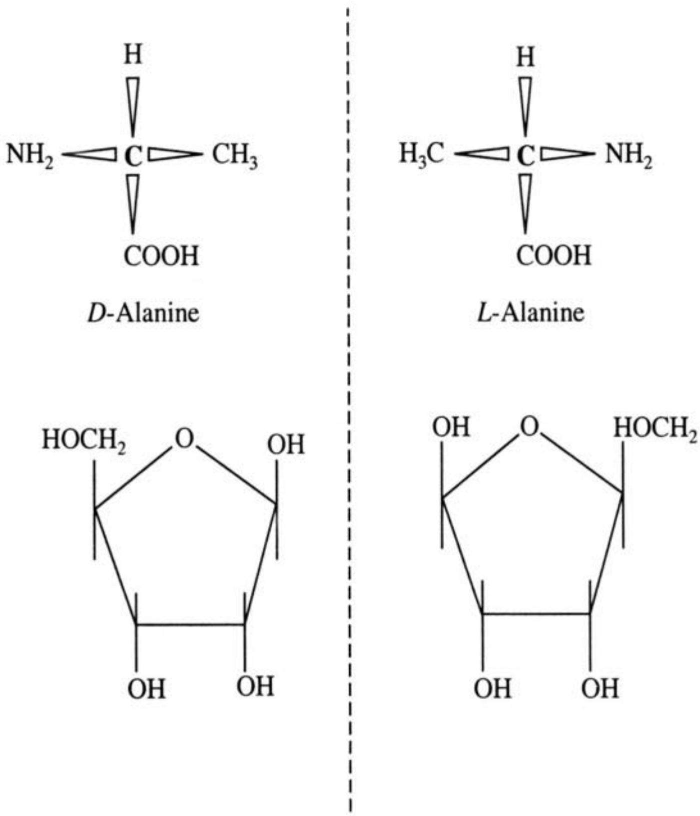


Fig. 14.8 α-β **and** α-γ **amide linkages**

or in some hot springs or in volcanoes. Other condensation mechanisms have also been proposed for condensation of amino acids into polypeptides.

The formation of short oligonucleotides from the monomers could also have occurred purely by thermal processes. However, the specificity that exists in the polymerisation of nucleotides in the present day world, viz. the $5' \rightarrow 3'$ direction of the polynucleotide chain, may not have been present under prebiotic conditions.

The prebiotic synthesis of fundamental molecules described above requires special conditions, and all of them may not have existed at the same time at the same place. For example the reactions which produce the monomers require very high concentrations of the reactants and the subsequent yield of biologically important monomers is low. It is believed that repeated evaporation and rehydration, freezing and thawing might have concentrated both the reactants and the products. The building blocks of life must have been condensed into still larger molecules by heating or chemical treatment. Thus, when the prebiotic solution of molecules (the prebiotic 'soup') cooled, it could have contained oligonucleotides, peptides and oligosaccharides. However, a mere collection of these molecules will not make a living system as we know it today.

To summarise, though the above experiments and arguments do not unequivocally establish the mechanisms for origin of life on eath, and for its subsequent evolution, they definitely show that the conditions on primitive earth were suitable for the emergence of some primitive form of life.

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